# Swelio

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# Table of Contents


## Symbol Reference 1

**Swelio.Engine Namespace** 1

Address Class 2

Address.Address Constructor 3

Address Methods 3

Address.Parse Method 3

Address Properties 3

Address.Municipality Property 3

Address.Street Property 4

Address.Zip Property 4

Card Class 4

Card.Card Constructor 5

Card Methods 6

DisplayCertificate Method 6

Dispose Method 6

Card.EncodeCertificate Method 6

EncodePhoto Method 7

Card.ReadAddress Method 7

Card.ReadCertificate Method 7

Card.ReadIdentity Method 7

Card.ReadPhoto Method 8

Save Method 8

Card.SaveCertificate Method 8

Card.SavePhoto Method 9

Card.SaveQRCode Method 9

Card.VerifyPinCode Method 9

Card Properties 9

Card.AppletVersion Property 9

Card.EidCard Property 10

Card.Inserted Property 10

Card.SerialNumber Property 10

Card.Tag Property 10

CardDate Class 10

CardDate Methods 11

CardDate.ToDate Method 11

CardDate.ToLongDateString Method 11

CardDate.ToShortDateString Method 12


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CardEventArgs Class 12
CardEventArgs.CardEventArgs Constructor 12
CardEventArgs Properties 13
CardEventArgs.ReaderNumber Property 13
CardNumber Class 13
CardNumber Methods 13
CardNumber.ToString Method 13
CardReader Class 14
CardReader.CardReader Constructor 15
CardReader Fields 15
CardReader.card Field 15
CardReader Methods 15
CardReader.ActivateCard Method 16
CardReader.DeactivateCard Method 16
Dispose Method 16
GetCard Method 16
CardReader Properties 17
CardReader.CardActivated Property 17
CardReader.CardPresent Property 17
CardReader.Index Property 17
CardReader.Name Property 17
CardReader.Selected Property 17
CardReader.Tag Property 17
DocumentContainer Class 18
DocumentContainer.DocumentContainer Constructor 18
DocumentContainer Methods 19
DocumentContainer.AddFile Method 19
DocumentContainer.Close Method 19
DocumentContainer.Save Method 19
SelectCertificate Method 19
EIDAddress Class 20
EIDAddress Fields 21
EIDAddress.municipality Field 21
EIDAddress.street Field 21
EIDAddress.zip Field 21
EIDCertificate Class 21
EIDCertificate Fields 22
EIDCertificate.certificate Field 22
EIDCertificate.certificateLength Field 22
EIDDataLength Class 22
EIDDataLength Fields 24

EIDDataLength.EidMaxBirthDateLen Field 24
EIDDataLength.EidMaxBirthplaceLen Field 24
EIDDataLength.EidMaxBrexitMention1Len Field 24
EIDDataLength.EidMaxBrexitMention2Len Field 24
EIDDataLength.EidMaxCardNumberLen Field 24
EIDDataLength.EidMaxCertLen Field 24
EIDDataLength.EidMaxChipNumberLen Field 25
EIDDataLength.EidMaxDateAndCountryOfProtectionLen Field 25
EIDDataLength.EidMaxDateBeginLen Field 25
EIDDataLength.EidMaxDateEndLen Field 25
EIDDataLength.EidMaxDeliveryMunicipalityLen Field 25
EIDDataLength.EidMaxDocumentTypeLen Field 25
EIDDataLength.EidMaxDuplicateLen Field 25
EIDDataLength.EidMaxFirstNameLen Field 25
EIDDataLength.EidMaxMemberOfFamilyLen Field 26
EIDDataLength.EidMaxMiddleNameLen Field 26
EIDDataLength.EidMaxMunicipalityName Field 26
EIDDataLength.EidMaxNameLen Field 26
EIDDataLength.EidMaxNationalityLen Field 26
EIDDataLength.EidMaxNationalNumberLen Field 26
EIDDataLength.EidMaxNobleConditionLen Field 26
EIDDataLength.EidMaxPictureLen Field 27
EIDDataLength.EidMaxRegionalFileNumberLen Field 27
EIDDataLength.EidMaxSexLen Field 27
EIDDataLength.EidMaxSpecialOrganizationLen Field 27
EIDDataLength.EidMaxSpecialStatusLen Field 27
EIDDataLength.EidMaxStreetNameLen Field 27
EIDDataLength.EidMaxVAT1Len Field 27
EIDDataLength.EidMaxVAT2Len Field 28
EIDDataLength.EidMaxWorkPermitTypeLen Field 28
EIDDataLength.EidMaxZipLen Field 28
EIDError Class 28
EIDError Fields 29
EIDError.Code Field 29
EIDError.Description Field 29
EIDIdentity Class 29
EIDIdentity Fields 31
EIDIdentity.birthDate Field 31
EIDIdentity.birthLocation Field 31
EIDIdentity.brexitMention1 Field 31
EIDIdentity.brexitMention2 Field 31

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EIDIdentity.cardNumber Field 31
EIDIdentity.chipNumber Field 31
EIDIdentity.dateAndCountryOfProtection Field 32
EIDIdentity.documentType Field 32
EIDIdentity.duplicate Field 32
EIDIdentity.extendedMinority Field 32
EIDIdentity.firstName1 Field 32
EIDIdentity.firstName2 Field 32
EIDIdentity.memberOfFamily Field 33
EIDIdentity.municipality Field 33
EIDIdentity.name Field 33
EIDIdentity.nationality Field 33
EIDIdentity.nationalNumber Field 33
EIDIdentity.nobleCondition Field 33
EIDIdentity.regionalFileNumber Field 33
EIDIdentity.sex Field 34
EIDIdentity.specialOrganization Field 34
EIDIdentity.validityDateBegin Field 34
EIDIdentity.validityDateEnd Field 34
EIDIdentity.vat1 Field 34
EIDIdentity.vat2 Field 34
EIDIdentity.whiteCane Field 35
EIDIdentity.workPermitType Field 35
EIDIdentity.yellowCane Field 35
EIDPicture Class 35
EIDPicture Fields 35
EIDPicture.picture Field 36
EIDPicture.pictureLength Field 36
Encryption Class 36
Encryption Methods 37
Encryption.CheckFileMD5 Method 37
Encryption.CheckFileSHA1 Method 37
CheckMD5 Method 37
CheckSHA1 Method 38
DecryptFile Method 38
EncryptFile Method 39
Encryption.GenerateAuthenticationSignature Method 39
Encryption.GenerateNonRepudiationSignature Method 40
Encryption.GetFileMD5 Method 40
Encryption.GetFileSHA1 Method 40
GetMD5 Method 40

GetSHA1 Method 41
Encryption.VerifySignature Method 41
FileOperations Class 42
FileOperations Methods 44
FileOperations.ClearFileAttributes Method 44
FileOperations.CopyFile Method 44
FileOperations.CreateUnicodeFile Method 44
FileOperations.DeleteFile Method 44
FileOperations.DeleteToRecycleBin Method 44
FileOperations.DirectoryExists Method 45
FileOperations.ExcludeTrailingPathDelimiter Method 45
FileOperations.FileExists Method 45
FileOperations.FileExtensionIs Method 45
FileOperations.FileIsAnimatedGif Method 45
FileOperations.FileIsExe Method 45
FileOperations.FileIsIcon Method 46
FileOperations.FileIsImage Method 46
FileOperations.FileIsLink Method 46
FileOperations.FileOrFolderExists Method 46
FileOperations.FullPath Method 46
FileOperations.GetExtension Method 47
FileOperations.GetFilesCount Method 47
FileOperations.GetFileSize Method 47
FileOperations.IncludeTrailingPathDelimiter Method 47
FileOperations.IsDirectory Method 47
FileOperations.IsUnicodeFile Method 48
FileOperations.IsValidFileName Method 48
FileOperations.IsValidPathName Method 48
FileOperations.RenameFile Method 48
FileOperations.ShellCopyFile Method 48
FileOperations.StripFileName Method 48
FileOperations.UnExpandPath Method 49
Identity Class 49
Identity Methods 51
Identity.Clear Method 51
Identity Properties 51
Identity.BirthDate Property 51
Identity.BirthLocation Property 51
Identity.BrexitMention1 Property 51
Identity.BrexitMention2 Property 51
Identity.CardNumber Property 52

Identity.ChipNumber Property 52
Identity.DateAndCountryOfProtection Property 52
Identity.DocumentType Property 52
Identity.Duplicate Property 52
Identity.ExtendedMinority Property 52
Identity.FirstName1 Property 53
Identity.FirstName2 Property 53
Identity.MemberOfFamily Property 53
Identity.Municipality Property 53
Identity.Nationality Property 53
Identity.NationalityISO Property 53
Identity.NationalNumber Property 54
Identity.NobleCondition Property 54
Identity.RegionalFileNumber Property 54
Identity.Sex Property 54
Identity.SpecialOrganization Property 54
Identity.Surname Property 54
Identity.ValidityDateBegin Property 54
Identity.ValidityDateEnd Property 55
Identity.Vat1 Property 55
Identity.Vat2 Property 55
Identity.WhiteCane Property 55
Identity.WorkPermitType Property 55
Identity.YellowCane Property 55
InteropHelper Class 56
InteropHelper Methods 57
InteropHelper.ClearUnusedMemory Method 57
InteropHelper.GetStartup Method 57
InteropHelper.SetStartup Method 57
InteropHelper.ShowScreenKeyboard Method 57
InteropHelper.ShowWindowsError Method 58
InteropHelper.TurnMonitorOff Method 58
InteropHelper.TurnMonitorOn Method 58
InteropHelper Properties 58
InteropHelper.IsMediaCenter Property 58
InteropHelper.IsMultiTouchReady Property 58
InteropHelper.IsTabletPC Property 58
InteropHelper.MetroActive Property 59
InteropHelper.RunningOnVista Property 59
InteropHelper.RunningOnWin10 Property 59
InteropHelper.RunningOnWin7 Property 59

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InteropHelper.RunningOnWin8 Property 59
InteropHelper.RunningOnXP Property 59
Manager Class 60
Manager.Manager Constructor 61
Manager Methods 61
Manager.Dispose Method 61
GetReader Method 61
Manager.GetReaderName Method 62
Manager.ListReaders Method 62
Manager.ReloadReadersList Method 62
Manager.SendAPDU Method 62
Manager.SetDefaultReader Method 62
Manager Properties 63
Manager.Active Property 63
Manager.ReaderCount Property 63
Manager.TraceEvents Property 63
Manager.TraceHardwareEvents Property 63
Manager.TraceServiceEvents Property 63
Manager Events 63
Manager.CardInserted Event 64
Manager.CardRemoved Event 64
Manager.ReadersListChanged Event 64
NationalNumber Class 64
NationalNumber Methods 64
NationalNumber.ToString Method 65
NativeMethods Class 65
NativeMethods Methods 73
NativeMethods.ActivateCardEx Method 73
NativeMethods.AddFileToContainer Method 73
NativeMethods.CardDecryptFile Method 73
NativeMethods.CardEncryptFile Method 74
NativeMethods.CardSignCadesT Method 74
NativeMethods.CardSignCadesTEx Method 74
NativeMethods.CardSignCMS Method 74
NativeMethods.CardSignCMSEx Method 74
NativeMethods.CertSignCadesT Method 74
NativeMethods.CertSignCadesTData Method 75
NativeMethods.CertSignCadesTEx Method 75
NativeMethods.CertSignCMS Method 75
NativeMethods.CertSignCMSData Method 75
NativeMethods.CertSignCMSEx Method 75

*Made with **Doc-O-Matic**.*

NativeMethods.CheckMD5 Method 75
NativeMethods.CheckSHA1 Method 76
NativeMethods.ClearFileAttributes Method 76
NativeMethods.ClearUnusedMemory Method 76
NativeMethods.ContainerCertificate Method 76
NativeMethods.ContainerEidCertificate Method 76
NativeMethods.ContainerPickCertificate Method 76
NativeMethods.CreateCardBuffer Method 77
NativeMethods.CreateUnicodeFile Method 77
NativeMethods.CurrentIPAddress Method 77
NativeMethods.DeactivateCardEx Method 77
NativeMethods.DecryptFileAES Method 77
NativeMethods.DeleteCardBuffer Method 77
NativeMethods.DeleteObject Method 78
NativeMethods.DeleteToRecycleBin Method 78
NativeMethods.DirectoryExists Method 78
NativeMethods.DisplayCertificate Method 78
NativeMethods.EmptyRecycleBin Method 78
NativeMethods.EncodeCertificate Method 78
NativeMethods.EncodePhoto Method 79
NativeMethods.EncryptFileAES Method 79
NativeMethods.FileCopy Method 79
NativeMethods.FileDelete Method 79
NativeMethods.FileExists Method 79
NativeMethods.FileExtensionIs Method 79
NativeMethods.FileGetSize Method 80
NativeMethods.FileIsExe Method 80
NativeMethods.FileIsIcon Method 80
NativeMethods.FileIsImage Method 80
NativeMethods.FileIsLink Method 80
NativeMethods.FileOrFolderExists Method 80
NativeMethods.FileRename Method 81
NativeMethods.FreeContainer Method 81
NativeMethods.FullPath Method 81
NativeMethods.GenerateAuthenticationSignatureEx Method 81
NativeMethods.GenerateBMP Method 81
NativeMethods.GenerateNonRepudiationSignatureEx Method 81
NativeMethods.GeneratePNG Method 82
NativeMethods.GenerateQRCodeEx Method 82
NativeMethods.GetCardBufferA Method 82
NativeMethods.GetCardBufferSize Method 82

*Made with **Doc-O-Matic**.*

NativeMethods.GetCardBufferW Method 82
NativeMethods.GetCardSerialNumber Method 82
NativeMethods.GetCardVersion Method 83
NativeMethods.GetEncodedCertificateSize Method 83
NativeMethods.GetEncodedPhotoSize Method 83
NativeMethods.GetFileMD5 Method 83
NativeMethods.GetFilesCount Method 83
NativeMethods.GetFileSHA1 Method 83
NativeMethods.GetHBitmap Method 84
NativeMethods.GetISOCode Method 84
NativeMethods.GetMD5 Method 84
NativeMethods.GetReaderIndex Method 84
NativeMethods.GetReaderName Method 84
NativeMethods.GetReaderNameLen Method 84
NativeMethods.GetReadersCount Method 85
NativeMethods.GetSelectedReaderIndex Method 85
NativeMethods.GetSHA1 Method 85
NativeMethods.GetStartup Method 85
NativeMethods.HibernateWindows Method 85
NativeMethods.IgnoreHardwareEvents Method 85
NativeMethods.IgnoreServiceEvents Method 86
NativeMethods.InitializeContainer Method 86
NativeMethods.IsAnimatedGIF Method 86
NativeMethods.IsCardActivatedEx Method 86
NativeMethods.IsCardPresentEx Method 86
NativeMethods.IsCardStillInsertedEx Method 86
NativeMethods.IsConnectedToInternet Method 87
NativeMethods.IsDirectory Method 87
NativeMethods.IsEIDCardEx Method 87
NativeMethods.IsEngineActive Method 87
NativeMethods.IsMale Method 87
NativeMethods.IsMediaCenter Method 87
NativeMethods.IsMetroActive Method 88
NativeMethods.IsMultiTouchReady Method 88
NativeMethods.IsTabletPC Method 88
NativeMethods.IsUnicodeFile Method 88
NativeMethods.IsValidFileName Method 88
NativeMethods.IsValidPathName Method 88
NativeMethods.IsWindows10 Method 89
NativeMethods.IsWindows7 Method 89
NativeMethods.IsWindows8 Method 89

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NativeMethods.IsWindowsVista Method 89
NativeMethods.IsWindowsXP Method 89
NativeMethods.IsWOW64 Method 89
NativeMethods.LoadCertificate Method 90
NativeMethods.LoadPhoto Method 90
NativeMethods.PathUnExpandEnvStrings Method 90
NativeMethods.PortAvailable Method 90
NativeMethods.ReadAddressEx Method 90
NativeMethods.ReadAuthenticationCertificate Method 90
NativeMethods.ReadCaCertificate Method 91
NativeMethods.ReadIdentityEx Method 91
NativeMethods.ReadNonRepudiationCertificate Method 91
NativeMethods.ReadPhotoEx Method 91
NativeMethods.ReadRootCaCertificate Method 91
NativeMethods.ReadRrnCertificate Method 91
NativeMethods.RecycleBinEmpty Method 92
NativeMethods.ReloadReadersList Method 92
NativeMethods.RemoveCallback Method 92
NativeMethods.RemoveStartup Method 92
NativeMethods.SaveAuthenticationCertificateEx Method 92
NativeMethods.SaveCaCertificateEx Method 92
NativeMethods.SaveCardToToXMLStreamExA Method 93
NativeMethods.SaveCardToToXMLStreamExW Method 93
NativeMethods.SaveCardToXmlEx Method 93
NativeMethods.SaveCardToXmlExAnsi Method 93
NativeMethods.SaveContainer Method 93
NativeMethods.SaveNonRepudiationCertificateEx Method 93
NativeMethods.SavePersonCsvToStreamA Method 94
NativeMethods.SavePersonCsvToStreamW Method 94
NativeMethods.SavePersonToCsvEx Method 94
NativeMethods.SavePersonToCsvExAnsi Method 94
NativeMethods.SavePhotoAsBitmapEx Method 94
NativeMethods.SavePhotoAsJpegEx Method 94
NativeMethods.SaveRootCaCertificateEx Method 95
NativeMethods.SaveRrnCertificateEx Method 95
NativeMethods.SelectReader Method 95
NativeMethods.SelectReaderByName Method 95
NativeMethods.SendAPDU Method 95
NativeMethods.SetCallback Method 95
NativeMethods.SetStartup Method 96
NativeMethods.ShellCopyFile Method 96

NativeMethods.ShowError Method 96
NativeMethods.ShutdownWindows Method 96
NativeMethods.StartEngine Method 96
NativeMethods.StopEngine Method 96
NativeMethods.StripFileName Method 97
NativeMethods.SuspendWindows Method 97
NativeMethods.TurnMonitorOff Method 97
NativeMethods.TurnMonitorOn Method 97
NativeMethods.VerifyPinEx Method 97
NativeMethods.VerifySignature Method 97
Network Class 98
Network Methods 98
Network.PortAvailable Method 98
Network Properties 98
Network.CurrentIPAddress Property 99
Network.IsConnectedToInternet Property 99
PostalAddress Class 99
PostalAddress.PostalAddress Constructor 100
PostalAddress Properties 100
PostalAddress.House Property 100
PostalAddress.Municipality Property 100
PostalAddress.Street Property 100
PostalAddress.Zip Property 100
QRCode Class 101
QRCode Methods 101
GenerateBitmap Method 101
GeneratePng Method 102
GetBitmap Method 102
RecycleBin Class 103
RecycleBin Methods 103
RecycleBin.Clear Method 104
RecycleBin.Send Method 104
RecycleBin.SendSilent Method 104
RecycleBin Properties 104
RecycleBin.IsEmpty Property 104
WindowsOperations Class 104
WindowsOperations Methods 105
WindowsOperations.Hibernate Method 105
WindowsOperations.Shutdown Method 105
WindowsOperations.Suspend Method 105
Swelio.Engine.CardEvent Enumeration 105

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Swelio.Engine.CertificateType Enumeration 106
Swelio.Engine.ChangeWindowMessageFilterFlags Enumeration 106
Swelio.Engine.DocumentType Enumeration 106
Swelio.Engine.ErrorCorrectionLevel Enumeration 107
Swelio.Engine.ExportFormat Enumeration 107
Swelio.Engine.FileEncoding Enumeration 107
Swelio.Engine.Gender Enumeration 108
Swelio.Engine.PhotoFormat Enumeration 108
Swelio.Engine.ShutdownOption Enumeration 108
Swelio.Engine.ReaderCallbackDelegate Type 109

**Files** **109**
Address.cs 112
Card.cs 113
CardDate.cs 113
CardEvent.cs 113
CardEventArgs.cs 113
CardNumber.cs 114
CardReader.cs 114
CertificateType.cs 114
DocumentContainer.cs 114
DocumentType.cs 114
Encryption.cs 115
ExportFormat.cs 115
FileEncoding.cs 115
FileOperations.cs 115
Gender.cs 116
Identity.cs 116
InteropHelper.cs 116
Manager.cs 116
NationalNumber.cs 117
NativeMethods.cs 117
NativeTypes.cs 117
Network.cs 117
PhotoFormat.cs 118
PostalAddress.cs 118
QRCode.cs 118
RecycleBin.cs 118
WindowsOperations.cs 118

# Index a

# 1 Symbol Reference

## 1.1 Swelio.Engine Namespace

This is namespace Swelio.Engine.

**Classes**

| | Name | Description |
|---|---|---|
| | Address (see page 2) | Represents a postal address of the person |
| | Card (see page 4) | This class represents a Belgian EID card |
| S | CardDate (see page 10) | Handles dates conversion between internal EID format and YYYYMMDD format |
| | CardEventArgs (see page 12) | This is class Swelio.Engine.CardEventArgs. |
| S | CardNumber (see page 13) | EID card number formatter. This class takes EID card number in raw form as input and return formatted EID card number |
| | CardReader (see page 14) | The physical smart card (see page 15) reader |
| | DocumentContainer (see page 18) | ASIC document container. Associated Signature Containers (ASiC) specifies the use of container structures to bind together one or more signed objects with either advanced electronic signatures or timestamp tokens into one single digital container. |
| | EIDAddress (see page 20) | EID address information, stored on the card |
| | EIDCertificate (see page 21) | Certificate, stored on EID card |
| S | EIDDataLength (see page 22) | Maximum length of EID fields |
| | EIDError (see page 28) | This is class Swelio.Engine.EIDError. |
| | EIDIdentity (see page 29) | Identity (see page 49) information stored on EID card |
| | EIDPicture (see page 35) | Raw picture (see page 36) data from EID card |
| S | Encryption (see page 36) | |
| S | FileOperations (see page 42) | Helper class for common file oprations |
| | Identity (see page 49) | Holds identity information that is stored on EID card |
| S | InteropHelper (see page 56) | Helper class for accessing native Windows API and resources Not recommeded to use it directly without deep understanding of Windows API |
| | Manager (see page 60) | The Swelio Engine Manager class used to activate and deactivate engine and gives an access to the smart card readers, connected to the PC |
| S | NationalNumber (see page 64) | National number formatter |
| S | NativeMethods (see page 65) | This is class Swelio.Engine.NativeMethods. |
| S | Network (see page 98) | This is class Swelio.Engine.Network. |
| | PostalAddress (see page 99) | This is class Swelio.Engine.PostalAddress. |
| S | QRCode (see page 101) | QR Code encoder class |
| S | RecycleBin (see page 103) | Send (see page 104) files directly to the recycle bin. |
| S | WindowsOperations (see page 104) | Windows power management class |

### Enumerations

| | Name | Description |
|---|---|---|
| | CardEvent (see page 105) | The code of the card event |
| | CertificateType (see page 106) | The type of the cerificate |
| | ChangeWindowMessageFilterFlags (see page 106) | This is record Swelio.Engine.ChangeWindowMessageFilterFlags. |
| | DocumentType (see page 106) | This is record Swelio.Engine.DocumentType. |
| | ErrorCorrectionLevel (see page 107) | This is record Swelio.Engine.ErrorCorrectionLevel. |
| | ExportFormat (see page 107) | This is record Swelio.Engine.ExportFormat. |
| | FileEncoding (see page 107) | This is record Swelio.Engine.FileEncoding. |
| | Gender (see page 108) | This is record Swelio.Engine.Gender. |
| | PhotoFormat (see page 108) | This is record Swelio.Engine.PhotoFormat. |
| | ShutdownOption (see page 108) | Windows shutdown options |

### Types

| Name | Description |
|---|---|
| ReaderCallbackDelegate (see page 109) | This is type Swelio.Engine.ReaderCallbackDelegate. |

# 1.1.1 Address Class

Represents a postal address of the person

### Inheritance Hierarchy

Swelio.Engine.Address

### C#

```
[Browsable(false)]
public class Address;
```

### File

Address.cs (see page 112)

### Members

### Methods

| | Name | Description |
|---|---|---|
| | Address (see page 3) | Initializes a new instance of the Address class. |

### Address Methods

| | Name | Description |
|---|---|---|
| | Parse (see page 3) | Parses adress string to street name and house number |

### Address Properties

| | Name | Description |
|---|---|---|
| R | Municipality (see page 3) | Municipality |
| R | Street (see page 4) | Street name |
| R | Zip (see page 4) | ZIP code |

**Methods**

| | Name | Description |
|---|---|---|
| | Address (see page 3) | Initializes a new instance of the Address class. |

**Address Methods**

| | Name | Description |
|---|---|---|
| | Parse (see page 3) | Parses adress string to street name and house number |

**Address Properties**

| | Name | Description |
|---|---|---|
| R | Municipality (see page 3) | Municipality |
| R | Street (see page 4) | Street name |
| R | Zip (see page 4) | ZIP code |

# 1.1.1.1 Address.Address Constructor

Initializes a new instance of the Address (see page 2) class.

**C#**

```
public Address(string street, string zip, string municipality);
```

**Parameters**

| Parameters | Description |
|---|---|
| string street | The street. |
| string zip | The zip code. |
| string municipality | The municipality. |

# 1.1.1.2 Address Methods

## 1.1.1.2.1 Address.Parse Method

Parses adress string to street name and house number

**C#**

```
public PostalAddress Parse();
```

**Returns**

The postal address

# 1.1.1.3 Address Properties

## 1.1.1.3.1 Address.Municipality Property

Municipality

**C#**

```
[Description("Municipality")]
public string Municipality;
```

### 1.1.1.3.2 Address.Street Property

Street name

**C#**

```
[Description("Street name")]
public string Street;
```

### 1.1.1.3.3 Address.Zip Property

ZIP code

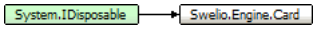
**C#**

```
[Description("ZIP code")]
public string Zip;
```

## 1.1.2 Card Class

This class represents a Belgian EID card

**Inheritance Hierarchy**




**C#**

```
public sealed class Card : IDisposable;
```

**File**

Card.cs (see page 113)

**Members**

**Methods**

| | Name | Description |
|---|---|---|
| | Card (see page 5) | Initializes a new instance of the Card class. |

**Card Methods**

| | Name | Description |
|---|---|---|
| | DisplayCertificate (see page 6) | Displays the certificate. |
| | Dispose (see page 6) | Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources. |
| | EncodeCertificate (see page 6) | Encodes the certificate. |
| | EncodePhoto (see page 7) | Encodes the photo. |
| | ReadAddress (see page 7) | Reads the address of the person from the card. |
| | ReadCertificate (see page 7) | Reads the certificate. |
| | ReadIdentity (see page 7) | This is ReadIdentity, a member of class Card. |
| | ReadPhoto (see page 8) | Reads the photo. |
| | Save (see page 8) | Saves information from the card to the string. |
| | SaveCertificate (see page 8) | Saves the certificate. |
| | SavePhoto (see page 9) | Saves the photo. |
| | SaveQRCode (see page 9) | Saves the QR code. |
| | VerifyPinCode (see page 9) | Verifies the pin code. |

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**Card Properties**

| | Name | Description |
|---|---|---|
| | AppletVersion (see page 9) | This is AppletVersion, a member of class Card. |
| | EidCard (see page 10) | Gets a value indicating whether the card is eid card. |
| | Inserted (see page 10) | Gets a value indicating whether the Card is inserted in the card reader. |
| | SerialNumber (see page 10) | This is SerialNumber, a member of class Card. |
| | Tag (see page 10) | This is Tag, a member of class Card. |

**Methods**

| | Name | Description |
|---|---|---|
| | Card (see page 5) | Initializes a new instance of the Card class. |

**Card Methods**

| | Name | Description |
|---|---|---|
| | DisplayCertificate (see page 6) | Displays the certificate. |
| | Dispose (see page 6) | Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources. |
| | EncodeCertificate (see page 6) | Encodes the certificate. |
| | EncodePhoto (see page 7) | Encodes the photo. |
| | ReadAddress (see page 7) | Reads the address of the person from the card. |
| | ReadCertificate (see page 7) | Reads the certificate. |
| | ReadIdentity (see page 7) | This is ReadIdentity, a member of class Card. |
| | ReadPhoto (see page 8) | Reads the photo. |
| | Save (see page 8) | Saves information from the card to the string. |
| | SaveCertificate (see page 8) | Saves the certificate. |
| | SavePhoto (see page 9) | Saves the photo. |
| | SaveQRCode (see page 9) | Saves the QR code. |
| | VerifyPinCode (see page 9) | Verifies the pin code. |

**Card Properties**

| | Name | Description |
|---|---|---|
| | AppletVersion (see page 9) | This is AppletVersion, a member of class Card. |
| | EidCard (see page 10) | Gets a value indicating whether the card is eid card. |
| | Inserted (see page 10) | Gets a value indicating whether the Card is inserted in the card reader. |
| | SerialNumber (see page 10) | This is SerialNumber, a member of class Card. |
| | Tag (see page 10) | This is Tag, a member of class Card. |

## 1.1.2.1 Card.Card Constructor

Initializes a new instance of the Card (see page 4) class.

**C#**

```
internal Card(CardReader reader);
```

**Parameters**

| Parameters | Description |
|---|---|
| CardReader reader | The reader. |

# 1.1.2.2 Card Methods

## 1.1.2.2.1 DisplayCertificate Method

### 1.1.2.2.1.1 Card.DisplayCertificate Method (CertificateType)

Displays the certificate.

**C#**

```
public void DisplayCertificate(CertificateType certificate);
```

**Parameters**

| Parameters | Description |
|---|---|
| CertificateType certificate | The certificate. |

### 1.1.2.2.1.2 Card.DisplayCertificate Method (string)

Displays the certificate.

**C#**

```
public void DisplayCertificate(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

## 1.1.2.2.2 Dispose Method

### 1.1.2.2.2.1 Card.Dispose Method ()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

**C#**

```
public void Dispose();
```

### 1.1.2.2.2.2 Card.Dispose Method (bool)

Releases unmanaged and - optionally - managed resources

**C#**

```
internal void Dispose(bool disposing);
```

**Parameters**

| Parameters | Description |
|---|---|
| bool disposing | `true` to release both managed and unmanaged resources; `false` to release only unmanaged resources. |

## 1.1.2.2.3 Card.EncodeCertificate Method

Encodes the certificate.

**C#**

```
public byte[] EncodeCertificate(CertificateType certificate);
```

**Parameters**

| Parameters | Description |
|---|---|
| CertificateType certificate | The certificate. |

## 1.1.2.2.4 EncodePhoto Method

### 1.1.2.2.4.1 Card.EncodePhoto Method ()

Encodes the photo.

**C#**

```
public byte[] EncodePhoto();
```

### 1.1.2.2.4.2 Card.EncodePhoto Method (string)

Encodes the photo.

**C#**

```
public byte[] EncodePhoto(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

## 1.1.2.2.5 Card.ReadAddress Method

Reads the address of the person from the card.

**C#**

```
public Address ReadAddress();
```

**Returns**

Returns the postal adress of the person

## 1.1.2.2.6 Card.ReadCertificate Method

Reads the certificate.

**C#**

```
public byte[] ReadCertificate(CertificateType certificate);
```

**Parameters**

| Parameters | Description |
|---|---|
| CertificateType certificate | The certificate. |

## 1.1.2.2.7 Card.ReadIdentity Method

**C#**

```
public Identity ReadIdentity();
```

**Remarks**

This is ReadIdentity, a member of class Card.

## 1.1.2.2.8 Card.ReadPhoto Method

Reads the photo.

**C#**

```
public Image ReadPhoto();
```

## 1.1.2.2.9 Save Method

### 1.1.2.2.9.1 Card.Save Method (ExportFormat)

Saves information from the card to the string.

**C#**

```
public string Save(ExportFormat format);
```

**Parameters**

| Parameters | Description |
|---|---|
| ExportFormat format | The format. |
| fileName | Name of the file. |
| encoding | The encoding. |

### 1.1.2.2.9.2 Card.Save Method (string, ExportFormat)

Saves information from the card to the file with specified file name.

**C#**

```
public bool Save(string fileName, ExportFormat format);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| ExportFormat format | The format of the export. |

### 1.1.2.2.9.3 Card.Save Method (string, ExportFormat, FileEncoding)

Saves information from the card to the file with specified file name.

**C#**

```
public bool Save(string fileName, ExportFormat format, FileEncoding encoding);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| ExportFormat format | The format. |
| FileEncoding encoding | The encoding. |

## 1.1.2.2.10 Card.SaveCertificate Method

Saves the certificate.

**C#**

```
public void SaveCertificate(string fileName, CertificateType certificate);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| CertificateType certificate | The certificate. |

## 1.1.2.2.11 Card.SavePhoto Method

Saves the photo.

**C#**

```
public bool SavePhoto(string fileName, PhotoFormat format);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| PhotoFormat format | The format. |

## 1.1.2.2.12 Card.SaveQRCode Method

Saves the QR code.

**C#**

```
public bool SaveQRCode(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

## 1.1.2.2.13 Card.VerifyPinCode Method

Verifies the pin code.

**C#**

```
public bool VerifyPinCode(string pinCode);
```

**Parameters**

| Parameters | Description |
|---|---|
| string pinCode | The pin code. |

# 1.1.2.3 Card Properties

## 1.1.2.3.1 Card.AppletVersion Property

**C#**

```
public int AppletVersion;
```

**Remarks**

This is AppletVersion, a member of class Card.

### 1.1.2.3.2 Card.EidCard Property

Gets a value indicating whether the card is eid card.

**C#**

```
public bool EidCard;
```

**Description**

`true` if the card is eid card; otherwise, `false`.

### 1.1.2.3.3 Card.Inserted Property

Gets a value indicating whether the Card (see page 4) is inserted in the card reader.

**C#**

```
public bool Inserted;
```

**Description**

`true` if inserted; otherwise, `false`.

### 1.1.2.3.4 Card.SerialNumber Property

**C#**

```
public string SerialNumber;
```

**Remarks**

This is SerialNumber, a member of class Card.

### 1.1.2.3.5 Card.Tag Property

**C#**

```
public int Tag;
```

**Remarks**

This is Tag, a member of class Card.

## 1.1.3 CardDate Class

Handles dates conversion between internal EID format and YYYYMMDD format

**Inheritance Hierarchy**

Swelio.Engine.CardDate

**C#**

```
public static class CardDate;
```

**File**

CardDate.cs (see page 113)

**Members**

**CardDate Methods**

| | Name | Description |
|---|---|---|
| S | ToDate (see page 11) | Converts from the internal EID representation to System.DateTime |
| S | ToLongDateString (see page 11) | Converts the value from YYYYMMDD format to its equivalent long date string representation |
| S | ToShortDateString (see page 12) | Converts the value from YYYYMMDD format to its equivalent short date string representation |

**CardDate Methods**

| | Name | Description |
|---|---|---|
| S | ToDate (see page 11) | Converts from the internal EID representation to System.DateTime |
| S | ToLongDateString (see page 11) | Converts the value from YYYYMMDD format to its equivalent long date string representation |
| S | ToShortDateString (see page 12) | Converts the value from YYYYMMDD format to its equivalent short date string representation |

# 1.1.3.1 CardDate Methods

## 1.1.3.1.1 CardDate.ToDate Method

Converts from the internal EID representation to System.DateTime

**C#**

```
public static DateTime ToDate(string date);
```

**Parameters**

| Parameters | Description |
|---|---|
| string date | The unformated date string |

**Returns**

System.DateTime

## 1.1.3.1.2 CardDate.ToLongDateString Method

Converts the value from YYYYMMDD format to its equivalent long date string representation

**C#**

```
public static string ToLongDateString(string date);
```

**Parameters**

| Parameters | Description |
|---|---|
| string date | The date. |

**Returns**

A string containing the name of the day of the week, the name of the month, the numeric day of the month, and the year equivalent to the date value

### 1.1.3.1.3 CardDate.ToShortDateString Method

Converts the value from YYYYMMDD format to its equivalent short date string representation

**C#**

```
public static string ToShortDateString(string date);
```

**Parameters**

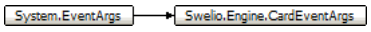
| Parameters | Description |
|---|---|
| string date | The date. |

**Returns**

A string containing the numeric month, the numeric day of the month, and the year equivalent to the date value

## 1.1.4 CardEventArgs Class

**Inheritance Hierarchy**




**C#**

```
public class CardEventArgs : EventArgs;
```

**File**

CardEventArgs.cs (see page 113)

**Remarks**

This is class Swelio.Engine.CardEventArgs.

**Members**

**Methods**

| | Name | Description |
|---|---|---|
| | CardEventArgs (see page 12) | Initializes a new instance of the CardEventArgs class. |

**CardEventArgs Properties**

| | Name | Description |
|---|---|---|
| | ReaderNumber (see page 13) | Gets the reader number. |

**Methods**

| | Name | Description |
|---|---|---|
| | CardEventArgs (see page 12) | Initializes a new instance of the CardEventArgs class. |

**CardEventArgs Properties**

| | Name | Description |
|---|---|---|
| | ReaderNumber (see page 13) | Gets the reader number. |

### 1.1.4.1 CardEventArgs.CardEventArgs Constructor

Initializes a new instance of the CardEventArgs (see page 12) class.

**C#**

```
public CardEventArgs(int readerNumber);
```

**Parameters**

| Parameters | Description |
|---|---|
| int readerNumber | The reader number. |

## 1.1.4.2 CardEventArgs Properties

### 1.1.4.2.1 CardEventArgs.ReaderNumber Property

Gets the reader number.

**C#**

```
public int ReaderNumber;
```

**Description**

The reader number.

## 1.1.5 CardNumber Class

EID card number formatter. This class takes EID card number in raw form as input and return formatted EID card number

**Inheritance Hierarchy**

Swelio.Engine.CardNumber

**C#**

```
public static class CardNumber;
```

**File**

CardNumber.cs (see page 114)

**Members**

**CardNumber Methods**

| | Name | Description |
|---|---|---|
| S | ToString (see page 13) | Converts the value of the card number to its String representation |

**CardNumber Methods**

| | Name | Description |
|---|---|---|
| S | ToString (see page 13) | Converts the value of the card number to its String representation |

## 1.1.5.1 CardNumber Methods

### 1.1.5.1.1 CardNumber.ToString Method

Converts the value of the card number to its String representation

**C#**

```
public static string ToString(string cardNumber);
```

**Parameters**

| Parameters | Description |
|---|---|
| string cardNumber | The unformatted card number |

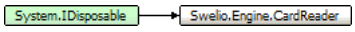
**Returns**

Formatted card number

# 1.1.6 CardReader Class

The physical smart card (see page 15) reader

**Inheritance Hierarchy**




**C#**

```
public sealed class CardReader : IDisposable;
```

**File**

CardReader.cs (see page 114)

**Members**

**Methods**

| | Name | Description |
|---|---|---|
| | CardReader (see page 15) | Initializes a new instance of the CardReader class. |

**CardReader Fields**

| | Name | Description |
|---|---|---|
| | card (see page 15) | This is card, a member of class CardReader. |

**CardReader Methods**

| | Name | Description |
|---|---|---|
| | ActivateCard (see page 16) | Activates the card (see page 15). |
| | DeactivateCard (see page 16) | Deactivates the card (see page 15). |
| | Dispose (see page 16) | Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources. |
| | GetCard (see page 16) | Gets the card (see page 15). |

**CardReader Properties**

| | Name | Description |
|---|---|---|
| R | CardActivated (see page 17) | Gets a value indicating whether [card (see page 15) activated]. |
| R | CardPresent (see page 17) | Gets a value indicating whether [card (see page 15) present]. |
| | Index (see page 17) | Gets the index of the smart card (see page 15) reader. |
| | Name (see page 17) | Gets the name of the smart card (see page 15) reader. |
| R | Selected (see page 17) | Gets a value indicating whether this CardReader is selected. |
| | Tag (see page 17) | Gets or sets the tag. |

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**Methods**

| | Name | Description |
|---|---|---|
| | CardReader (see page 15) | Initializes a new instance of the CardReader class. |

**CardReader Fields**

| | Name | Description |
|---|---|---|
| | card (see page 15) | This is card, a member of class CardReader. |

**CardReader Methods**

| | Name | Description |
|---|---|---|
| | ActivateCard (see page 16) | Activates the card (see page 15). |
| | DeactivateCard (see page 16) | Deactivates the card (see page 15). |
| | Dispose (see page 16) | Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources. |
| | GetCard (see page 16) | Gets the card (see page 15). |

**CardReader Properties**

| | Name | Description |
|---|---|---|
| R | CardActivated (see page 17) | Gets a value indicating whether [card (see page 15) activated]. |
| R | CardPresent (see page 17) | Gets a value indicating whether [card (see page 15) present]. |
| | Index (see page 17) | Gets the index of the smart card (see page 15) reader. |
| | Name (see page 17) | Gets the name of the smart card (see page 15) reader. |
| R | Selected (see page 17) | Gets a value indicating whether this CardReader is selected. |
| | Tag (see page 17) | Gets or sets the tag. |

## 1.1.6.1 CardReader.CardReader Constructor

Initializes a new instance of the CardReader (see page 14) class.

**C#**

```
internal CardReader();
```

## 1.1.6.2 CardReader Fields

### 1.1.6.2.1 CardReader.card Field

**C#**

```
internal Card card;
```

**Remarks**

This is card, a member of class CardReader.

## 1.1.6.3 CardReader Methods

## 1.1.6.3.1 CardReader.ActivateCard Method

Activates the card (see page 15).

**C#**

```
public bool ActivateCard();
```

## 1.1.6.3.2 CardReader.DeactivateCard Method

Deactivates the card (see page 15).

**C#**

```
public void DeactivateCard();
```

## 1.1.6.3.3 Dispose Method

### 1.1.6.3.3.1 CardReader.Dispose Method ()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

**C#**

```
public void Dispose();
```

### 1.1.6.3.3.2 CardReader.Dispose Method (bool)

Releases unmanaged and - optionally - managed resources

**C#**

```
internal void Dispose(bool disposing);
```

**Parameters**

| Parameters | Description |
|---|---|
| bool disposing | `true` to release both managed and unmanaged resources; `false` to release only unmanaged resources. |

## 1.1.6.3.4 GetCard Method

### 1.1.6.3.4.1 CardReader.GetCard Method ()

Gets the card (see page 15).

**C#**

```
public Card GetCard();
```

### 1.1.6.3.4.2 CardReader.GetCard Method (bool)

Gets the card (see page 15).

**C#**

```
public Card GetCard(bool activate);
```

**Parameters**

| Parameters | Description |
|---|---|
| bool activate | if set to `true` [activate]. |

# 1.1.6.4 CardReader Properties

## 1.1.6.4.1 CardReader.CardActivated Property

Gets a value indicating whether [card (see page 15) activated].

**C#**

```
public bool CardActivated;
```

**Description**

`true` if [card (see page 15) activated]; otherwise, `false`.

## 1.1.6.4.2 CardReader.CardPresent Property

Gets a value indicating whether [card (see page 15) present].

**C#**

```
public bool CardPresent;
```

**Description**

`true` if [card (see page 15) present]; otherwise, `false`.

## 1.1.6.4.3 CardReader.Index Property

Gets the index of the smart card (see page 15) reader.

**C#**

```
public int Index;
```

## 1.1.6.4.4 CardReader.Name Property

Gets the name of the smart card (see page 15) reader.

**C#**

```
public string Name;
```

## 1.1.6.4.5 CardReader.Selected Property

Gets a value indicating whether this CardReader (see page 14) is selected.

**C#**

```
public bool Selected;
```

**Description**

`true` if selected; otherwise, `false`.

## 1.1.6.4.6 CardReader.Tag Property

Gets or sets the tag.

**C#**

```
public int Tag;
```

**Description**

The tag.

# 1.1.7 DocumentContainer Class

ASIC document container. Associated Signature Containers (ASiC) specifies the use of container structures to bind together one or more signed objects with either advanced electronic signatures or timestamp tokens into one single digital container.

**Inheritance Hierarchy**

Swelio.Engine.DocumentContainer

**C#**

```
public class DocumentContainer;
```

**File**

DocumentContainer.cs (see page 114)

**Members**

**Methods**

| | Name | Description |
|---|---|---|
| | DocumentContainer (see page 18) | Initializes a new instance of the DocumentContainer class. |

**DocumentContainer Methods**

| | Name | Description |
|---|---|---|
| | AddFile (see page 19) | Add file to the container |
| | Close (see page 19) | Closes container. No any future calls are possible after closing container |
| | Save (see page 19) | Saves container to the file with specified file name. By default the file extension for ASIC containers is asice. |
| | SelectCertificate (see page 19) | Selects the certificate. |

**Methods**

| | Name | Description |
|---|---|---|
| | DocumentContainer (see page 18) | Initializes a new instance of the DocumentContainer class. |

**DocumentContainer Methods**

| | Name | Description |
|---|---|---|
| | AddFile (see page 19) | Add file to the container |
| | Close (see page 19) | Closes container. No any future calls are possible after closing container |
| | Save (see page 19) | Saves container to the file with specified file name. By default the file extension for ASIC containers is asice. |
| | SelectCertificate (see page 19) | Selects the certificate. |

# 1.1.7.1 DocumentContainer.DocumentContainer Constructor

Initializes a new instance of the DocumentContainer (see page 18) class.

**C#**

```
public DocumentContainer();
```

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# 1.1.7.2 DocumentContainer Methods

## 1.1.7.2.1 DocumentContainer.AddFile Method

Add file to the container

**C#**

```
public bool AddFile(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | The name of the file |

## 1.1.7.2.2 DocumentContainer.Close Method

Closes container. No any future calls are possible after closing container

**C#**

```
public void Close();
```

## 1.1.7.2.3 DocumentContainer.Save Method

Saves container to the file with specified file name. By default the file extension for ASIC containers is asice.

**C#**

```
public bool Save(string fileName, int readerNumber);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| int readerNumber | The EID card reader number. |

**Returns**

true, if success; otherwise - false.

## 1.1.7.2.4 SelectCertificate Method

### 1.1.7.2.4.1 DocumentContainer.SelectCertificate Method ()

Selects the certificate.

**C#**

```
public bool SelectCertificate();
```

**Returns**

True - if certificate is selected, otherwise - false

### 1.1.7.2.4.2 DocumentContainer.SelectCertificate Method (int)

Selects the certificate.

**C#**

```
public bool SelectCertificate(int readerNumber);
```

**Parameters**

| Parameters | Description |
|---|---|
| int readerNumber | The reader number. |

**Returns**

True - if certificate is selected, otherwise - false

### 1.1.7.2.4.3 DocumentContainer.SelectCertificate Method (string, string)

Selects the certificate.

**C#**

```
public bool SelectCertificate(string fileName, string password);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the certificate file in pfx format |
| string password | The password. |

**Returns**

True - if certificate is selected, otherwise - false

## 1.1.8 EIDAddress Class

EID address information, stored on the card

**Inheritance Hierarchy**

Swelio.Engine.EIDAddress

**C#**

```
[StructLayout(LayoutKind.Sequential, CharSet = CharSet.Unicode)]
[BestFitMapping(false, ThrowOnUnmappableChar = true)]
internal class EIDAddress;
```

**File**

NativeTypes.cs (see page 117)

**Members**

**EIDAddress Fields**

| | Name | Description |
|---|---|---|
| | municipality (see page 21) | Municipality |
| | street (see page 21) | Street name |
| | zip (see page 21) | ZIP code |

**EIDAddress Fields**

| | Name | Description |
|---|---|---|
| | municipality (see page 21) | Municipality |
| | street (see page 21) | Street name |

| | | |
|---|---|---|
| | zip (see page 21) | ZIP code |

## 1.1.8.1 EIDAddress Fields

### 1.1.8.1.1 EIDAddress.municipality Field

Municipality

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxMunicipalityName + 1)]
internal string municipality;
```

### 1.1.8.1.2 EIDAddress.street Field

Street name

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxStreetNameLen + 1)]
internal string street;
```

### 1.1.8.1.3 EIDAddress.zip Field

ZIP code

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxZipLen + 1)]
internal string zip;
```

## 1.1.9 EIDCertificate Class

Certificate, stored on EID card

**Inheritance Hierarchy**

Swelio.Engine.EIDCertificate

**C#**

```
[StructLayout(LayoutKind.Sequential, CharSet = CharSet.Unicode)]
internal class EIDCertificate;
```

**File**

NativeTypes.cs (see page 117)

**Members**

**EIDCertificate Fields**

| | Name | Description |
|---|---|---|
| | certificate (see page 22) | Certificate raw data buffer |
| | certificateLength (see page 22) | Certificate data length |

**EIDCertificate Fields**

| | Name | Description |
|---|---|---|
| | certificate (see page 22) | Certificate raw data buffer |

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| | | |
|---|---|---|
| | certificateLength (see page 22) | Certificate data length |

### 1.1.9.1 EIDCertificate Fields

#### 1.1.9.1.1 EIDCertificate.certificate Field

Certificate raw data buffer

**C#**

```
[MarshalAs(UnmanagedType.ByValArray, SizeConst = EIDDataLength.EidMaxCertLen + 1)]
internal byte[] certificate;
```

#### 1.1.9.1.2 EIDCertificate.certificateLength Field

Certificate data length

**C#**

```
[MarshalAs(UnmanagedType.I4)]
internal int certificateLength;
```

## 1.1.10 EIDDataLength Class

Maximum length of EID fields

**Inheritance Hierarchy**

Swelio.Engine.EIDDataLength

**C#**

```
internal static class EIDDataLength;
```

**File**

NativeTypes.cs (see page 117)

**Members**

**EIDDataLength Fields**

| | Name | Description |
|---|---|---|
| | EidMaxBirthDateLen (see page 24) | Max length of the birthdate |
| | EidMaxBirthplaceLen (see page 24) | Max length of the birthplace |
| | EidMaxBrexitMention1Len (see page 24) | BREXIT mention |
| | EidMaxBrexitMention2Len (see page 24) | BREXIT mention |
| | EidMaxCardNumberLen (see page 24) | Max length of the card number field |
| | EidMaxCertLen (see page 24) | Max length of the certificate data |
| | EidMaxChipNumberLen (see page 25) | Max length of the chip number field |
| | EidMaxDateAndCountryOfProtectionLen (see page 25) | Date and country of protection |
| | EidMaxDateBeginLen (see page 25) | Max length of the begin date field |
| | EidMaxDateEndLen (see page 25) | Max length of the end date field |
| | EidMaxDeliveryMunicipalityLen (see page 25) | Max length of the name of the devivery municipality |
| | EidMaxDocumentTypeLen (see page 25) | Max length of the document type field |

| | | |
|---|---|---|
| | EidMaxDuplicateLen (see page 25) | Duplicate |
| | EidMaxFirstNameLen (see page 25) | Max length of the first name |
| | EidMaxMemberOfFamilyLen (see page 26) | Member of family |
| | EidMaxMiddleNameLen (see page 26) | Max length of the first name |
| | EidMaxMunicipalityName (see page 26) | Max length of the municipality name field |
| | EidMaxNameLen (see page 26) | Max length of the surname |
| | EidMaxNationalityLen (see page 26) | Max length of the nationality |
| | EidMaxNationalNumberLen (see page 26) | Max length of the national number |
| | EidMaxNobleConditionLen (see page 26) | Max length of the noble condition field |
| | EidMaxPictureLen (see page 27) | Max length of the picture data |
| | EidMaxRegionalFileNumberLen (see page 27) | Regional file number |
| | EidMaxSexLen (see page 27) | Max length of the sex field |
| | EidMaxSpecialOrganizationLen (see page 27) | Special organization |
| | EidMaxSpecialStatusLen (see page 27) | Max length of the special status field |
| | EidMaxStreetNameLen (see page 27) | Max length of the street name field |
| | EidMaxVAT1Len (see page 27) | VAT1 |
| | EidMaxVAT2Len (see page 28) | VAT2 |
| | EidMaxWorkPermitTypeLen (see page 28) | The type of the workpermit |
| | EidMaxZipLen (see page 28) | Max length of the ZIP code field |

**EIDDataLength Fields**

| | Name | Description |
|---|---|---|
| | EidMaxBirthDateLen (see page 24) | Max length of the birthdate |
| | EidMaxBirthplaceLen (see page 24) | Max length of the birthplace |
| | EidMaxBrexitMention1Len (see page 24) | BREXIT mention |
| | EidMaxBrexitMention2Len (see page 24) | BREXIT mention |
| | EidMaxCardNumberLen (see page 24) | Max length of the card number field |
| | EidMaxCertLen (see page 24) | Max length of the certificate data |
| | EidMaxChipNumberLen (see page 25) | Max length of the chip number field |
| | EidMaxDateAndCountryOfProtectionLen (see page 25) | Date and country of protection |
| | EidMaxDateBeginLen (see page 25) | Max length of the begin date field |
| | EidMaxDateEndLen (see page 25) | Max length of the end date field |
| | EidMaxDeliveryMunicipalityLen (see page 25) | Max length of the name of the devivery municipality |
| | EidMaxDocumentTypeLen (see page 25) | Max length of the document type field |
| | EidMaxDuplicateLen (see page 25) | Duplicate |
| | EidMaxFirstNameLen (see page 25) | Max length of the first name |
| | EidMaxMemberOfFamilyLen (see page 26) | Member of family |
| | EidMaxMiddleNameLen (see page 26) | Max length of the first name |
| | EidMaxMunicipalityName (see page 26) | Max length of the municipality name field |
| | EidMaxNameLen (see page 26) | Max length of the surname |
| | EidMaxNationalityLen (see page 26) | Max length of the nationality |
| | EidMaxNationalNumberLen (see page 26) | Max length of the national number |
| | EidMaxNobleConditionLen (see page 26) | Max length of the noble condition field |
| | EidMaxPictureLen (see page 27) | Max length of the picture data |
| | EidMaxRegionalFileNumberLen (see page 27) | Regional file number |
| | EidMaxSexLen (see page 27) | Max length of the sex field |
| | EidMaxSpecialOrganizationLen (see page 27) | Special organization |

| | | |
|---|---|---|
| | EidMaxSpecialStatusLen (see page 27) | Max length of the special status field |
| | EidMaxStreetNameLen (see page 27) | Max length of the street name field |
| | EidMaxVAT1Len (see page 27) | VAT1 |
| | EidMaxVAT2Len (see page 28) | VAT2 |
| | EidMaxWorkPermitTypeLen (see page 28) | The type of the workpermit |
| | EidMaxZipLen (see page 28) | Max length of the ZIP code field |

# 1.1.10.1 EIDDataLength Fields

## 1.1.10.1.1 EIDDataLength.EidMaxBirthDateLen Field

Max length of the birthdate

**C#**

```
public const int EidMaxBirthDateLen = 12;
```

## 1.1.10.1.2 EIDDataLength.EidMaxBirthplaceLen Field

Max length of the birthplace

**C#**

```
public const int EidMaxBirthplaceLen = 80;
```

## 1.1.10.1.3 EIDDataLength.EidMaxBrexitMention1Len Field

**C#**

```
public const int EidMaxBrexitMention1Len = 1;
```

**Remarks**

BREXIT mention

## 1.1.10.1.4 EIDDataLength.EidMaxBrexitMention2Len Field

**C#**

```
public const int EidMaxBrexitMention2Len = 1;
```

**Remarks**

BREXIT mention

## 1.1.10.1.5 EIDDataLength.EidMaxCardNumberLen Field

Max length of the card number field

**C#**

```
public const int EidMaxCardNumberLen = 12;
```

## 1.1.10.1.6 EIDDataLength.EidMaxCertLen Field

Max length of the certificate data

**C#**

```
public const int EidMaxCertLen = 2048;
```

### 1.1.10.1.7 EIDDataLength.EidMaxChipNumberLen Field

Max length of the chip number field

**C#**

```
public const int EidMaxChipNumberLen = 32;
```

### 1.1.10.1.8 EIDDataLength.EidMaxDateAndCountryOfProtectionLen Field

**C#**

```
public const int EidMaxDateAndCountryOfProtectionLen = 13;
```

**Remarks**

Date and country of protection

### 1.1.10.1.9 EIDDataLength.EidMaxDateBeginLen Field

Max length of the begin date field

**C#**

```
public const int EidMaxDateBeginLen = 10;
```

### 1.1.10.1.10 EIDDataLength.EidMaxDateEndLen Field

Max length of the end date field

**C#**

```
public const int EidMaxDateEndLen = 10;
```

### 1.1.10.1.11 EIDDataLength.EidMaxDeliveryMunicipalityLen Field

Max length of the name of the devivery municipality

**C#**

```
public const int EidMaxDeliveryMunicipalityLen = 80;
```

### 1.1.10.1.12 EIDDataLength.EidMaxDocumentTypeLen Field

Max length of the document type field

**C#**

```
public const int EidMaxDocumentTypeLen = 2;
```

### 1.1.10.1.13 EIDDataLength.EidMaxDuplicateLen Field

**C#**

```
public const int EidMaxDuplicateLen = 2;
```

**Remarks**

Duplicate

### 1.1.10.1.14 EIDDataLength.EidMaxFirstNameLen Field

Max length of the first name

**C#**

```
public const int EidMaxFirstNameLen = 95;
```

## 1.1.10.1.15 EIDDataLength.EidMaxMemberOfFamilyLen Field

**C#**

```
public const int EidMaxMemberOfFamilyLen = 1;
```

**Remarks**

Member of family

## 1.1.10.1.16 EIDDataLength.EidMaxMiddleNameLen Field

Max length of the first name

**C#**

```
public const int EidMaxMiddleNameLen = 3;
```

## 1.1.10.1.17 EIDDataLength.EidMaxMunicipalityName Field

Max length of the municipality name field

**C#**

```
public const int EidMaxMunicipalityName = 67;
```

## 1.1.10.1.18 EIDDataLength.EidMaxNameLen Field

Max length of the surname

**C#**

```
public const int EidMaxNameLen = 110;
```

## 1.1.10.1.19 EIDDataLength.EidMaxNationalityLen Field

Max length of the nationality

**C#**

```
public const int EidMaxNationalityLen = 85;
```

## 1.1.10.1.20 EIDDataLength.EidMaxNationalNumberLen Field

Max length of the national number

**C#**

```
public const int EidMaxNationalNumberLen = 11;
```

## 1.1.10.1.21 EIDDataLength.EidMaxNobleConditionLen Field

Max length of the noble condition field

**C#**

```
public const int EidMaxNobleConditionLen = 50;
```

## 1.1.10.1.22 EIDDataLength.EidMaxPictureLen Field

Max length of the picture data

**C#**

```
public const int EidMaxPictureLen = 4096;
```

## 1.1.10.1.23 EIDDataLength.EidMaxRegionalFileNumberLen Field

**C#**

```
public const int EidMaxRegionalFileNumberLen = 18;
```

**Remarks**

Regional file number

## 1.1.10.1.24 EIDDataLength.EidMaxSexLen Field

Max length of the sex field

**C#**

```
public const int EidMaxSexLen = 1;
```

## 1.1.10.1.25 EIDDataLength.EidMaxSpecialOrganizationLen Field

**C#**

```
public const int EidMaxSpecialOrganizationLen = 1;
```

**Remarks**

Special organization

## 1.1.10.1.26 EIDDataLength.EidMaxSpecialStatusLen Field

Max length of the special status field

**C#**

```
public const int EidMaxSpecialStatusLen = 2;
```

## 1.1.10.1.27 EIDDataLength.EidMaxStreetNameLen Field

Max length of the street name field

**C#**

```
public const int EidMaxStreetNameLen = 80;
```

## 1.1.10.1.28 EIDDataLength.EidMaxVAT1Len Field

**C#**

```
public const int EidMaxVAT1Len = 13;
```

**Remarks**

VAT1

### 1.1.10.1.29 EIDDataLength.EidMaxVAT2Len Field

**C#**

```
public const int EidMaxVAT2Len = 13;
```

**Remarks**

VAT2

### 1.1.10.1.30 EIDDataLength.EidMaxWorkPermitTypeLen Field

**C#**

```
public const int EidMaxWorkPermitTypeLen = 1;
```

**Remarks**

The type of the workpermit

### 1.1.10.1.31 EIDDataLength.EidMaxZipLen Field

Max length of the ZIP code field

**C#**

```
public const int EidMaxZipLen = 4;
```

## 1.1.11 EIDError Class

**Inheritance Hierarchy**

Swelio.Engine.EIDError

**C#**

```
[StructLayout(LayoutKind.Sequential, CharSet = CharSet.Unicode)]
internal class EIDError;
```

**File**

NativeTypes.cs (see page 117)

**Remarks**

This is class Swelio.Engine.EIDError.

**Members**

**EIDError Fields**

| | Name | Description |
|---|---|---|
| | Code (see page 29) | This is Code, a member of class EIDError. |
| | Description (see page 29) | This is Description, a member of class EIDError. |

**EIDError Fields**

| | Name | Description |
|---|---|---|
| | Code (see page 29) | This is Code, a member of class EIDError. |
| | Description (see page 29) | This is Description, a member of class EIDError. |

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# 1.1.11.1 EIDError Fields

## 1.1.11.1.1 EIDError.Code Field

**C#**

```
internal int Code;
```

**Remarks**

This is Code, a member of class EIDError.

## 1.1.11.1.2 EIDError.Description Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = 255)]
internal string Description;
```

**Remarks**

This is Description, a member of class EIDError.

# 1.1.12 EIDIdentity Class

Identity (see page 49) information stored on EID card

**Inheritance Hierarchy**

Swelio.Engine.EIDIdentity

**C#**

```
[StructLayout(LayoutKind.Sequential, CharSet = CharSet.Unicode)]
[BestFitMapping(false, ThrowOnUnmappableChar = true)]
internal class EIDIdentity;
```

**File**

NativeTypes.cs (see page 117)

**Members**

**EIDIdentity Fields**

| | Name | Description |
|---|---|---|
| | birthDate (see page 31) | Birth date |
| | birthLocation (see page 31) | Birth location |
| | brexitMention1 (see page 31) | This is brexitMention1, a member of class EIDIdentity. |
| | brexitMention2 (see page 31) | This is brexitMention2, a member of class EIDIdentity. |
| | cardNumber (see page 31) | Electronic ID card number |
| | chipNumber (see page 31) | Electronic ID card physical chip number |
| | dateAndCountryOfProtection (see page 32) | This is dateAndCountryOfProtection, a member of class EIDIdentity. |
| | documentType (see page 32) | Document type (Belgian citizen, European Community citizen, ...) |
| | duplicate (see page 32) | This is duplicate, a member of class EIDIdentity. |

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| | | |
|---|---|---|
| | extendedMinority (see page 32) | Extended minority |
| | firstName1 (see page 32) | First name (see page 33) |
| | firstName2 (see page 32) | First name (see page 33) |
| | memberOfFamily (see page 33) | This is memberOfFamily, a member of class EIDIdentity. |
| | municipality (see page 33) | Card (see page 4) delivery municipality |
| | name (see page 33) | Surname |
| | nationality (see page 33) | Nationality |
| | nationalNumber (see page 33) | National number |
| | nobleCondition (see page 33) | Noble condition |
| | regionalFileNumber (see page 33) | This is regionalFileNumber, a member of class EIDIdentity. |
| | sex (see page 34) | Sex |
| | specialOrganization (see page 34) | This is specialOrganization, a member of class EIDIdentity. |
| | validityDateBegin (see page 34) | Card (see page 4) validity date begin |
| | validityDateEnd (see page 34) | Card (see page 4) validity date end |
| | vat1 (see page 34) | This is vat1, a member of class EIDIdentity. |
| | vat2 (see page 34) | This is vat2, a member of class EIDIdentity. |
| | whiteCane (see page 35) | White cane allowed (blind people) |
| | workPermitType (see page 35) | This is workPermitType, a member of class EIDIdentity. |
| | yellowCane (see page 35) | Yellow cane allowed (partially sighted people) |

**EIDIdentity Fields**

| | Name | Description |
|---|---|---|
| | birthDate (see page 31) | Birth date |
| | birthLocation (see page 31) | Birth location |
| | brexitMention1 (see page 31) | This is brexitMention1, a member of class EIDIdentity. |
| | brexitMention2 (see page 31) | This is brexitMention2, a member of class EIDIdentity. |
| | cardNumber (see page 31) | Electronic ID card number |
| | chipNumber (see page 31) | Electronic ID card physical chip number |
| | dateAndCountryOfProtection (see page 32) | This is dateAndCountryOfProtection, a member of class EIDIdentity. |
| | documentType (see page 32) | Document type (Belgian citizen, European Community citizen, ...) |
| | duplicate (see page 32) | This is duplicate, a member of class EIDIdentity. |
| | extendedMinority (see page 32) | Extended minority |
| | firstName1 (see page 32) | First name (see page 33) |
| | firstName2 (see page 32) | First name (see page 33) |
| | memberOfFamily (see page 33) | This is memberOfFamily, a member of class EIDIdentity. |
| | municipality (see page 33) | Card (see page 4) delivery municipality |
| | name (see page 33) | Surname |
| | nationality (see page 33) | Nationality |
| | nationalNumber (see page 33) | National number |
| | nobleCondition (see page 33) | Noble condition |
| | regionalFileNumber (see page 33) | This is regionalFileNumber, a member of class EIDIdentity. |
| | sex (see page 34) | Sex |
| | specialOrganization (see page 34) | This is specialOrganization, a member of class EIDIdentity. |
| | validityDateBegin (see page 34) | Card (see page 4) validity date begin |
| | validityDateEnd (see page 34) | Card (see page 4) validity date end |
| | vat1 (see page 34) | This is vat1, a member of class EIDIdentity. |

| | | |
|---|---|---|
| | vat2 (see page 34) | This is vat2, a member of class EIDIdentity. |
| | whiteCane (see page 35) | White cane allowed (blind people) |
| | workPermitType (see page 35) | This is workPermitType, a member of class EIDIdentity. |
| | yellowCane (see page 35) | Yellow cane allowed (partially sighted people) |

# 1.1.12.1 EIDIdentity Fields

## 1.1.12.1.1 EIDIdentity.birthDate Field

Birth date

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxBirthDateLen + 1)]
internal string birthDate;
```

## 1.1.12.1.2 EIDIdentity.birthLocation Field

Birth location

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxBirthplaceLen + 1)]
internal string birthLocation;
```

## 1.1.12.1.3 EIDIdentity.brexitMention1 Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxBrexitMention1Len + 1)]
internal string brexitMention1;
```

**Remarks**

This is brexitMention1, a member of class EIDIdentity.

## 1.1.12.1.4 EIDIdentity.brexitMention2 Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxBrexitMention2Len + 1)]
internal string brexitMention2;
```

**Remarks**

This is brexitMention2, a member of class EIDIdentity.

## 1.1.12.1.5 EIDIdentity.cardNumber Field

Electronic ID card number

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxCardNumberLen + 1)]
internal string cardNumber;
```

## 1.1.12.1.6 EIDIdentity.chipNumber Field

Electronic ID card physical chip number

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxChipNumberLen + 1)]
internal string chipNumber;
```

### 1.1.12.1.7 EIDIdentity.dateAndCountryOfProtection Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst =
EIDDataLength.EidMaxDateAndCountryOfProtectionLen + 1)]
internal string dateAndCountryOfProtection;
```

**Remarks**

This is dateAndCountryOfProtection, a member of class EIDIdentity.

### 1.1.12.1.8 EIDIdentity.documentType Field

Document type (Belgian citizen, European Community citizen, ...)

**C#**

```
[MarshalAs(UnmanagedType.I4)]
internal int documentType;
```

### 1.1.12.1.9 EIDIdentity.duplicate Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxDuplicateLen + 1)]
internal string duplicate;
```

**Remarks**

This is duplicate, a member of class EIDIdentity.

### 1.1.12.1.10 EIDIdentity.extendedMinority Field

Extended minority

**C#**

```
[MarshalAs(UnmanagedType.Bool)]
internal bool extendedMinority;
```

### 1.1.12.1.11 EIDIdentity.firstName1 Field

First name (see page 33)

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxFirstNameLen + 1)]
internal string firstName1;
```

### 1.1.12.1.12 EIDIdentity.firstName2 Field

First name (see page 33)

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxMiddleNameLen + 1)]
internal string firstName2;
```

## 1.1.12.1.13 EIDIdentity.memberOfFamily Field

**C#**

```
[MarshalAs(UnmanagedType.Bool)]
internal bool memberOfFamily;
```

**Remarks**

This is memberOfFamily, a member of class EIDIdentity.

## 1.1.12.1.14 EIDIdentity.municipality Field

Card (see page 4) delivery municipality

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxDeliveryMunicipalityLen
+ 1)]
internal string municipality;
```

## 1.1.12.1.15 EIDIdentity.name Field

Surname

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxNameLen + 1)]
internal string name;
```

## 1.1.12.1.16 EIDIdentity.nationality Field

Nationality

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxNationalityLen + 1)]
internal string nationality;
```

## 1.1.12.1.17 EIDIdentity.nationalNumber Field

National number

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxNationalNumberLen + 1)]
internal string nationalNumber;
```

## 1.1.12.1.18 EIDIdentity.nobleCondition Field

Noble condition

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxNobleConditionLen + 1)]
internal string nobleCondition;
```

## 1.1.12.1.19 EIDIdentity.regionalFileNumber Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxRegionalFileNumberLen +
1)]
internal string regionalFileNumber;
```

**Remarks**

This is regionalFileNumber, a member of class EIDIdentity.

## 1.1.12.1.20 EIDIdentity.sex Field

Sex

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxSexLen + 1)]
internal string sex;
```

## 1.1.12.1.21 EIDIdentity.specialOrganization Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxSpecialOrganizationLen
+ 1)]
internal string specialOrganization;
```

**Remarks**

This is specialOrganization, a member of class EIDIdentity.

## 1.1.12.1.22 EIDIdentity.validityDateBegin Field

Card (see page 4) validity date begin

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxDateBeginLen + 1)]
internal string validityDateBegin;
```

## 1.1.12.1.23 EIDIdentity.validityDateEnd Field

Card (see page 4) validity date end

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxDateEndLen + 1)]
internal string validityDateEnd;
```

## 1.1.12.1.24 EIDIdentity.vat1 Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxVAT1Len + 1)]
internal string vat1;
```

**Remarks**

This is vat1, a member of class EIDIdentity.

## 1.1.12.1.25 EIDIdentity.vat2 Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxVAT2Len + 1)]
internal string vat2;
```

**Remarks**

This is vat2, a member of class EIDIdentity.

### 1.1.12.1.26 EIDIdentity.whiteCane Field

White cane allowed (blind people)

**C#**

```
[MarshalAs(UnmanagedType.Bool)]
internal bool whiteCane;
```

### 1.1.12.1.27 EIDIdentity.workPermitType Field

**C#**

```
[MarshalAs(UnmanagedType.ByValTStr, SizeConst = EIDDataLength.EidMaxWorkPermitTypeLen + 1)]
internal string workPermitType;
```

**Remarks**

This is workPermitType, a member of class EIDIdentity.

### 1.1.12.1.28 EIDIdentity.yellowCane Field

Yellow cane allowed (partially sighted people)

**C#**

```
[MarshalAs(UnmanagedType.Bool)]
internal bool yellowCane;
```

## 1.1.13 EIDPicture Class

Raw picture (see page 36) data from EID card

**Inheritance Hierarchy**

Swelio.Engine.EIDPicture

**C#**

```
[StructLayout(LayoutKind.Sequential, CharSet = CharSet.Unicode)]
internal class EIDPicture;
```

**File**

NativeTypes.cs (see page 117)

**Members**

**EIDPicture Fields**

| | Name | Description |
|---|---|---|
| | picture (see page 36) | Picture raw data buffer |
| | pictureLength (see page 36) | Picture raw data buffer length |

**EIDPicture Fields**

| | Name | Description |
|---|---|---|
| | picture (see page 36) | Picture raw data buffer |
| | pictureLength (see page 36) | Picture raw data buffer length |

### 1.1.13.1 EIDPicture Fields

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### 1.1.13.1.1 EIDPicture.picture Field

Picture raw data buffer

**C#**

```
[MarshalAs(UnmanagedType.ByValArray, SizeConst = EIDDataLength.EidMaxPictureLen + 1)]
internal byte[] picture;
```

### 1.1.13.1.2 EIDPicture.pictureLength Field

Picture raw data buffer length

**C#**

```
[MarshalAs(UnmanagedType.I4)]
internal int pictureLength;
```

## 1.1.14 Encryption Class

**Inheritance Hierarchy**

Swelio.Engine.Encryption

**C#**

```
public static class Encryption;
```

**File**

Encryption.cs (see page 115)

**Members**

**Encryption Methods**

| | Name | Description |
|---|---|---|
| S | CheckFileMD5 (see page 37) | Checks the file M d5. |
| S | CheckFileSHA1 (see page 37) | Checks the file SH a1. |
| S | CheckMD5 (see page 37) | Checks the M d5. |
| S | CheckSHA1 (see page 38) | Checks the SH a1. |
| S | DecryptFile (see page 38) | Decrypts the file. |
| S | EncryptFile (see page 39) | Encrypts the file. |
| S | GenerateAuthenticationSignature (see page 39) | Generates the authentication signature. |
| S | GenerateNonRepudiationSignature (see page 40) | Generates the non repudiation signature. |
| S | GetFileMD5 (see page 40) | Gets the file M d5. |
| S | GetFileSHA1 (see page 40) | Gets the file SH a1. |
| S | GetMD5 (see page 40) | Gets the M d5. |
| S | GetSHA1 (see page 41) | Gets the SH a1. |
| S | VerifySignature (see page 41) | Verifies the signature. |

**Encryption Methods**

| | Name | Description |
|---|---|---|
| S | CheckFileMD5 (see page 37) | Checks the file M d5. |
| S | CheckFileSHA1 (see page 37) | Checks the file SH a1. |

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| | | |
|---|---|---|
| | CheckMD5 (see page 37) | Checks the M d5. |
| | CheckSHA1 (see page 38) | Checks the SH a1. |
| | DecryptFile (see page 38) | Decrypts the file. |
| | EncryptFile (see page 39) | Encrypts the file. |
| | GenerateAuthenticationSignature (see page 39) | Generates the authentication signature. |
| | GenerateNonRepudiationSignature (see page 40) | Generates the non repudiation signature. |
| | GetFileMD5 (see page 40) | Gets the file M d5. |
| | GetFileSHA1 (see page 40) | Gets the file SH a1. |
| | GetMD5 (see page 40) | Gets the M d5. |
| | GetSHA1 (see page 41) | Gets the SH a1. |
| | VerifySignature (see page 41) | Verifies the signature. |

# 1.1.14.1 Encryption Methods

## 1.1.14.1.1 Encryption.CheckFileMD5 Method

Checks the file M d5.

**C#**

```
public static bool CheckFileMD5(string fileName, byte[] md5sum);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| byte[] md5sum | The md5sum. |

## 1.1.14.1.2 Encryption.CheckFileSHA1 Method

Checks the file SH a1.

**C#**

```
public static bool CheckFileSHA1(string fileName, byte[] sha1sum);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| byte[] sha1sum | The sha1sum. |

## 1.1.14.1.3 CheckMD5 Method

### 1.1.14.1.3.1 Encryption.CheckMD5 Method (byte[], byte[])

Checks the M d5.

**C#**

```
public static bool CheckMD5(byte[] message, byte[] md5sum);
```

**Parameters**

| Parameters | Description |
|---|---|
| byte[] message | The message. |
| byte[] md5sum | The md5sum. |

### 1.1.14.1.3.2 Encryption.CheckMD5 Method (string, byte[])

Checks the M d5.

**C#**

```
public static bool CheckMD5(string message, byte[] md5sum);
```

**Parameters**

| Parameters | Description |
|---|---|
| string message | The message. |
| byte[] md5sum | The md5sum. |

## 1.1.14.1.4 CheckSHA1 Method

### 1.1.14.1.4.1 Encryption.CheckSHA1 Method (byte[], byte[])

Checks the SH a1.

**C#**

```
public static bool CheckSHA1(byte[] message, byte[] sha1sum);
```

**Parameters**

| Parameters | Description |
|---|---|
| byte[] message | The message. |
| byte[] sha1sum | The sha1sum. |

### 1.1.14.1.4.2 Encryption.CheckSHA1 Method (string, byte[])

Checks the SH a1.

**C#**

```
public static bool CheckSHA1(string message, byte[] sha1sum);
```

**Parameters**

| Parameters | Description |
|---|---|
| string message | The message. |
| byte[] sha1sum | The sha1sum. |

## 1.1.14.1.5 DecryptFile Method

### 1.1.14.1.5.1 Encryption.DecryptFile Method (CardReader, string, string)

Decrypts the file.

**C#**

```
public static bool DecryptFile(CardReader reader, string source, string destination);
```

**Parameters**

| Parameters | Description |
|---|---|
| CardReader reader | The reader. |
| string source | The source. |
| string destination | The destination. |

### 1.1.14.1.5.2 Encryption.DecryptFile Method (string, string, string)

Decrypts the file.

**C#**

```
public static bool DecryptFile(string sourceFileName, string destinationFileName, string password);
```

**Parameters**

| Parameters | Description |
|---|---|
| string sourceFileName | Name of the source file. |
| string destinationFileName | Name of the destination file. |
| string password | The password. |

## 1.1.14.1.6 EncryptFile Method

### 1.1.14.1.6.1 Encryption.EncryptFile Method (CardReader, string, string)

Encrypts the file.

**C#**

```
public static bool EncryptFile(CardReader reader, string source, string destination);
```

**Parameters**

| Parameters | Description |
|---|---|
| CardReader reader | The reader. |
| string source | The source. |
| string destination | The destination. |

### 1.1.14.1.6.2 Encryption.EncryptFile Method (string, string, string)

Encrypts the file.

**C#**

```
public static bool EncryptFile(string sourceFileName, string destinationFileName, string password);
```

**Parameters**

| Parameters | Description |
|---|---|
| string sourceFileName | Name of the source file. |
| string destinationFileName | Name of the destination file. |
| string password | The password. |

## 1.1.14.1.7 Encryption.GenerateAuthenticationSignature Method

Generates the authentication signature.

**C#**

```
public static byte[] GenerateAuthenticationSignature(CardReader reader, string pinCode,
byte[] dataHash, int hashSize);
```

**Parameters**

| Parameters | Description |
|---|---|
| CardReader reader | The reader. |
| string pinCode | The pin code. |
| byte[] dataHash | The data hash. |
| int hashSize | Size of the hash. |

## 1.1.14.1.8 Encryption.GenerateNonRepudiationSignature Method

Generates the non repudiation signature.

**C#**

```
public static byte[] GenerateNonRepudiationSignature(CardReader reader, string pinCode,
byte[] dataHash, int hashSize);
```

**Parameters**

| Parameters | Description |
|---|---|
| CardReader reader | The reader. |
| string pinCode | The pin code. |
| byte[] dataHash | The data hash. |
| int hashSize | Size of the hash. |

## 1.1.14.1.9 Encryption.GetFileMD5 Method

Gets the file M d5.

**C#**

```
public static byte[] GetFileMD5(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

## 1.1.14.1.10 Encryption.GetFileSHA1 Method

Gets the file SH a1.

**C#**

```
public static byte[] GetFileSHA1(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

## 1.1.14.1.11 GetMD5 Method

### 1.1.14.1.11.1 Encryption.GetMD5 Method (byte[])

Gets the M d5.

**C#**

```
public static byte[] GetMD5(byte[] message);
```

**Parameters**

| Parameters | Description |
|---|---|
| byte[] message | The message. |

### 1.1.14.1.11.2 Encryption.GetMD5 Method (string)

Gets the M d5.

**C#**

```
public static byte[] GetMD5(string message);
```

**Parameters**

| Parameters | Description |
|---|---|
| string message | The message. |

## 1.1.14.1.12 GetSHA1 Method

### 1.1.14.1.12.1 Encryption.GetSHA1 Method (byte[])

Gets the SH a1.

**C#**

```
public static byte[] GetSHA1(byte[] message);
```

**Parameters**

| Parameters | Description |
|---|---|
| byte[] message | The message. |

### 1.1.14.1.12.2 Encryption.GetSHA1 Method (string)

Gets the SH a1.

**C#**

```
public static byte[] GetSHA1(string message);
```

**Parameters**

| Parameters | Description |
|---|---|
| string message | The message. |

## 1.1.14.1.13 Encryption.VerifySignature Method

Verifies the signature.

**C#**

```
public static bool VerifySignature(byte[] certificate, byte[] dataHash, int hashSize,
byte[] signature, int signatureSize);
```

**Parameters**

| Parameters | Description |
|---|---|
| byte[] certificate | The certificate. |
| byte[] dataHash | The data hash. |

| | |
|---|---|
| int hashSize | Size of the hash. |
| byte[] signature | The signature. |
| int signatureSize | Size of the signature. |

# 1.1.15 FileOperations Class

Helper class for common file oprations

**Inheritance Hierarchy**

Swelio.Engine.FileOperations

**C#**

```
public static class FileOperations;
```

**File**

FileOperations.cs (see page 115)

**Members**

**FileOperations Methods**

| | Name | Description |
|---|---|---|
| S | ClearFileAttributes (see page 44) | This is ClearFileAttributes, a member of class FileOperations. |
| S | CopyFile (see page 44) | This is CopyFile, a member of class FileOperations. |
| S | CreateUnicodeFile (see page 44) | This is CreateUnicodeFile, a member of class FileOperations. |
| S | DeleteFile (see page 44) | Deletes the file. |
| S | DeleteToRecycleBin (see page 44) | Deletes file to recycle bin. |
| S | DirectoryExists (see page 45) | This is DirectoryExists, a member of class FileOperations. |
| S | ExcludeTrailingPathDelimiter (see page 45) | This is ExcludeTrailingPathDelimiter, a member of class FileOperations. |
| S | FileExists (see page 45) | Checks if file exists |
| S | FileExtensionIs (see page 45) | This is FileExtensionIs, a member of class FileOperations. |
| S | FileIsAnimatedGif (see page 45) | This is FileIsAnimatedGif, a member of class FileOperations. |
| S | FileIsExe (see page 45) | This is FileIsExe, a member of class FileOperations. |
| S | FileIsIcon (see page 46) | Checks if the file is a Windows icon file |
| S | FileIsImage (see page 46) | The file is an image (jpeg, bmp, gif, png or ico) |
| S | FileIsLink (see page 46) | This is FileIsLink, a member of class FileOperations. |
| S | FileOrFolderExists (see page 46) | Check if file or folder with given name exists |
| S | FullPath (see page 46) | If file exists in the current folder then this method simply returns the file name. if not, it will search the file with provided name in every folder, specified in the system search path. |
| S | GetExtension (see page 47) | This is GetExtension, a member of class FileOperations. |
| S | GetFilesCount (see page 47) | This is GetFilesCount, a member of class FileOperations. |
| S | GetFileSize (see page 47) | This is GetFileSize, a member of class FileOperations. |
| S | IncludeTrailingPathDelimiter (see page 47) | This is IncludeTrailingPathDelimiter, a member of class FileOperations. |
| S | IsDirectory (see page 47) | This is IsDirectory, a member of class FileOperations. |
| S | IsUnicodeFile (see page 48) | This is IsUnicodeFile, a member of class FileOperations. |
| S | IsValidFileName (see page 48) | This is IsValidFileName, a member of class FileOperations. |
| S | IsValidPathName (see page 48) | This is IsValidPathName, a member of class FileOperations. |

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| | | |
|---|---|---|
| | RenameFile (see page 48) | This is RenameFile, a member of class FileOperations. |
| | ShellCopyFile (see page 48) | This is ShellCopyFile, a member of class FileOperations. |
| | StripFileName (see page 48) | Expand environment values and relace it with the full value. This method removes environment varaibles from the file name. To insert environment variable to the file name use UnExpandPath (see page 49) method |
| | UnExpandPath (see page 49) | Replace known path by environment variables. This function does the opposite work to StripFileName (see page 48) function |

**FileOperations Methods**

| | Name | Description |
|---|---|---|
| | ClearFileAttributes (see page 44) | This is ClearFileAttributes, a member of class FileOperations. |
| | CopyFile (see page 44) | This is CopyFile, a member of class FileOperations. |
| | CreateUnicodeFile (see page 44) | This is CreateUnicodeFile, a member of class FileOperations. |
| | DeleteFile (see page 44) | Deletes the file. |
| | DeleteToRecycleBin (see page 44) | Deletes file to recycle bin. |
| | DirectoryExists (see page 45) | This is DirectoryExists, a member of class FileOperations. |
| | ExcludeTrailingPathDelimiter (see page 45) | This is ExcludeTrailingPathDelimiter, a member of class FileOperations. |
| | FileExists (see page 45) | Checks if file exists |
| | FileExtensionIs (see page 45) | This is FileExtensionIs, a member of class FileOperations. |
| | FileIsAnimatedGif (see page 45) | This is FileIsAnimatedGif, a member of class FileOperations. |
| | FileIsExe (see page 45) | This is FileIsExe, a member of class FileOperations. |
| | FileIsIcon (see page 46) | Checks if the file is a Windows icon file |
| | FileIsImage (see page 46) | The file is an image (jpeg, bmp, gif, png or ico) |
| | FileIsLink (see page 46) | This is FileIsLink, a member of class FileOperations. |
| | FileOrFolderExists (see page 46) | Check if file or folder with given name exists |
| | FullPath (see page 46) | If file exists in the current folder then this method simply returns the file name. if not, it will search the file with provided name in every folder, specified in the system search path. |
| | GetExtension (see page 47) | This is GetExtension, a member of class FileOperations. |
| | GetFilesCount (see page 47) | This is GetFilesCount, a member of class FileOperations. |
| | GetFileSize (see page 47) | This is GetFileSize, a member of class FileOperations. |
| | IncludeTrailingPathDelimiter (see page 47) | This is IncludeTrailingPathDelimiter, a member of class FileOperations. |
| | IsDirectory (see page 47) | This is IsDirectory, a member of class FileOperations. |
| | IsUnicodeFile (see page 48) | This is IsUnicodeFile, a member of class FileOperations. |
| | IsValidFileName (see page 48) | This is IsValidFileName, a member of class FileOperations. |
| | IsValidPathName (see page 48) | This is IsValidPathName, a member of class FileOperations. |
| | RenameFile (see page 48) | This is RenameFile, a member of class FileOperations. |
| | ShellCopyFile (see page 48) | This is ShellCopyFile, a member of class FileOperations. |
| | StripFileName (see page 48) | Expand environment values and relace it with the full value. This method removes environment varaibles from the file name. To insert environment variable to the file name use UnExpandPath (see page 49) method |
| | UnExpandPath (see page 49) | Replace known path by environment variables. This function does the opposite work to StripFileName (see page 48) function |

# 1.1.15.1 FileOperations Methods

## 1.1.15.1.1 FileOperations.ClearFileAttributes Method

**C#**

```
public static void ClearFileAttributes(string fileName);
```

**Remarks**

This is ClearFileAttributes, a member of class FileOperations.

## 1.1.15.1.2 FileOperations.CopyFile Method

**C#**

```
public static void CopyFile(string sourceFileName, string destinationFileName);
```

**Remarks**

This is CopyFile, a member of class FileOperations.

## 1.1.15.1.3 FileOperations.CreateUnicodeFile Method

**C#**

```
public static void CreateUnicodeFile(string fileName);
```

**Remarks**

This is CreateUnicodeFile, a member of class FileOperations.

## 1.1.15.1.4 FileOperations.DeleteFile Method

Deletes the file.

**C#**

```
public static void DeleteFile(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

## 1.1.15.1.5 FileOperations.DeleteToRecycleBin Method

Deletes file to recycle bin.

**C#**

```
public static void DeleteToRecycleBin(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

## 1.1.15.1.6 FileOperations.DirectoryExists Method

**C#**

```
public static bool DirectoryExists(string directoryName);
```

**Remarks**

This is DirectoryExists, a member of class FileOperations.

## 1.1.15.1.7 FileOperations.ExcludeTrailingPathDelimiter Method

**C#**

```
public static string ExcludeTrailingPathDelimiter(string fullPath);
```

**Remarks**

This is ExcludeTrailingPathDelimiter, a member of class FileOperations.

## 1.1.15.1.8 FileOperations.FileExists Method

Checks if file exists

**C#**

```
public static bool FileExists(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| name | The file name. |

**Returns**

True, if file exists, otherwise false

## 1.1.15.1.9 FileOperations.FileExtensionIs Method

**C#**

```
public static bool FileExtensionIs(string fileName, string extension);
```

**Remarks**

This is FileExtensionIs, a member of class FileOperations.

## 1.1.15.1.10 FileOperations.FileIsAnimatedGif Method

**C#**

```
public static bool FileIsAnimatedGif(string fileName);
```

**Remarks**

This is FileIsAnimatedGif, a member of class FileOperations.

## 1.1.15.1.11 FileOperations.FileIsExe Method

**C#**

```
public static bool FileIsExe(string fileName);
```

**Remarks**

This is FileIsExe, a member of class FileOperations.

## 1.1.15.1.12 FileOperations.FileIsIcon Method

Checks if the file is a Windows icon file

**C#**

```
public static bool FileIsIcon(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

**Returns**

Returns true if the file is a Windows icon file; otherwise returns false

## 1.1.15.1.13 FileOperations.FileIsImage Method

The file is an image (jpeg, bmp, gif, png or ico)

**C#**

```
public static bool FileIsImage(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

## 1.1.15.1.14 FileOperations.FileIsLink Method

**C#**

```
public static bool FileIsLink(string fileName);
```

**Remarks**

This is FileIsLink, a member of class FileOperations.

## 1.1.15.1.15 FileOperations.FileOrFolderExists Method

Check if file or folder with given name exists

**C#**

```
public static bool FileOrFolderExists(string name);
```

**Parameters**

| Parameters | Description |
|---|---|
| string name | The name. |

## 1.1.15.1.16 FileOperations.FullPath Method

If file exists in the current folder then this method simply returns the file name. if not, it will search the file with provided name in every folder, specified in the system search path.

**C#**

```
public static string FullPath(string fileName);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |

**Returns**

Returns the full file name including the full path to it

## 1.1.15.1.17 FileOperations.GetExtension Method

**C#**

```
public static string GetExtension(string path, bool includeDot);
```

**Remarks**

This is GetExtension, a member of class FileOperations.

## 1.1.15.1.18 FileOperations.GetFilesCount Method

**C#**

```
public static int GetFilesCount(string folderName);
```

**Remarks**

This is GetFilesCount, a member of class FileOperations.

## 1.1.15.1.19 FileOperations.GetFileSize Method

**C#**

```
public static int GetFileSize(string fileName);
```

**Remarks**

This is GetFileSize, a member of class FileOperations.

## 1.1.15.1.20 FileOperations.IncludeTrailingPathDelimiter Method

**C#**

```
public static string IncludeTrailingPathDelimiter(string fullPath);
```

**Remarks**

This is IncludeTrailingPathDelimiter, a member of class FileOperations.

## 1.1.15.1.21 FileOperations.IsDirectory Method

**C#**

```
public static bool IsDirectory(string directoryName);
```

**Remarks**

This is IsDirectory, a member of class FileOperations.

## 1.1.15.1.22 FileOperations.IsUnicodeFile Method

**C#**

```
public static bool IsUnicodeFile(string fileName);
```

**Remarks**

This is IsUnicodeFile, a member of class FileOperations.

## 1.1.15.1.23 FileOperations.IsValidFileName Method

**C#**

```
public static bool IsValidFileName(string path);
```

**Remarks**

This is IsValidFileName, a member of class FileOperations.

## 1.1.15.1.24 FileOperations.IsValidPathName Method

**C#**

```
public static bool IsValidPathName(string path);
```

**Remarks**

This is IsValidPathName, a member of class FileOperations.

## 1.1.15.1.25 FileOperations.RenameFile Method

**C#**

```
public static void RenameFile(string oldFileName, string newFileName);
```

**Remarks**

This is RenameFile, a member of class FileOperations.

## 1.1.15.1.26 FileOperations.ShellCopyFile Method

**C#**

```
public static void ShellCopyFile(string oldName, string newName);
```

**Remarks**

This is ShellCopyFile, a member of class FileOperations.

## 1.1.15.1.27 FileOperations.StripFileName Method

Expand environment values and relace it with the full value. This method removes environment varaibles from the file name. To insert environment variable to the file name use UnExpandPath (see page 49) method

**C#**

```
public static string StripFileName(string name);
```

**Parameters**

| Parameters | Description |
|---|---|
| string name | The name of the file. |

**Returns**

A string with each environment variable replaced by its value.

### 1.1.15.1.28 FileOperations.UnExpandPath Method

Replace known path by environment variables. This function does the opposite work to StripFileName (see page 48) function

**C#**

```
public static string UnExpandPath(string path);
```

**Parameters**

| Parameters | Description |
|---|---|
| string path | The path. |

## 1.1.16 Identity Class

Holds identity information that is stored on EID card

**Inheritance Hierarchy**

Swelio.Engine.Identity

**C#**

```
[Browsable(false)]
public class Identity;
```

**File**

Identity.cs (see page 116)

**Members**

**Identity Methods**

| | Name | Description |
|---|---|---|
| | Clear (see page 51) | Clear all stored information |

**Identity Properties**

| | Name | Description |
|---|---|---|
| | BirthDate (see page 51) | Birth date in YYYYMMDD format |
| | BirthLocation (see page 51) | Birth location |
| | BrexitMention1 (see page 51) | This is BrexitMention1, a member of class Identity. |
| | BrexitMention2 (see page 51) | This is BrexitMention2, a member of class Identity. |
| | CardNumber (see page 52) | Electronic ID card number |
| | ChipNumber (see page 52) | Electronic ID card physical chip number |
| | DateAndCountryOfProtection (see page 52) | This is DateAndCountryOfProtection, a member of class Identity. |
| | DocumentType (see page 52) | Document type (Belgian citizen, European Community citizen, ...) |
| | Duplicate (see page 52) | This is Duplicate, a member of class Identity. |
| | ExtendedMinority (see page 52) | Extended minority |
| | FirstName1 (see page 53) | First name |
| | FirstName2 (see page 53) | First name |

| | | |
|---|---|---|
| | MemberOfFamily (see page 53) | This is MemberOfFamily, a member of class Identity. |
| | Municipality (see page 53) | Card (see page 4) delivery municipality |
| | Nationality (see page 53) | Nationality |
| | NationalityISO (see page 53) | Gets the nationality ISO 3166 code. |
| | NationalNumber (see page 54) | National number |
| | NobleCondition (see page 54) | Noble condition |
| | RegionalFileNumber (see page 54) | This is RegionalFileNumber, a member of class Identity. |
| | Sex (see page 54) | Sex |
| | SpecialOrganization (see page 54) | This is SpecialOrganization, a member of class Identity. |
| | Surname (see page 54) | Surname |
| | ValidityDateBegin (see page 54) | Card (see page 4) validity date begin in YYYYMMDD format |
| | ValidityDateEnd (see page 55) | Card (see page 4) validity date end in YYYYMMDD format |
| | Vat1 (see page 55) | This is Vat1, a member of class Identity. |
| | Vat2 (see page 55) | This is Vat2, a member of class Identity. |
| | WhiteCane (see page 55) | White cane allowed (blind people) |
| | WorkPermitType (see page 55) | This is WorkPermitType, a member of class Identity. |
| | YellowCane (see page 55) | Yellow cane allowed (partially sighted people) |

**Identity Methods**

| | Name | Description |
|---|---|---|
| | Clear (see page 51) | Clear all stored information |

**Identity Properties**

| | Name | Description |
|---|---|---|
| | BirthDate (see page 51) | Birth date in YYYYMMDD format |
| | BirthLocation (see page 51) | Birth location |
| | BrexitMention1 (see page 51) | This is BrexitMention1, a member of class Identity. |
| | BrexitMention2 (see page 51) | This is BrexitMention2, a member of class Identity. |
| | CardNumber (see page 52) | Electronic ID card number |
| | ChipNumber (see page 52) | Electronic ID card physical chip number |
| | DateAndCountryOfProtection (see page 52) | This is DateAndCountryOfProtection, a member of class Identity. |
| | DocumentType (see page 52) | Document type (Belgian citizen, European Community citizen, ...) |
| | Duplicate (see page 52) | This is Duplicate, a member of class Identity. |
| | ExtendedMinority (see page 52) | Extended minority |
| | FirstName1 (see page 53) | First name |
| | FirstName2 (see page 53) | First name |
| | MemberOfFamily (see page 53) | This is MemberOfFamily, a member of class Identity. |
| | Municipality (see page 53) | Card (see page 4) delivery municipality |
| | Nationality (see page 53) | Nationality |
| | NationalityISO (see page 53) | Gets the nationality ISO 3166 code. |
| | NationalNumber (see page 54) | National number |
| | NobleCondition (see page 54) | Noble condition |
| | RegionalFileNumber (see page 54) | This is RegionalFileNumber, a member of class Identity. |
| | Sex (see page 54) | Sex |
| | SpecialOrganization (see page 54) | This is SpecialOrganization, a member of class Identity. |
| | Surname (see page 54) | Surname |

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| | | |
|---|---|---|
| | ValidityDateBegin (see page 54) | Card (see page 4) validity date begin in YYYYMMDD format |
| | ValidityDateEnd (see page 55) | Card (see page 4) validity date end in YYYYMMDD format |
| | Vat1 (see page 55) | This is Vat1, a member of class Identity. |
| | Vat2 (see page 55) | This is Vat2, a member of class Identity. |
| | WhiteCane (see page 55) | White cane allowed (blind people) |
| | WorkPermitType (see page 55) | This is WorkPermitType, a member of class Identity. |
| | YellowCane (see page 55) | Yellow cane allowed (partially sighted people) |

# 1.1.16.1 Identity Methods

## 1.1.16.1.1 Identity.Clear Method

Clear all stored information

**C#**

```
public void Clear();
```

# 1.1.16.2 Identity Properties

## 1.1.16.2.1 Identity.BirthDate Property

Birth date in YYYYMMDD format

**C#**

```
[Description("Birth date")]
[ReadOnly(true)]
public DateTime BirthDate;
```

## 1.1.16.2.2 Identity.BirthLocation Property

Birth location

**C#**

```
[Description("Birth location")]
[ReadOnly(true)]
public string BirthLocation;
```

## 1.1.16.2.3 Identity.BrexitMention1 Property

**C#**

```
public string BrexitMention1;
```

**Remarks**

This is BrexitMention1, a member of class Identity.

## 1.1.16.2.4 Identity.BrexitMention2 Property

**C#**

```
public string BrexitMention2;
```

**Remarks**

This is BrexitMention2, a member of class Identity.

## 1.1.16.2.5 Identity.CardNumber Property

Electronic ID card number

**C#**

```
[Description("Electronic ID card number")]
[ReadOnly(true)]
public string CardNumber;
```

## 1.1.16.2.6 Identity.ChipNumber Property

Electronic ID card physical chip number

**C#**

```
[Description("Electronic ID card physical chip number")]
[ReadOnly(true)]
public string ChipNumber;
```

## 1.1.16.2.7 Identity.DateAndCountryOfProtection Property

**C#**

```
public string DateAndCountryOfProtection;
```

**Remarks**

This is DateAndCountryOfProtection, a member of class Identity.

## 1.1.16.2.8 Identity.DocumentType Property

Document type (Belgian citizen, European Community citizen, ...)

**C#**

```
[Description("Document type (Belgian citizen, European Community citizen, ...)")]
[ReadOnly(true)]
public DocumentType DocumentType;
```

## 1.1.16.2.9 Identity.Duplicate Property

**C#**

```
public string Duplicate;
```

**Remarks**

This is Duplicate, a member of class Identity.

## 1.1.16.2.10 Identity.ExtendedMinority Property

Extended minority

**C#**

```
[Description("Extended minotity")]
[ReadOnly(true)]
public bool ExtendedMinority;
```

## 1.1.16.2.11 Identity.FirstName1 Property

First name

**C#**

```
[Description("First name of the card owner")]
[ReadOnly(true)]
public string FirstName1;
```

## 1.1.16.2.12 Identity.FirstName2 Property

First name

**C#**

```
[Description("First name of the card owner")]
[ReadOnly(true)]
public string FirstName2;
```

## 1.1.16.2.13 Identity.MemberOfFamily Property

**C#**

```
public bool MemberOfFamily;
```

**Remarks**

This is MemberOfFamily, a member of class Identity.

## 1.1.16.2.14 Identity.Municipality Property

Card (see page 4) delivery municipality

**C#**

```
[Description("Card delivery municipality")]
[ReadOnly(true)]
public string Municipality;
```

## 1.1.16.2.15 Identity.Nationality Property

Nationality

**C#**

```
[Description("Nationality of the card owner")]
[ReadOnly(true)]
public string Nationality;
```

## 1.1.16.2.16 Identity.NationalityISO Property

Gets the nationality ISO 3166 code.

**C#**

```
public string NationalityISO;
```

**Description**

The nationality ISO 3166 code.

## 1.1.16.2.17 Identity.NationalNumber Property

National number

**C#**

```
[Description("Unique national number")]
[ReadOnly(true)]
public string NationalNumber;
```

## 1.1.16.2.18 Identity.NobleCondition Property

Noble condition

**C#**

```
[Description("Noble condition")]
[ReadOnly(true)]
public string NobleCondition;
```

## 1.1.16.2.19 Identity.RegionalFileNumber Property

**C#**

```
public string RegionalFileNumber;
```

**Remarks**

This is RegionalFileNumber, a member of class Identity.

## 1.1.16.2.20 Identity.Sex Property

Sex

**C#**

```
[Description("Sex of the card owner")]
[ReadOnly(true)]
public Gender Sex;
```

## 1.1.16.2.21 Identity.SpecialOrganization Property

**C#**

```
public string SpecialOrganization;
```

**Remarks**

This is SpecialOrganization, a member of class Identity.

## 1.1.16.2.22 Identity.Surname Property

Surname

**C#**

```
[Description("Surname of the card owner")]
[ReadOnly(true)]
public string Surname;
```

## 1.1.16.2.23 Identity.ValidityDateBegin Property

Card (see page 4) validity date begin in YYYYMMDD format

**C#**

```
[Description("Card validity date begin")]
[ReadOnly(true)]
public DateTime ValidityDateBegin;
```

## 1.1.16.2.24 Identity.ValidityDateEnd Property

Card (see page 4) validity date end in YYYYMMDD format

**C#**

```
[Description("Card validity date end")]
[ReadOnly(true)]
public DateTime ValidityDateEnd;
```

## 1.1.16.2.25 Identity.Vat1 Property

**C#**

```
public string Vat1;
```

**Remarks**

This is Vat1, a member of class Identity.

## 1.1.16.2.26 Identity.Vat2 Property

**C#**

```
public string Vat2;
```

**Remarks**

This is Vat2, a member of class Identity.

## 1.1.16.2.27 Identity.WhiteCane Property

White cane allowed (blind people)

**C#**

```
[Description("White cane allowed (blind people)")]
[ReadOnly(true)]
public bool WhiteCane;
```

## 1.1.16.2.28 Identity.WorkPermitType Property

**C#**

```
public string WorkPermitType;
```

**Remarks**

This is WorkPermitType, a member of class Identity.

## 1.1.16.2.29 Identity.YellowCane Property

Yellow cane allowed (partially sighted people)

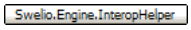
**C#**

```
[Description("Yellow cane allowed (partially sighted people)")]
[ReadOnly(true)]
public bool YellowCane;
```

# 1.1.17 InteropHelper Class

Helper class for accessing native Windows API and resources Not recommeded to use it directly without deep understanding of Windows API

**Inheritance Hierarchy**




**C#**

```
public static class InteropHelper;
```

**File**

InteropHelper.cs (see page 116)

**Members**

**InteropHelper Methods**

| | Name | Description |
|---|---|---|
| | ClearUnusedMemory (see page 57) | Clears the unused memory. |
| | GetStartup (see page 57) | This is GetStartup, a member of class InteropHelper. |
| | SetStartup (see page 57) | Add/Remove registry entries for windows startup. Only for current user |
| | ShowScreenKeyboard (see page 57) | Shows the screen keyboard. |
| | ShowWindowsError (see page 58) | Shows the windows error. |
| | TurnMonitorOff (see page 58) | Turns the monitor off. |
| | TurnMonitorOn (see page 58) | Turns the monitor on. |

**InteropHelper Properties**

| | Name | Description |
|---|---|---|
| | IsMediaCenter (see page 58) | This is IsMediaCenter, a member of class InteropHelper. |
| | IsMultiTouchReady (see page 58) | This is IsMultiTouchReady, a member of class InteropHelper. |
| | IsTabletPC (see page 58) | This is IsTabletPC, a member of class InteropHelper. |
| | MetroActive (see page 59) | This is MetroActive, a member of class InteropHelper. |
| | RunningOnVista (see page 59) | Gets a value indicating whether [running on vista]. |
| | RunningOnWin10 (see page 59) | Determines if the application is running on Windows 10 |
| | RunningOnWin7 (see page 59) | Determines if the application is running on Windows 7 |
| | RunningOnWin8 (see page 59) | This is RunningOnWin8, a member of class InteropHelper. |
| | RunningOnXP (see page 59) | Gets a value indicating whether [running on XP]. |

**InteropHelper Methods**

| | Name | Description |
|---|---|---|
| | ClearUnusedMemory (see page 57) | Clears the unused memory. |
| | GetStartup (see page 57) | This is GetStartup, a member of class InteropHelper. |
| | SetStartup (see page 57) | Add/Remove registry entries for windows startup. Only for current user |
| | ShowScreenKeyboard (see page 57) | Shows the screen keyboard. |
| | ShowWindowsError (see page 58) | Shows the windows error. |
| | TurnMonitorOff (see page 58) | Turns the monitor off. |
| | TurnMonitorOn (see page 58) | Turns the monitor on. |

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**InteropHelper Properties**

| | Name | Description |
|---|---|---|
| | IsMediaCenter (see page 58) | This is IsMediaCenter, a member of class InteropHelper. |
| | IsMultiTouchReady (see page 58) | This is IsMultiTouchReady, a member of class InteropHelper. |
| | IsTabletPC (see page 58) | This is IsTabletPC, a member of class InteropHelper. |
| | MetroActive (see page 59) | This is MetroActive, a member of class InteropHelper. |
| | RunningOnVista (see page 59) | Gets a value indicating whether [running on vista]. |
| | RunningOnWin10 (see page 59) | Determines if the application is running on Windows 10 |
| | RunningOnWin7 (see page 59) | Determines if the application is running on Windows 7 |
| | RunningOnWin8 (see page 59) | This is RunningOnWin8, a member of class InteropHelper. |
| | RunningOnXP (see page 59) | Gets a value indicating whether [running on XP]. |

# 1.1.17.1 InteropHelper Methods

## 1.1.17.1.1 InteropHelper.ClearUnusedMemory Method

Clears the unused memory.

**C#**

```
public static void ClearUnusedMemory();
```

## 1.1.17.1.2 InteropHelper.GetStartup Method

**C#**

```
public static bool GetStartup(string appName);
```

**Remarks**

This is GetStartup, a member of class InteropHelper.

## 1.1.17.1.3 InteropHelper.SetStartup Method

Add/Remove registry entries for windows startup. Only for current user

**C#**

```
public static void SetStartup(string appName, string appPath, bool enable);
```

**Parameters**

| Parameters | Description |
|---|---|
| string appName | Name of the application. |
| string appPath | The application path. |
| bool enable | if set to `true` [enable]. |

## 1.1.17.1.4 InteropHelper.ShowScreenKeyboard Method

Shows the screen keyboard.

**C#**

```
[SecurityPermissionAttribute(SecurityAction.LinkDemand, Flags =
SecurityPermissionFlag.UnmanagedCode)]
public static void ShowScreenKeyboard();
```

### 1.1.17.1.5 InteropHelper.ShowWindowsError Method

Shows the windows error.

**C#**

```
public static void ShowWindowsError(int errorCode);
```

**Parameters**

| Parameters | Description |
|---|---|
| int errorCode | The error code. |

### 1.1.17.1.6 InteropHelper.TurnMonitorOff Method

Turns the monitor off.

**C#**

```
public static void TurnMonitorOff();
```

### 1.1.17.1.7 InteropHelper.TurnMonitorOn Method

Turns the monitor on.

**C#**

```
public static void TurnMonitorOn();
```

## 1.1.17.2 InteropHelper Properties

### 1.1.17.2.1 InteropHelper.IsMediaCenter Property

**C#**

```
public static bool IsMediaCenter;
```

**Remarks**

This is IsMediaCenter, a member of class InteropHelper.

### 1.1.17.2.2 InteropHelper.IsMultiTouchReady Property

**C#**

```
public static bool IsMultiTouchReady;
```

**Remarks**

This is IsMultiTouchReady, a member of class InteropHelper.

### 1.1.17.2.3 InteropHelper.IsTabletPC Property

**C#**

```
public static bool IsTabletPC;
```

**Remarks**

This is IsTabletPC, a member of class InteropHelper.

## 1.1.17.2.4 InteropHelper.MetroActive Property

**C#**

```
public static bool MetroActive;
```

**Remarks**

This is MetroActive, a member of class InteropHelper.

## 1.1.17.2.5 InteropHelper.RunningOnVista Property

Gets a value indicating whether [running on vista].

**C#**

```
public static bool RunningOnVista;
```

**Description**

`true` if [running on vista]; otherwise, `false`.

## 1.1.17.2.6 InteropHelper.RunningOnWin10 Property

Determines if the application is running on Windows 10

**C#**

```
public static bool RunningOnWin10;
```

## 1.1.17.2.7 InteropHelper.RunningOnWin7 Property

Determines if the application is running on Windows 7

**C#**

```
public static bool RunningOnWin7;
```

## 1.1.17.2.8 InteropHelper.RunningOnWin8 Property

**C#**

```
public static bool RunningOnWin8;
```

**Remarks**

This is RunningOnWin8, a member of class InteropHelper.

## 1.1.17.2.9 InteropHelper.RunningOnXP Property

Gets a value indicating whether [running on XP].

**C#**

```
public static bool RunningOnXP;
```

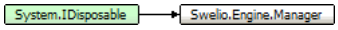
**Description**

`true` if [running on XP]; otherwise, `false`.

# 1.1.18 Manager Class

The Swelio Engine Manager class used to activate and deactivate engine and gives an access to the smart card readers, connected to the PC

**Inheritance Hierarchy**




**C#**

```
public sealed class Manager : IDisposable;
```

**File**

Manager.cs (see page 116)

**Members**

**Methods**

| | Name | Description |
|---|---|---|
| | Manager (see page 61) | Initializes a new instance of the Manager class. |

**Manager Methods**

| | Name | Description |
|---|---|---|
| | Dispose (see page 61) | Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources. |
| | GetReader (see page 61) | Gets the default card reader. |
| | GetReaderName (see page 62) | Gets the name of the reader. |
| | ListReaders (see page 62) | Returns the array with the names of the attached card readers |
| | ReloadReadersList (see page 62) | Reloads the readers list. |
| | SendAPDU (see page 62) | This is SendAPDU, a member of class Manager. |
| | SetDefaultReader (see page 62) | Sets the default card reader. |

**Manager Properties**

| | Name | Description |
|---|---|---|
| | Active (see page 63) | Gets or sets a value indicating whether this Manager is active. |
| R | ReaderCount (see page 63) | Gets the reader count. |
| | TraceEvents (see page 63) | Gets or sets a value indicating whether the card events are traced. |
| | TraceHardwareEvents (see page 63) | This is TraceHardwareEvents, a member of class Manager. |
| | TraceServiceEvents (see page 63) | This is TraceServiceEvents, a member of class Manager. |

**Manager Events**

| | Name | Description |
|---|---|---|
| | CardInserted (see page 64) | This is CardInserted, a member of class Manager. |
| | CardRemoved (see page 64) | This is CardRemoved, a member of class Manager. |
| | ReadersListChanged (see page 64) | This is ReadersListChanged, a member of class Manager. |

**Methods**

| | Name | Description |
|---|---|---|
| | Manager (see page 61) | Initializes a new instance of the Manager class. |

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**Manager Events**

| | Name | Description |
|---|---|---|
| | CardInserted (see page 64) | This is CardInserted, a member of class Manager. |
| | CardRemoved (see page 64) | This is CardRemoved, a member of class Manager. |
| | ReadersListChanged (see page 64) | This is ReadersListChanged, a member of class Manager. |

**Manager Methods**

| | Name | Description |
|---|---|---|
| | Dispose (see page 61) | Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources. |
| | GetReader (see page 61) | Gets the default card reader. |
| | GetReaderName (see page 62) | Gets the name of the reader. |
| | ListReaders (see page 62) | Returns the array with the names of the attached card readers |
| | ReloadReadersList (see page 62) | Reloads the readers list. |
| | SendAPDU (see page 62) | This is SendAPDU, a member of class Manager. |
| | SetDefaultReader (see page 62) | Sets the default card reader. |

**Manager Properties**

| | Name | Description |
|---|---|---|
| | Active (see page 63) | Gets or sets a value indicating whether this Manager is active. |
| R | ReaderCount (see page 63) | Gets the reader count. |
| | TraceEvents (see page 63) | Gets or sets a value indicating whether the card events are traced. |
| | TraceHardwareEvents (see page 63) | This is TraceHardwareEvents, a member of class Manager. |
| | TraceServiceEvents (see page 63) | This is TraceServiceEvents, a member of class Manager. |

# 1.1.18.1 Manager.Manager Constructor

Initializes a new instance of the Manager (see page 60) class.

**C#**

```
public Manager();
```

# 1.1.18.2 Manager Methods

## 1.1.18.2.1 Manager.Dispose Method

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

**C#**

```
public void Dispose();
```

## 1.1.18.2.2 GetReader Method

### 1.1.18.2.2.1 Manager.GetReader Method ()

Gets the default card reader.

**C#**

```
public CardReader GetReader(int readerNumber);
public CardReader GetReader();
```

### 1.1.18.2.2.2 Manager.GetReader Method (string)

Gets the reader.

**C#**

```
public CardReader GetReader(string name);
```

**Parameters**

| Parameters | Description |
|---|---|
| string name | The name. |

## 1.1.18.2.3 Manager.GetReaderName Method

Gets the name of the reader.

**C#**

```
public string GetReaderName(int readerNumber);
```

**Parameters**

| Parameters | Description |
|---|---|
| int readerNumber | The reader number. |

## 1.1.18.2.4 Manager.ListReaders Method

Returns the array with the names of the attached card readers

**C#**

```
public string[] ListReaders();
```

## 1.1.18.2.5 Manager.ReloadReadersList Method

Reloads the readers list.

**C#**

```
public void ReloadReadersList();
```

## 1.1.18.2.6 Manager.SendAPDU Method

**C#**

```
public bool SendAPDU(int readerNumber, byte[] apdu, int apduLen, byte[] result, ref int
len);
```

**Remarks**

This is SendAPDU, a member of class Manager.

## 1.1.18.2.7 Manager.SetDefaultReader Method

Sets the default card reader.

**C#**

```
public void SetDefaultReader(int readerNumber);
```

**Parameters**

| Parameters | Description |
|---|---|
| int readerNumber | The reader number. |

# 1.1.18.3 Manager Properties

## 1.1.18.3.1 Manager.Active Property

Gets or sets a value indicating whether this Manager (see page 60) is active.

**C#**

```
public bool Active;
```

**Description**

`true` if active; otherwise, `false`.

## 1.1.18.3.2 Manager.ReaderCount Property

Gets the reader count.

**C#**

```
public int ReaderCount;
```

## 1.1.18.3.3 Manager.TraceEvents Property

Gets or sets a value indicating whether the card events are traced.

**C#**

```
public bool TraceEvents;
```

**Description**

`true` if trace card events; otherwise, `false`.

## 1.1.18.3.4 Manager.TraceHardwareEvents Property

**C#**

```
public bool TraceHardwareEvents;
```

**Remarks**

This is TraceHardwareEvents, a member of class Manager.

## 1.1.18.3.5 Manager.TraceServiceEvents Property

**C#**

```
public bool TraceServiceEvents;
```

**Remarks**

This is TraceServiceEvents, a member of class Manager.

# 1.1.18.4 Manager Events

### 1.1.18.4.1 Manager.CardInserted Event

**C#**

```
public event EventHandler<CardEventArgs> CardInserted;
```

**Remarks**

This is CardInserted, a member of class Manager.

### 1.1.18.4.2 Manager.CardRemoved Event

**C#**

```
public event EventHandler<CardEventArgs> CardRemoved;
```

**Remarks**

This is CardRemoved, a member of class Manager.

### 1.1.18.4.3 Manager.ReadersListChanged Event

**C#**

```
public event EventHandler ReadersListChanged;
```

**Remarks**

This is ReadersListChanged, a member of class Manager.

## 1.1.19 NationalNumber Class

National number formatter

**Inheritance Hierarchy**

Swelio.Engine.NationalNumber

**C#**

```
public static class NationalNumber;
```

**File**

NationalNumber.cs (see page 117)

**Members**

**NationalNumber Methods**

| | Name | Description |
|---|---|---|
| S | ToString (see page 65) | Converts the natinal number value to its formatted String representation |

**NationalNumber Methods**

| | Name | Description |
|---|---|---|
| S | ToString (see page 65) | Converts the natinal number value to its formatted String representation |

### 1.1.19.1 NationalNumber Methods

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### 1.1.19.1.1 NationalNumber.ToString Method

Converts the natinal number value to its formatted String representation

**C#**

```
public static string ToString(string nationalNumber);
```

**Parameters**

| Parameters | Description |
|---|---|
| string nationalNumber | The unformatted national number |

## 1.1.20 NativeMethods Class

**Inheritance Hierarchy**

Swelio.Engine.NativeMethods

**C#**

```
internal static class NativeMethods;
```

**File**

NativeMethods.cs (see page 117)

**Remarks**

This is class Swelio.Engine.NativeMethods.

**Members**

**NativeMethods Methods**

| | Name | Description |
|---|---|---|
| | ActivateCardEx (see page 73) | This is ActivateCardEx, a member of class NativeMethods. |
| | AddFileToContainer (see page 73) | This is AddFileToContainer, a member of class NativeMethods. |
| | CardDecryptFile (see page 73) | This is CardDecryptFile, a member of class NativeMethods. |
| | CardEncryptFile (see page 74) | This is CardEncryptFile, a member of class NativeMethods. |
| | CardSignCadesT (see page 74) | This is CardSignCadesT, a member of class NativeMethods. |
| | CardSignCadesTEx (see page 74) | This is CardSignCadesTEx, a member of class NativeMethods. |
| | CardSignCMS (see page 74) | This is CardSignCMS, a member of class NativeMethods. |
| | CardSignCMSEx (see page 74) | This is CardSignCMSEx, a member of class NativeMethods. |
| | CertSignCadesT (see page 74) | This is CertSignCadesT, a member of class NativeMethods. |
| | CertSignCadesTData (see page 75) | This is CertSignCadesTData, a member of class NativeMethods. |
| | CertSignCadesTEx (see page 75) | This is CertSignCadesTEx, a member of class NativeMethods. |
| | CertSignCMS (see page 75) | This is CertSignCMS, a member of class NativeMethods. |
| | CertSignCMSData (see page 75) | This is CertSignCMSData, a member of class NativeMethods. |
| | CertSignCMSEx (see page 75) | This is CertSignCMSEx, a member of class NativeMethods. |
| | CheckMD5 (see page 75) | This is CheckMD5, a member of class NativeMethods. |
| | CheckSHA1 (see page 76) | This is CheckSHA1, a member of class NativeMethods. |

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| | | |
|---|---|---|
| | ClearFileAttributes (see page 76) | This is ClearFileAttributes, a member of class NativeMethods. |
| | ClearUnusedMemory (see page 76) | This is ClearUnusedMemory, a member of class NativeMethods. |
| | ContainerCertificate (see page 76) | This is ContainerCertificate, a member of class NativeMethods. |
| | ContainerEidCertificate (see page 76) | This is ContainerEidCertificate, a member of class NativeMethods. |
| | ContainerPickCertificate (see page 76) | This is ContainerPickCertificate, a member of class NativeMethods. |
| | CreateCardBuffer (see page 77) | This is CreateCardBuffer, a member of class NativeMethods. |
| | CreateUnicodeFile (see page 77) | This is CreateUnicodeFile, a member of class NativeMethods. |
| | CurrentIPAddress (see page 77) | This is CurrentIPAddress, a member of class NativeMethods. |
| | DeactivateCardEx (see page 77) | This is DeactivateCardEx, a member of class NativeMethods. |
| | DecryptFileAES (see page 77) | This is DecryptFileAES, a member of class NativeMethods. |
| | DeleteCardBuffer (see page 77) | This is DeleteCardBuffer, a member of class NativeMethods. |
| | DeleteObject (see page 78) | This is DeleteObject, a member of class NativeMethods. |
| | DeleteToRecycleBin (see page 78) | This is DeleteToRecycleBin, a member of class NativeMethods. |
| | DirectoryExists (see page 78) | This is DirectoryExists, a member of class NativeMethods. |
| | DisplayCertificate (see page 78) | This is DisplayCertificate, a member of class NativeMethods. |
| | EmptyRecycleBin (see page 78) | This is EmptyRecycleBin, a member of class NativeMethods. |
| | EncodeCertificate (see page 78) | This is EncodeCertificate, a member of class NativeMethods. |
| | EncodePhoto (see page 79) | This is EncodePhoto, a member of class NativeMethods. |
| | EncryptFileAES (see page 79) | This is EncryptFileAES, a member of class NativeMethods. |
| | FileCopy (see page 79) | This is FileCopy, a member of class NativeMethods. |
| | FileDelete (see page 79) | This is FileDelete, a member of class NativeMethods. |
| | FileExists (see page 79) | This is FileExists, a member of class NativeMethods. |
| | FileExtensionIs (see page 79) | This is FileExtensionIs, a member of class NativeMethods. |
| | FileGetSize (see page 80) | This is FileGetSize, a member of class NativeMethods. |
| | FileIsExe (see page 80) | This is FileIsExe, a member of class NativeMethods. |
| | FileIsIcon (see page 80) | This is FileIsIcon, a member of class NativeMethods. |
| | FileIsImage (see page 80) | This is FileIsImage, a member of class NativeMethods. |
| | FileIsLink (see page 80) | This is FileIsLink, a member of class NativeMethods. |
| | FileOrFolderExists (see page 80) | This is FileOrFolderExists, a member of class NativeMethods. |
| | FileRename (see page 81) | This is FileRename, a member of class NativeMethods. |
| | FreeContainer (see page 81) | This is FreeContainer, a member of class NativeMethods. |
| | FullPath (see page 81) | This is FullPath, a member of class NativeMethods. |
| | GenerateAuthenticationSignatureEx (see page 81) | This is GenerateAuthenticationSignatureEx, a member of class NativeMethods. |
| | GenerateBMP (see page 81) | This is GenerateBMP, a member of class NativeMethods. |
| | GenerateNonRepudiationSignatureEx (see page 81) | This is GenerateNonRepudiationSignatureEx, a member of class NativeMethods. |
| | GeneratePNG (see page 82) | This is GeneratePNG, a member of class NativeMethods. |
| | GenerateQRCodeEx (see page 82) | This is GenerateQRCodeEx, a member of class NativeMethods. |
| | GetCardBufferA (see page 82) | This is GetCardBufferA, a member of class NativeMethods. |

| | | |
|---|---|---|
| | GetCardBufferSize (see page 82) | This is GetCardBufferSize, a member of class NativeMethods. |
| | GetCardBufferW (see page 82) | This is GetCardBufferW, a member of class NativeMethods. |
| | GetCardSerialNumber (see page 82) | This is GetCardSerialNumber, a member of class NativeMethods. |
| | GetCardVersion (see page 83) | This is GetCardVersion, a member of class NativeMethods. |
| | GetEncodedCertificateSize (see page 83) | This is GetEncodedCertificateSize, a member of class NativeMethods. |
| | GetEncodedPhotoSize (see page 83) | This is GetEncodedPhotoSize, a member of class NativeMethods. |
| | GetFileMD5 (see page 83) | This is GetFileMD5, a member of class NativeMethods. |
| | GetFilesCount (see page 83) | This is GetFilesCount, a member of class NativeMethods. |
| | GetFileSHA1 (see page 83) | This is GetFileSHA1, a member of class NativeMethods. |
| | GetHBitmap (see page 84) | This is GetHBitmap, a member of class NativeMethods. |
| | GetISOCode (see page 84) | This is GetISOCode, a member of class NativeMethods. |
| | GetMD5 (see page 84) | This is GetMD5, a member of class NativeMethods. |
| | GetReaderIndex (see page 84) | This is GetReaderIndex, a member of class NativeMethods. |
| | GetReaderName (see page 84) | This is GetReaderName, a member of class NativeMethods. |
| | GetReaderNameLen (see page 84) | This is GetReaderNameLen, a member of class NativeMethods. |
| | GetReadersCount (see page 85) | This is GetReadersCount, a member of class NativeMethods. |
| | GetSelectedReaderIndex (see page 85) | This is GetSelectedReaderIndex, a member of class NativeMethods. |
| | GetSHA1 (see page 85) | This is GetSHA1, a member of class NativeMethods. |
| | GetStartup (see page 85) | This is GetStartup, a member of class NativeMethods. |
| | HibernateWindows (see page 85) | This is HibernateWindows, a member of class NativeMethods. |
| | IgnoreHardwareEvents (see page 85) | This is IgnoreHardwareEvents, a member of class NativeMethods. |
| | IgnoreServiceEvents (see page 86) | This is IgnoreServiceEvents, a member of class NativeMethods. |
| | InitializeContainer (see page 86) | This is InitializeContainer, a member of class NativeMethods. |
| | IsAnimatedGIF (see page 86) | This is IsAnimatedGIF, a member of class NativeMethods. |
| | IsCardActivatedEx (see page 86) | This is IsCardActivatedEx, a member of class NativeMethods. |
| | IsCardPresentEx (see page 86) | This is IsCardPresentEx, a member of class NativeMethods. |
| | IsCardStillInsertedEx (see page 86) | This is IsCardStillInsertedEx, a member of class NativeMethods. |
| | IsConnectedToInternet (see page 87) | This is IsConnectedToInternet, a member of class NativeMethods. |
| | IsDirectory (see page 87) | This is IsDirectory, a member of class NativeMethods. |
| | IsEIDCardEx (see page 87) | This is IsEIDCardEx, a member of class NativeMethods. |
| | IsEngineActive (see page 87) | This is IsEngineActive, a member of class NativeMethods. |
| | IsMale (see page 87) | This is IsMale, a member of class NativeMethods. |
| | IsMediaCenter (see page 87) | This is IsMediaCenter, a member of class NativeMethods. |
| | IsMetroActive (see page 88) | This is IsMetroActive, a member of class NativeMethods. |
| | IsMultiTouchReady (see page 88) | This is IsMultiTouchReady, a member of class NativeMethods. |
| | IsTabletPC (see page 88) | This is IsTabletPC, a member of class NativeMethods. |
| | IsUnicodeFile (see page 88) | This is IsUnicodeFile, a member of class NativeMethods. |
| | IsValidFileName (see page 88) | This is IsValidFileName, a member of class NativeMethods. |

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| | | |
|---|---|---|
| | IsValidPathName (see page 88) | This is IsValidPathName, a member of class NativeMethods. |
| | IsWindows10 (see page 89) | This is IsWindows10, a member of class NativeMethods. |
| | IsWindows7 (see page 89) | This is IsWindows7, a member of class NativeMethods. |
| | IsWindows8 (see page 89) | This is IsWindows8, a member of class NativeMethods. |
| | IsWindowsVista (see page 89) | This is IsWindowsVista, a member of class NativeMethods. |
| | IsWindowsXP (see page 89) | This is IsWindowsXP, a member of class NativeMethods. |
| | IsWOW64 (see page 89) | This is IsWOW64, a member of class NativeMethods. |
| | LoadCertificate (see page 90) | This is LoadCertificate, a member of class NativeMethods. |
| | LoadPhoto (see page 90) | This is LoadPhoto, a member of class NativeMethods. |
| | PathUnExpandEnvStrings (see page 90) | This is PathUnExpandEnvStrings, a member of class NativeMethods. |
| | PortAvailable (see page 90) | This is PortAvailable, a member of class NativeMethods. |
| | ReadAddressEx (see page 90) | This is ReadAddressEx, a member of class NativeMethods. |
| | ReadAuthenticationCertificate (see page 90) | This is ReadAuthenticationCertificate, a member of class NativeMethods. |
| | ReadCaCertificate (see page 91) | This is ReadCaCertificate, a member of class NativeMethods. |
| | ReadIdentityEx (see page 91) | This is ReadIdentityEx, a member of class NativeMethods. |
| | ReadNonRepudiationCertificate (see page 91) | This is ReadNonRepudiationCertificate, a member of class NativeMethods. |
| | ReadPhotoEx (see page 91) | This is ReadPhotoEx, a member of class NativeMethods. |
| | ReadRootCaCertificate (see page 91) | This is ReadRootCaCertificate, a member of class NativeMethods. |
| | ReadRrnCertificate (see page 91) | This is ReadRrnCertificate, a member of class NativeMethods. |
| | RecycleBinEmpty (see page 92) | This is RecycleBinEmpty, a member of class NativeMethods. |
| | ReloadReadersList (see page 92) | This is ReloadReadersList, a member of class NativeMethods. |
| | RemoveCallback (see page 92) | This is RemoveCallback, a member of class NativeMethods. |
| | RemoveStartup (see page 92) | This is RemoveStartup, a member of class NativeMethods. |
| | SaveAuthenticationCertificateEx (see page 92) | This is SaveAuthenticationCertificateEx, a member of class NativeMethods. |
| | SaveCaCertificateEx (see page 92) | This is SaveCaCertificateEx, a member of class NativeMethods. |
| | SaveCardToToXMLStreamExA (see page 93) | This is SaveCardToToXMLStreamExA, a member of class NativeMethods. |
| | SaveCardToToXMLStreamExW (see page 93) | This is SaveCardToToXMLStreamExW, a member of class NativeMethods. |
| | SaveCardToXmlEx (see page 93) | This is SaveCardToXmlEx, a member of class NativeMethods. |
| | SaveCardToXmlExAnsi (see page 93) | This is SaveCardToXmlExAnsi, a member of class NativeMethods. |
| | SaveContainer (see page 93) | This is SaveContainer, a member of class NativeMethods. |
| | SaveNonRepudiationCertificateEx (see page 93) | This is SaveNonRepudiationCertificateEx, a member of class NativeMethods. |
| | SavePersonCsvToStreamA (see page 94) | This is SavePersonCsvToStreamA, a member of class NativeMethods. |
| | SavePersonCsvToStreamW (see page 94) | This is SavePersonCsvToStreamW, a member of class NativeMethods. |
| | SavePersonToCsvEx (see page 94) | This is SavePersonToCsvEx, a member of class NativeMethods. |
| | SavePersonToCsvExAnsi (see page 94) | This is SavePersonToCsvExAnsi, a member of class NativeMethods. |

| | | |
|---|---|---|
| | SavePhotoAsBitmapEx (see page 94) | This is SavePhotoAsBitmapEx, a member of class NativeMethods. |
| | SavePhotoAsJpegEx (see page 94) | This is SavePhotoAsJpegEx, a member of class NativeMethods. |
| | SaveRootCaCertificateEx (see page 95) | This is SaveRootCaCertificateEx, a member of class NativeMethods. |
| | SaveRrnCertificateEx (see page 95) | This is SaveRrnCertificateEx, a member of class NativeMethods. |
| | SelectReader (see page 95) | This is SelectReader, a member of class NativeMethods. |
| | SelectReaderByName (see page 95) | This is SelectReaderByName, a member of class NativeMethods. |
| | SendAPDU (see page 95) | This is SendAPDU, a member of class NativeMethods. |
| | SetCallback (see page 95) | This is SetCallback, a member of class NativeMethods. |
| | SetStartup (see page 96) | This is SetStartup, a member of class NativeMethods. |
| | ShellCopyFile (see page 96) | This is ShellCopyFile, a member of class NativeMethods. |
| | ShowError (see page 96) | This is ShowError, a member of class NativeMethods. |
| | ShutdownWindows (see page 96) | This is ShutdownWindows, a member of class NativeMethods. |
| | StartEngine (see page 96) | This is StartEngine, a member of class NativeMethods. |
| | StopEngine (see page 96) | This is StopEngine, a member of class NativeMethods. |
| | StripFileName (see page 97) | This is StripFileName, a member of class NativeMethods. |
| | SuspendWindows (see page 97) | This is SuspendWindows, a member of class NativeMethods. |
| | TurnMonitorOff (see page 97) | This is TurnMonitorOff, a member of class NativeMethods. |
| | TurnMonitorOn (see page 97) | This is TurnMonitorOn, a member of class NativeMethods. |
| | VerifyPinEx (see page 97) | This is VerifyPinEx, a member of class NativeMethods. |
| | VerifySignature (see page 97) | This is VerifySignature, a member of class NativeMethods. |

**NativeMethods Methods**

| | Name | Description |
|---|---|---|
| | ActivateCardEx (see page 73) | This is ActivateCardEx, a member of class NativeMethods. |
| | AddFileToContainer (see page 73) | This is AddFileToContainer, a member of class NativeMethods. |
| | CardDecryptFile (see page 73) | This is CardDecryptFile, a member of class NativeMethods. |
| | CardEncryptFile (see page 74) | This is CardEncryptFile, a member of class NativeMethods. |
| | CardSignCadesT (see page 74) | This is CardSignCadesT, a member of class NativeMethods. |
| | CardSignCadesTEx (see page 74) | This is CardSignCadesTEx, a member of class NativeMethods. |
| | CardSignCMS (see page 74) | This is CardSignCMS, a member of class NativeMethods. |
| | CardSignCMSEx (see page 74) | This is CardSignCMSEx, a member of class NativeMethods. |
| | CertSignCadesT (see page 74) | This is CertSignCadesT, a member of class NativeMethods. |
| | CertSignCadesTData (see page 75) | This is CertSignCadesTData, a member of class NativeMethods. |
| | CertSignCadesTEx (see page 75) | This is CertSignCadesTEx, a member of class NativeMethods. |
| | CertSignCMS (see page 75) | This is CertSignCMS, a member of class NativeMethods. |
| | CertSignCMSData (see page 75) | This is CertSignCMSData, a member of class NativeMethods. |
| | CertSignCMSEx (see page 75) | This is CertSignCMSEx, a member of class NativeMethods. |
| | CheckMD5 (see page 75) | This is CheckMD5, a member of class NativeMethods. |
| | CheckSHA1 (see page 76) | This is CheckSHA1, a member of class NativeMethods. |

| | | |
|---|---|---|
| | ClearFileAttributes (see page 76) | This is ClearFileAttributes, a member of class NativeMethods. |
| | ClearUnusedMemory (see page 76) | This is ClearUnusedMemory, a member of class NativeMethods. |
| | ContainerCertificate (see page 76) | This is ContainerCertificate, a member of class NativeMethods. |
| | ContainerEidCertificate (see page 76) | This is ContainerEidCertificate, a member of class NativeMethods. |
| | ContainerPickCertificate (see page 76) | This is ContainerPickCertificate, a member of class NativeMethods. |
| | CreateCardBuffer (see page 77) | This is CreateCardBuffer, a member of class NativeMethods. |
| | CreateUnicodeFile (see page 77) | This is CreateUnicodeFile, a member of class NativeMethods. |
| | CurrentIPAddress (see page 77) | This is CurrentIPAddress, a member of class NativeMethods. |
| | DeactivateCardEx (see page 77) | This is DeactivateCardEx, a member of class NativeMethods. |
| | DecryptFileAES (see page 77) | This is DecryptFileAES, a member of class NativeMethods. |
| | DeleteCardBuffer (see page 77) | This is DeleteCardBuffer, a member of class NativeMethods. |
| | DeleteObject (see page 78) | This is DeleteObject, a member of class NativeMethods. |
| | DeleteToRecycleBin (see page 78) | This is DeleteToRecycleBin, a member of class NativeMethods. |
| | DirectoryExists (see page 78) | This is DirectoryExists, a member of class NativeMethods. |
| | DisplayCertificate (see page 78) | This is DisplayCertificate, a member of class NativeMethods. |
| | EmptyRecycleBin (see page 78) | This is EmptyRecycleBin, a member of class NativeMethods. |
| | EncodeCertificate (see page 78) | This is EncodeCertificate, a member of class NativeMethods. |
| | EncodePhoto (see page 79) | This is EncodePhoto, a member of class NativeMethods. |
| | EncryptFileAES (see page 79) | This is EncryptFileAES, a member of class NativeMethods. |
| | FileCopy (see page 79) | This is FileCopy, a member of class NativeMethods. |
| | FileDelete (see page 79) | This is FileDelete, a member of class NativeMethods. |
| | FileExists (see page 79) | This is FileExists, a member of class NativeMethods. |
| | FileExtensionIs (see page 79) | This is FileExtensionIs, a member of class NativeMethods. |
| | FileGetSize (see page 80) | This is FileGetSize, a member of class NativeMethods. |
| | FileIsExe (see page 80) | This is FileIsExe, a member of class NativeMethods. |
| | FileIsIcon (see page 80) | This is FileIsIcon, a member of class NativeMethods. |
| | FileIsImage (see page 80) | This is FileIsImage, a member of class NativeMethods. |
| | FileIsLink (see page 80) | This is FileIsLink, a member of class NativeMethods. |
| | FileOrFolderExists (see page 80) | This is FileOrFolderExists, a member of class NativeMethods. |
| | FileRename (see page 81) | This is FileRename, a member of class NativeMethods. |
| | FreeContainer (see page 81) | This is FreeContainer, a member of class NativeMethods. |
| | FullPath (see page 81) | This is FullPath, a member of class NativeMethods. |
| | GenerateAuthenticationSignatureEx (see page 81) | This is GenerateAuthenticationSignatureEx, a member of class NativeMethods. |
| | GenerateBMP (see page 81) | This is GenerateBMP, a member of class NativeMethods. |
| | GenerateNonRepudiationSignatureEx (see page 81) | This is GenerateNonRepudiationSignatureEx, a member of class NativeMethods. |
| | GeneratePNG (see page 82) | This is GeneratePNG, a member of class NativeMethods. |
| | GenerateQRCodeEx (see page 82) | This is GenerateQRCodeEx, a member of class NativeMethods. |
| | GetCardBufferA (see page 82) | This is GetCardBufferA, a member of class NativeMethods. |

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|---|---|---|
| | GetCardBufferSize (see page 82) | This is GetCardBufferSize, a member of class NativeMethods. |
| | GetCardBufferW (see page 82) | This is GetCardBufferW, a member of class NativeMethods. |
| | GetCardSerialNumber (see page 82) | This is GetCardSerialNumber, a member of class NativeMethods. |
| | GetCardVersion (see page 83) | This is GetCardVersion, a member of class NativeMethods. |
| | GetEncodedCertificateSize (see page 83) | This is GetEncodedCertificateSize, a member of class NativeMethods. |
| | GetEncodedPhotoSize (see page 83) | This is GetEncodedPhotoSize, a member of class NativeMethods. |
| | GetFileMD5 (see page 83) | This is GetFileMD5, a member of class NativeMethods. |
| | GetFilesCount (see page 83) | This is GetFilesCount, a member of class NativeMethods. |
| | GetFileSHA1 (see page 83) | This is GetFileSHA1, a member of class NativeMethods. |
| | GetHBitmap (see page 84) | This is GetHBitmap, a member of class NativeMethods. |
| | GetISOCode (see page 84) | This is GetISOCode, a member of class NativeMethods. |
| | GetMD5 (see page 84) | This is GetMD5, a member of class NativeMethods. |
| | GetReaderIndex (see page 84) | This is GetReaderIndex, a member of class NativeMethods. |
| | GetReaderName (see page 84) | This is GetReaderName, a member of class NativeMethods. |
| | GetReaderNameLen (see page 84) | This is GetReaderNameLen, a member of class NativeMethods. |
| | GetReadersCount (see page 85) | This is GetReadersCount, a member of class NativeMethods. |
| | GetSelectedReaderIndex (see page 85) | This is GetSelectedReaderIndex, a member of class NativeMethods. |
| | GetSHA1 (see page 85) | This is GetSHA1, a member of class NativeMethods. |
| | GetStartup (see page 85) | This is GetStartup, a member of class NativeMethods. |
| | HibernateWindows (see page 85) | This is HibernateWindows, a member of class NativeMethods. |
| | IgnoreHardwareEvents (see page 85) | This is IgnoreHardwareEvents, a member of class NativeMethods. |
| | IgnoreServiceEvents (see page 86) | This is IgnoreServiceEvents, a member of class NativeMethods. |
| | InitializeContainer (see page 86) | This is InitializeContainer, a member of class NativeMethods. |
| | IsAnimatedGIF (see page 86) | This is IsAnimatedGIF, a member of class NativeMethods. |
| | IsCardActivatedEx (see page 86) | This is IsCardActivatedEx, a member of class NativeMethods. |
| | IsCardPresentEx (see page 86) | This is IsCardPresentEx, a member of class NativeMethods. |
| | IsCardStillInsertedEx (see page 86) | This is IsCardStillInsertedEx, a member of class NativeMethods. |
| | IsConnectedToInternet (see page 87) | This is IsConnectedToInternet, a member of class NativeMethods. |
| | IsDirectory (see page 87) | This is IsDirectory, a member of class NativeMethods. |
| | IsEIDCardEx (see page 87) | This is IsEIDCardEx, a member of class NativeMethods. |
| | IsEngineActive (see page 87) | This is IsEngineActive, a member of class NativeMethods. |
| | IsMale (see page 87) | This is IsMale, a member of class NativeMethods. |
| | IsMediaCenter (see page 87) | This is IsMediaCenter, a member of class NativeMethods. |
| | IsMetroActive (see page 88) | This is IsMetroActive, a member of class NativeMethods. |
| | IsMultiTouchReady (see page 88) | This is IsMultiTouchReady, a member of class NativeMethods. |
| | IsTabletPC (see page 88) | This is IsTabletPC, a member of class NativeMethods. |
| | IsUnicodeFile (see page 88) | This is IsUnicodeFile, a member of class NativeMethods. |
| | IsValidFileName (see page 88) | This is IsValidFileName, a member of class NativeMethods. |

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|---|---|---|
| | IsValidPathName (see page 88) | This is IsValidPathName, a member of class NativeMethods. |
| | IsWindows10 (see page 89) | This is IsWindows10, a member of class NativeMethods. |
| | IsWindows7 (see page 89) | This is IsWindows7, a member of class NativeMethods. |
| | IsWindows8 (see page 89) | This is IsWindows8, a member of class NativeMethods. |
| | IsWindowsVista (see page 89) | This is IsWindowsVista, a member of class NativeMethods. |
| | IsWindowsXP (see page 89) | This is IsWindowsXP, a member of class NativeMethods. |
| | IsWOW64 (see page 89) | This is IsWOW64, a member of class NativeMethods. |
| | LoadCertificate (see page 90) | This is LoadCertificate, a member of class NativeMethods. |
| | LoadPhoto (see page 90) | This is LoadPhoto, a member of class NativeMethods. |
| | PathUnExpandEnvStrings (see page 90) | This is PathUnExpandEnvStrings, a member of class NativeMethods. |
| | PortAvailable (see page 90) | This is PortAvailable, a member of class NativeMethods. |
| | ReadAddressEx (see page 90) | This is ReadAddressEx, a member of class NativeMethods. |
| | ReadAuthenticationCertificate (see page 90) | This is ReadAuthenticationCertificate, a member of class NativeMethods. |
| | ReadCaCertificate (see page 91) | This is ReadCaCertificate, a member of class NativeMethods. |
| | ReadIdentityEx (see page 91) | This is ReadIdentityEx, a member of class NativeMethods. |
| | ReadNonRepudiationCertificate (see page 91) | This is ReadNonRepudiationCertificate, a member of class NativeMethods. |
| | ReadPhotoEx (see page 91) | This is ReadPhotoEx, a member of class NativeMethods. |
| | ReadRootCaCertificate (see page 91) | This is ReadRootCaCertificate, a member of class NativeMethods. |
| | ReadRrnCertificate (see page 91) | This is ReadRrnCertificate, a member of class NativeMethods. |
| | RecycleBinEmpty (see page 92) | This is RecycleBinEmpty, a member of class NativeMethods. |
| | ReloadReadersList (see page 92) | This is ReloadReadersList, a member of class NativeMethods. |
| | RemoveCallback (see page 92) | This is RemoveCallback, a member of class NativeMethods. |
| | RemoveStartup (see page 92) | This is RemoveStartup, a member of class NativeMethods. |
| | SaveAuthenticationCertificateEx (see page 92) | This is SaveAuthenticationCertificateEx, a member of class NativeMethods. |
| | SaveCaCertificateEx (see page 92) | This is SaveCaCertificateEx, a member of class NativeMethods. |
| | SaveCardToToXMLStreamExA (see page 93) | This is SaveCardToToXMLStreamExA, a member of class NativeMethods. |
| | SaveCardToToXMLStreamExW (see page 93) | This is SaveCardToToXMLStreamExW, a member of class NativeMethods. |
| | SaveCardToXmlEx (see page 93) | This is SaveCardToXmlEx, a member of class NativeMethods. |
| | SaveCardToXmlExAnsi (see page 93) | This is SaveCardToXmlExAnsi, a member of class NativeMethods. |
| | SaveContainer (see page 93) | This is SaveContainer, a member of class NativeMethods. |
| | SaveNonRepudiationCertificateEx (see page 93) | This is SaveNonRepudiationCertificateEx, a member of class NativeMethods. |
| | SavePersonCsvToStreamA (see page 94) | This is SavePersonCsvToStreamA, a member of class NativeMethods. |
| | SavePersonCsvToStreamW (see page 94) | This is SavePersonCsvToStreamW, a member of class NativeMethods. |
| | SavePersonToCsvEx (see page 94) | This is SavePersonToCsvEx, a member of class NativeMethods. |
| | SavePersonToCsvExAnsi (see page 94) | This is SavePersonToCsvExAnsi, a member of class NativeMethods. |

| | | |
|---|---|---|
| | SavePhotoAsBitmapEx (see page 94) | This is SavePhotoAsBitmapEx, a member of class NativeMethods. |
| | SavePhotoAsJpegEx (see page 94) | This is SavePhotoAsJpegEx, a member of class NativeMethods. |
| | SaveRootCaCertificateEx (see page 95) | This is SaveRootCaCertificateEx, a member of class NativeMethods. |
| | SaveRrnCertificateEx (see page 95) | This is SaveRrnCertificateEx, a member of class NativeMethods. |
| | SelectReader (see page 95) | This is SelectReader, a member of class NativeMethods. |
| | SelectReaderByName (see page 95) | This is SelectReaderByName, a member of class NativeMethods. |
| | SendAPDU (see page 95) | This is SendAPDU, a member of class NativeMethods. |
| | SetCallback (see page 95) | This is SetCallback, a member of class NativeMethods. |
| | SetStartup (see page 96) | This is SetStartup, a member of class NativeMethods. |
| | ShellCopyFile (see page 96) | This is ShellCopyFile, a member of class NativeMethods. |
| | ShowError (see page 96) | This is ShowError, a member of class NativeMethods. |
| | ShutdownWindows (see page 96) | This is ShutdownWindows, a member of class NativeMethods. |
| | StartEngine (see page 96) | This is StartEngine, a member of class NativeMethods. |
| | StopEngine (see page 96) | This is StopEngine, a member of class NativeMethods. |
| | StripFileName (see page 97) | This is StripFileName, a member of class NativeMethods. |
| | SuspendWindows (see page 97) | This is SuspendWindows, a member of class NativeMethods. |
| | TurnMonitorOff (see page 97) | This is TurnMonitorOff, a member of class NativeMethods. |
| | TurnMonitorOn (see page 97) | This is TurnMonitorOn, a member of class NativeMethods. |
| | VerifyPinEx (see page 97) | This is VerifyPinEx, a member of class NativeMethods. |
| | VerifySignature (see page 97) | This is VerifySignature, a member of class NativeMethods. |

# 1.1.20.1 NativeMethods Methods

## 1.1.20.1.1 NativeMethods.ActivateCardEx Method

**C#**

```
public static bool ActivateCardEx(int readerNumber);
```

**Remarks**

This is ActivateCardEx, a member of class NativeMethods.

## 1.1.20.1.2 NativeMethods.AddFileToContainer Method

**C#**

```
public static bool AddFileToContainer(IntPtr container, string fileName);
```

**Remarks**

This is AddFileToContainer, a member of class NativeMethods.

## 1.1.20.1.3 NativeMethods.CardDecryptFile Method

**C#**

```
public static bool CardDecryptFile(string source, string destination);
```

**Remarks**

This is CardDecryptFile, a member of class NativeMethods.

## 1.1.20.1.4 NativeMethods.CardEncryptFile Method

**C#**

```
public static bool CardEncryptFile(string source, string destination);
```

**Remarks**

This is CardEncryptFile, a member of class NativeMethods.

## 1.1.20.1.5 NativeMethods.CardSignCadesT Method

**C#**

```
public static bool CardSignCadesT(int readerNumber, byte[] data, int dataLen, byte[]
signature, ref int signatureLen);
```

**Remarks**

This is CardSignCadesT, a member of class NativeMethods.

## 1.1.20.1.6 NativeMethods.CardSignCadesTEx Method

**C#**

```
public static bool CardSignCadesTEx(int readerNumber, byte[] data, int dataLen, byte[]
signature, ref int signatureLen, [In, Out] EIDError error);
```

**Remarks**

This is CardSignCadesTEx, a member of class NativeMethods.

## 1.1.20.1.7 NativeMethods.CardSignCMS Method

**C#**

```
public static bool CardSignCMS(int readerNumber, byte[] data, int dataLen, byte[]
signature, ref int signatureLen);
```

**Remarks**

This is CardSignCMS, a member of class NativeMethods.

## 1.1.20.1.8 NativeMethods.CardSignCMSEx Method

**C#**

```
public static bool CardSignCMSEx(int readerNumber, byte[] data, int dataLen, byte[]
signature, ref int signatureLen, [In, Out] EIDError error);
```

**Remarks**

This is CardSignCMSEx, a member of class NativeMethods.

## 1.1.20.1.9 NativeMethods.CertSignCadesT Method

**C#**

```
public static bool CertSignCadesT(string certificate, string password, byte[] data, int
dataLen, byte[] signature, ref int signatureLen);
```

**Remarks**

This is CertSignCadesT, a member of class NativeMethods.

## 1.1.20.1.10 NativeMethods.CertSignCadesTData Method

**C#**

```
public static bool CertSignCadesTData(byte[] certificate, int certLen, string password,
byte[] data, int dataLen, byte[] signature, ref int signatureLen, [In, Out] EIDError error);
```

**Remarks**

This is CertSignCadesTData, a member of class NativeMethods.

## 1.1.20.1.11 NativeMethods.CertSignCadesTEx Method

**C#**

```
public static bool CertSignCadesTEx(string certificate, string password, byte[] data, int
dataLen, byte[] signature, ref int signatureLen, [In, Out] EIDError error);
```

**Remarks**

This is CertSignCadesTEx, a member of class NativeMethods.

## 1.1.20.1.12 NativeMethods.CertSignCMS Method

**C#**

```
public static bool CertSignCMS(string certificate, string password, byte[] data, int
dataLen, byte[] signature, ref int signatureLen);
```

**Remarks**

This is CertSignCMS, a member of class NativeMethods.

## 1.1.20.1.13 NativeMethods.CertSignCMSData Method

**C#**

```
public static bool CertSignCMSData(byte[] certificate, int certLen, string password, byte[]
data, int dataLen, byte[] signature, ref int signatureLen, [In, Out] EIDError error);
```

**Remarks**

This is CertSignCMSData, a member of class NativeMethods.

## 1.1.20.1.14 NativeMethods.CertSignCMSEx Method

**C#**

```
public static bool CertSignCMSEx(string certificate, string password, byte[] data, int
dataLen, byte[] signature, ref int signatureLen, [In, Out] EIDError error);
```

**Remarks**

This is CertSignCMSEx, a member of class NativeMethods.

## 1.1.20.1.15 NativeMethods.CheckMD5 Method

**C#**

```
public static bool CheckMD5(byte[] source, int sourceSize, byte[] buffer, int bufferSize);
```

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**Remarks**

This is CheckMD5, a member of class NativeMethods.

## 1.1.20.1.16 NativeMethods.CheckSHA1 Method

**C#**

```
public static bool CheckSHA1(byte[] source, int sourceSize, byte[] buffer, int bufferSize);
```

**Remarks**

This is CheckSHA1, a member of class NativeMethods.

## 1.1.20.1.17 NativeMethods.ClearFileAttributes Method

**C#**

```
public static void ClearFileAttributes(string fileName);
```

**Remarks**

This is ClearFileAttributes, a member of class NativeMethods.

## 1.1.20.1.18 NativeMethods.ClearUnusedMemory Method

**C#**

```
public static void ClearUnusedMemory();
```

**Remarks**

This is ClearUnusedMemory, a member of class NativeMethods.

## 1.1.20.1.19 NativeMethods.ContainerCertificate Method

**C#**

```
public static bool ContainerCertificate(IntPtr container, string fileName, string password);
```

**Remarks**

This is ContainerCertificate, a member of class NativeMethods.

## 1.1.20.1.20 NativeMethods.ContainerEidCertificate Method

**C#**

```
public static bool ContainerEidCertificate(IntPtr container, int readerNumber);
```

**Remarks**

This is ContainerEidCertificate, a member of class NativeMethods.

## 1.1.20.1.21 NativeMethods.ContainerPickCertificate Method

**C#**

```
public static bool ContainerPickCertificate(IntPtr container);
```

**Remarks**

This is ContainerPickCertificate, a member of class NativeMethods.

## 1.1.20.1.22 NativeMethods.CreateCardBuffer Method

**C#**

```
public static IntPtr CreateCardBuffer();
```

**Remarks**

This is CreateCardBuffer, a member of class NativeMethods.

## 1.1.20.1.23 NativeMethods.CreateUnicodeFile Method

**C#**

```
public static void CreateUnicodeFile(string fileName);
```

**Remarks**

This is CreateUnicodeFile, a member of class NativeMethods.

## 1.1.20.1.24 NativeMethods.CurrentIPAddress Method

**C#**

```
public static void CurrentIPAddress(StringBuilder address, uint bufferSize);
```

**Remarks**

This is CurrentIPAddress, a member of class NativeMethods.

## 1.1.20.1.25 NativeMethods.DeactivateCardEx Method

**C#**

```
public static void DeactivateCardEx(int readerNumber);
```

**Remarks**

This is DeactivateCardEx, a member of class NativeMethods.

## 1.1.20.1.26 NativeMethods.DecryptFileAES Method

**C#**

```
public static bool DecryptFileAES(string source, string destination, string password);
```

**Remarks**

This is DecryptFileAES, a member of class NativeMethods.

## 1.1.20.1.27 NativeMethods.DeleteCardBuffer Method

**C#**

```
public static void DeleteCardBuffer(IntPtr buffer);
```

**Remarks**

This is DeleteCardBuffer, a member of class NativeMethods.

## 1.1.20.1.28 NativeMethods.DeleteObject Method

**C#**

```
[DllImport("gdi32.dll", CharSet = CharSet.Auto, CallingConvention =
CallingConvention.StdCall, SetLastError = true)]
[return: MarshalAs(UnmanagedType.Bool)]
public static extern bool DeleteObject(IntPtr hObject);
```

**Remarks**

This is DeleteObject, a member of class NativeMethods.

## 1.1.20.1.29 NativeMethods.DeleteToRecycleBin Method

**C#**

```
public static void DeleteToRecycleBin(string fileName, bool silent);
```

**Remarks**

This is DeleteToRecycleBin, a member of class NativeMethods.

## 1.1.20.1.30 NativeMethods.DirectoryExists Method

**C#**

```
public static bool DirectoryExists(string fileName);
```

**Remarks**

This is DirectoryExists, a member of class NativeMethods.

## 1.1.20.1.31 NativeMethods.DisplayCertificate Method

**C#**

```
public static void DisplayCertificate([In, Out] EIDCertificate certificate);
```

**Remarks**

This is DisplayCertificate, a member of class NativeMethods.

## 1.1.20.1.32 NativeMethods.EmptyRecycleBin Method

**C#**

```
public static void EmptyRecycleBin();
```

**Remarks**

This is EmptyRecycleBin, a member of class NativeMethods.

## 1.1.20.1.33 NativeMethods.EncodeCertificate Method

**C#**

```
public static int EncodeCertificate([In, Out] EIDCertificate certificate, byte[] buffer,
int bufferSize);
```

**Remarks**

This is EncodeCertificate, a member of class NativeMethods.

## 1.1.20.1.34 NativeMethods.EncodePhoto Method

**C#**

```
public static int EncodePhoto([In, Out] EIDPicture photo, byte[] buffer, int bufferSize);
```

**Remarks**

This is EncodePhoto, a member of class NativeMethods.

## 1.1.20.1.35 NativeMethods.EncryptFileAES Method

**C#**

```
public static bool EncryptFileAES(string source, string destination, string password);
```

**Remarks**

This is EncryptFileAES, a member of class NativeMethods.

## 1.1.20.1.36 NativeMethods.FileCopy Method

**C#**

```
public static bool FileCopy(string oldName, string newName);
```

**Remarks**

This is FileCopy, a member of class NativeMethods.

## 1.1.20.1.37 NativeMethods.FileDelete Method

**C#**

```
public static void FileDelete(string fileName);
```

**Remarks**

This is FileDelete, a member of class NativeMethods.

## 1.1.20.1.38 NativeMethods.FileExists Method

**C#**

```
public static bool FileExists(string fileName);
```

**Remarks**

This is FileExists, a member of class NativeMethods.

## 1.1.20.1.39 NativeMethods.FileExtensionIs Method

**C#**

```
public static bool FileExtensionIs(string fileName, string extension);
```

**Remarks**

This is FileExtensionIs, a member of class NativeMethods.

## 1.1.20.1.40 NativeMethods.FileGetSize Method

**C#**

```
public static int FileGetSize(string fileName);
```

**Remarks**

This is FileGetSize, a member of class NativeMethods.

## 1.1.20.1.41 NativeMethods.FileIsExe Method

**C#**

```
public static bool FileIsExe(string fileName);
```

**Remarks**

This is FileIsExe, a member of class NativeMethods.

## 1.1.20.1.42 NativeMethods.FileIsIcon Method

**C#**

```
public static bool FileIsIcon(string fileName);
```

**Remarks**

This is FileIsIcon, a member of class NativeMethods.

## 1.1.20.1.43 NativeMethods.FileIsImage Method

**C#**

```
public static bool FileIsImage(string fileName);
```

**Remarks**

This is FileIsImage, a member of class NativeMethods.

## 1.1.20.1.44 NativeMethods.FileIsLink Method

**C#**

```
public static bool FileIsLink(string fileName);
```

**Remarks**

This is FileIsLink, a member of class NativeMethods.

## 1.1.20.1.45 NativeMethods.FileOrFolderExists Method

**C#**

```
public static bool FileOrFolderExists(string fileName);
```

**Remarks**

This is FileOrFolderExists, a member of class NativeMethods.

## 1.1.20.1.46 NativeMethods.FileRename Method

**C#**

```
public static bool FileRename(string oldName, string newName);
```

**Remarks**

This is FileRename, a member of class NativeMethods.

## 1.1.20.1.47 NativeMethods.FreeContainer Method

**C#**

```
public static void FreeContainer(IntPtr container);
```

**Remarks**

This is FreeContainer, a member of class NativeMethods.

## 1.1.20.1.48 NativeMethods.FullPath Method

**C#**

```
public static bool FullPath(string fileName, StringBuilder sb);
```

**Remarks**

This is FullPath, a member of class NativeMethods.

## 1.1.20.1.49 NativeMethods.GenerateAuthenticationSignatureEx Method

**C#**

```
public static bool GenerateAuthenticationSignatureEx(int readerNumber, string pinCode,
byte[] dataHash, int hashSize, byte[] signature, ref int signatureSize);
```

**Remarks**

This is GenerateAuthenticationSignatureEx, a member of class NativeMethods.

## 1.1.20.1.50 NativeMethods.GenerateBMP Method

**C#**

```
public static void GenerateBMP(string fileName, string Text, int Margin, int Size, int
Level);
```

**Remarks**

This is GenerateBMP, a member of class NativeMethods.

## 1.1.20.1.51 NativeMethods.GenerateNonRepudiationSignatureEx Method

**C#**

```
public static bool GenerateNonRepudiationSignatureEx(int readerNumber, string pinCode,
byte[] dataHash, int hashSize, byte[] signature, ref int signatureSize);
```

**Remarks**

This is GenerateNonRepudiationSignatureEx, a member of class NativeMethods.

## 1.1.20.1.52 NativeMethods.GeneratePNG Method

**C#**

```
public static void GeneratePNG(string fileName, string Text, int Margin, int Size, int
Level);
```

**Remarks**

This is GeneratePNG, a member of class NativeMethods.

## 1.1.20.1.53 NativeMethods.GenerateQRCodeEx Method

**C#**

```
public static bool GenerateQRCodeEx(int readerNumber, string fileName);
```

**Remarks**

This is GenerateQRCodeEx, a member of class NativeMethods.

## 1.1.20.1.54 NativeMethods.GetCardBufferA Method

**C#**

```
public static void GetCardBufferA(IntPtr buffer, StringBuilder strDest, int count);
```

**Remarks**

This is GetCardBufferA, a member of class NativeMethods.

## 1.1.20.1.55 NativeMethods.GetCardBufferSize Method

**C#**

```
public static int GetCardBufferSize(IntPtr buffer);
```

**Remarks**

This is GetCardBufferSize, a member of class NativeMethods.

## 1.1.20.1.56 NativeMethods.GetCardBufferW Method

**C#**

```
public static void GetCardBufferW(IntPtr buffer, StringBuilder strDest, int count);
```

**Remarks**

This is GetCardBufferW, a member of class NativeMethods.

## 1.1.20.1.57 NativeMethods.GetCardSerialNumber Method

**C#**

```
public static bool GetCardSerialNumber(int readerNumber, byte[] serialNumber, ref int
serialNumberSize);
```

**Remarks**

This is GetCardSerialNumber, a member of class NativeMethods.

## 1.1.20.1.58 NativeMethods.GetCardVersion Method

**C#**

```
public static int GetCardVersion(int readerNumber);
```

**Remarks**

This is GetCardVersion, a member of class NativeMethods.

## 1.1.20.1.59 NativeMethods.GetEncodedCertificateSize Method

**C#**

```
public static int GetEncodedCertificateSize([In, Out] EIDCertificate Certificate);
```

**Remarks**

This is GetEncodedCertificateSize, a member of class NativeMethods.

## 1.1.20.1.60 NativeMethods.GetEncodedPhotoSize Method

**C#**

```
public static int GetEncodedPhotoSize([In, Out] EIDPicture Photo);
```

**Remarks**

This is GetEncodedPhotoSize, a member of class NativeMethods.

## 1.1.20.1.61 NativeMethods.GetFileMD5 Method

**C#**

```
public static bool GetFileMD5(string fileName, byte[] buffer, int bufferSize);
```

**Remarks**

This is GetFileMD5, a member of class NativeMethods.

## 1.1.20.1.62 NativeMethods.GetFilesCount Method

**C#**

```
public static int GetFilesCount(string FolderName);
```

**Remarks**

This is GetFilesCount, a member of class NativeMethods.

## 1.1.20.1.63 NativeMethods.GetFileSHA1 Method

**C#**

```
public static bool GetFileSHA1(string fileName, byte[] buffer, int bufferSize);
```

**Remarks**

This is GetFileSHA1, a member of class NativeMethods.

## 1.1.20.1.64 NativeMethods.GetHBitmap Method

**C#**

```
public static IntPtr GetHBitmap(string Text, int Margin, int Size, int Level);
```

**Remarks**

This is GetHBitmap, a member of class NativeMethods.

## 1.1.20.1.65 NativeMethods.GetISOCode Method

**C#**

```
public static bool GetISOCode(string Nationality, StringBuilder IsoCode, int BufferSize);
```

**Remarks**

This is GetISOCode, a member of class NativeMethods.

## 1.1.20.1.66 NativeMethods.GetMD5 Method

**C#**

```
public static bool GetMD5(byte[] source, int sourceSize, byte[] buffer, int bufferSize);
```

**Remarks**

This is GetMD5, a member of class NativeMethods.

## 1.1.20.1.67 NativeMethods.GetReaderIndex Method

**C#**

```
public static int GetReaderIndex(string ReaderName);
```

**Remarks**

This is GetReaderIndex, a member of class NativeMethods.

## 1.1.20.1.68 NativeMethods.GetReaderName Method

**C#**

```
public static int GetReaderName(int readerNumber, StringBuilder StrDest, int Count);
```

**Remarks**

This is GetReaderName, a member of class NativeMethods.

## 1.1.20.1.69 NativeMethods.GetReaderNameLen Method

**C#**

```
public static int GetReaderNameLen(int readerNumber);
```

**Remarks**

This is GetReaderNameLen, a member of class NativeMethods.

## 1.1.20.1.70 NativeMethods.GetReadersCount Method

**C#**

```
public static int GetReadersCount();
```

**Remarks**

This is GetReadersCount, a member of class NativeMethods.

## 1.1.20.1.71 NativeMethods.GetSelectedReaderIndex Method

**C#**

```
public static int GetSelectedReaderIndex();
```

**Remarks**

This is GetSelectedReaderIndex, a member of class NativeMethods.

## 1.1.20.1.72 NativeMethods.GetSHA1 Method

**C#**

```
public static bool GetSHA1(byte[] source, int sourceSize, byte[] buffer, int bufferSize);
```

**Remarks**

This is GetSHA1, a member of class NativeMethods.

## 1.1.20.1.73 NativeMethods.GetStartup Method

**C#**

```
public static bool GetStartup(string AppName);
```

**Remarks**

This is GetStartup, a member of class NativeMethods.

## 1.1.20.1.74 NativeMethods.HibernateWindows Method

**C#**

```
public static bool HibernateWindows();
```

**Remarks**

This is HibernateWindows, a member of class NativeMethods.

## 1.1.20.1.75 NativeMethods.IgnoreHardwareEvents Method

**C#**

```
public static void IgnoreHardwareEvents(bool value);
```

**Remarks**

This is IgnoreHardwareEvents, a member of class NativeMethods.

## 1.1.20.1.76 NativeMethods.IgnoreServiceEvents Method

**C#**

```
public static void IgnoreServiceEvents(bool value);
```

**Remarks**

This is IgnoreServiceEvents, a member of class NativeMethods.

## 1.1.20.1.77 NativeMethods.InitializeContainer Method

**C#**

```
public static IntPtr InitializeContainer();
```

**Remarks**

This is InitializeContainer, a member of class NativeMethods.

## 1.1.20.1.78 NativeMethods.IsAnimatedGIF Method

**C#**

```
public static bool IsAnimatedGIF(string fileName);
```

**Remarks**

This is IsAnimatedGIF, a member of class NativeMethods.

## 1.1.20.1.79 NativeMethods.IsCardActivatedEx Method

**C#**

```
public static bool IsCardActivatedEx(int readerNumber);
```

**Remarks**

This is IsCardActivatedEx, a member of class NativeMethods.

## 1.1.20.1.80 NativeMethods.IsCardPresentEx Method

**C#**

```
public static bool IsCardPresentEx(int readerNumber);
```

**Remarks**

This is IsCardPresentEx, a member of class NativeMethods.

## 1.1.20.1.81 NativeMethods.IsCardStillInsertedEx Method

**C#**

```
public static bool IsCardStillInsertedEx(int readerNumber);
```

**Remarks**

This is IsCardStillInsertedEx, a member of class NativeMethods.

## 1.1.20.1.82 NativeMethods.IsConnectedToInternet Method

**C#**

```
public static bool IsConnectedToInternet();
```

**Remarks**

This is IsConnectedToInternet, a member of class NativeMethods.

## 1.1.20.1.83 NativeMethods.IsDirectory Method

**C#**

```
public static bool IsDirectory(string FolderName);
```

**Remarks**

This is IsDirectory, a member of class NativeMethods.

## 1.1.20.1.84 NativeMethods.IsEIDCardEx Method

**C#**

```
public static bool IsEIDCardEx(int readerNumber);
```

**Remarks**

This is IsEIDCardEx, a member of class NativeMethods.

## 1.1.20.1.85 NativeMethods.IsEngineActive Method

**C#**

```
public static bool IsEngineActive();
```

**Remarks**

This is IsEngineActive, a member of class NativeMethods.

## 1.1.20.1.86 NativeMethods.IsMale Method

**C#**

```
public static bool IsMale([In, Out] EIDIdentity Identity);
```

**Remarks**

This is IsMale, a member of class NativeMethods.

## 1.1.20.1.87 NativeMethods.IsMediaCenter Method

**C#**

```
public static bool IsMediaCenter();
```

**Remarks**

This is IsMediaCenter, a member of class NativeMethods.

## 1.1.20.1.88 NativeMethods.IsMetroActive Method

**C#**

```
public static bool IsMetroActive();
```

**Remarks**

This is IsMetroActive, a member of class NativeMethods.

## 1.1.20.1.89 NativeMethods.IsMultiTouchReady Method

**C#**

```
public static bool IsMultiTouchReady();
```

**Remarks**

This is IsMultiTouchReady, a member of class NativeMethods.

## 1.1.20.1.90 NativeMethods.IsTabletPC Method

**C#**

```
public static bool IsTabletPC();
```

**Remarks**

This is IsTabletPC, a member of class NativeMethods.

## 1.1.20.1.91 NativeMethods.IsUnicodeFile Method

**C#**

```
public static bool IsUnicodeFile(string fileName);
```

**Remarks**

This is IsUnicodeFile, a member of class NativeMethods.

## 1.1.20.1.92 NativeMethods.IsValidFileName Method

**C#**

```
public static bool IsValidFileName(string fileName);
```

**Remarks**

This is IsValidFileName, a member of class NativeMethods.

## 1.1.20.1.93 NativeMethods.IsValidPathName Method

**C#**

```
public static bool IsValidPathName(string fileName);
```

**Remarks**

This is IsValidPathName, a member of class NativeMethods.

## 1.1.20.1.94 NativeMethods.IsWindows10 Method

**C#**

```
public static bool IsWindows10();
```

**Remarks**

This is IsWindows10, a member of class NativeMethods.

## 1.1.20.1.95 NativeMethods.IsWindows7 Method

**C#**

```
public static bool IsWindows7();
```

**Remarks**

This is IsWindows7, a member of class NativeMethods.

## 1.1.20.1.96 NativeMethods.IsWindows8 Method

**C#**

```
public static bool IsWindows8();
```

**Remarks**

This is IsWindows8, a member of class NativeMethods.

## 1.1.20.1.97 NativeMethods.IsWindowsVista Method

**C#**

```
public static bool IsWindowsVista();
```

**Remarks**

This is IsWindowsVista, a member of class NativeMethods.

## 1.1.20.1.98 NativeMethods.IsWindowsXP Method

**C#**

```
public static bool IsWindowsXP();
```

**Remarks**

This is IsWindowsXP, a member of class NativeMethods.

## 1.1.20.1.99 NativeMethods.IsWOW64 Method

**C#**

```
public static bool IsWOW64();
```

**Remarks**

This is IsWOW64, a member of class NativeMethods.

## 1.1.20.1.100 NativeMethods.LoadCertificate Method

**C#**

```
public static void LoadCertificate(string fileName, [In, Out] EIDCertificate Certificate);
```

**Remarks**

This is LoadCertificate, a member of class NativeMethods.

## 1.1.20.1.101 NativeMethods.LoadPhoto Method

**C#**

```
public static void LoadPhoto([In, Out] EIDPicture photo, string fileName);
```

**Remarks**

This is LoadPhoto, a member of class NativeMethods.

## 1.1.20.1.102 NativeMethods.PathUnExpandEnvStrings Method

**C#**

```
[DllImport("shlwapi.dll", CharSet = CharSet.Unicode)]
[return: MarshalAs(UnmanagedType.Bool)]
public static extern bool PathUnExpandEnvStrings([MarshalAs(UnmanagedType.LPWStr)] string
pszPath, [Out] StringBuilder pszBuf, int cchBuf);
```

**Remarks**

This is PathUnExpandEnvStrings, a member of class NativeMethods.

## 1.1.20.1.103 NativeMethods.PortAvailable Method

**C#**

```
public static bool PortAvailable(int portNumber);
```

**Remarks**

This is PortAvailable, a member of class NativeMethods.

## 1.1.20.1.104 NativeMethods.ReadAddressEx Method

**C#**

```
public static bool ReadAddressEx(int readerNumber, [In, Out] EIDAddress address);
```

**Remarks**

This is ReadAddressEx, a member of class NativeMethods.

## 1.1.20.1.105 NativeMethods.ReadAuthenticationCertificate Method

**C#**

```
public static bool ReadAuthenticationCertificate([In, Out] EIDCertificate certificate);
```

**Remarks**

This is ReadAuthenticationCertificate, a member of class NativeMethods.

## 1.1.20.1.106 NativeMethods.ReadCaCertificate Method

**C#**

```
public static bool ReadCaCertificate([In, Out] EIDCertificate certificate);
```

**Remarks**

This is ReadCaCertificate, a member of class NativeMethods.

## 1.1.20.1.107 NativeMethods.ReadIdentityEx Method

**C#**

```
public static bool ReadIdentityEx(int readerNumber, [In, Out] EIDIdentity identity);
```

**Remarks**

This is ReadIdentityEx, a member of class NativeMethods.

## 1.1.20.1.108 NativeMethods.ReadNonRepudiationCertificate Method

**C#**

```
public static bool ReadNonRepudiationCertificate([In, Out] EIDCertificate certificate);
```

**Remarks**

This is ReadNonRepudiationCertificate, a member of class NativeMethods.

## 1.1.20.1.109 NativeMethods.ReadPhotoEx Method

**C#**

```
public static bool ReadPhotoEx(int readerNumber, [In, Out] EIDPicture Photo);
```

**Remarks**

This is ReadPhotoEx, a member of class NativeMethods.

## 1.1.20.1.110 NativeMethods.ReadRootCaCertificate Method

**C#**

```
public static bool ReadRootCaCertificate([In, Out] EIDCertificate certificate);
```

**Remarks**

This is ReadRootCaCertificate, a member of class NativeMethods.

## 1.1.20.1.111 NativeMethods.ReadRrnCertificate Method

**C#**

```
public static bool ReadRrnCertificate([In, Out] EIDCertificate certificate);
```

**Remarks**

This is ReadRrnCertificate, a member of class NativeMethods.

## 1.1.20.1.112 NativeMethods.RecycleBinEmpty Method

**C#**

```
public static bool RecycleBinEmpty();
```

**Remarks**

This is RecycleBinEmpty, a member of class NativeMethods.

## 1.1.20.1.113 NativeMethods.ReloadReadersList Method

**C#**

```
public static void ReloadReadersList();
```

**Remarks**

This is ReloadReadersList, a member of class NativeMethods.

## 1.1.20.1.114 NativeMethods.RemoveCallback Method

**C#**

```
public static void RemoveCallback();
```

**Remarks**

This is RemoveCallback, a member of class NativeMethods.

## 1.1.20.1.115 NativeMethods.RemoveStartup Method

**C#**

```
public static void RemoveStartup(string AppName);
```

**Remarks**

This is RemoveStartup, a member of class NativeMethods.

## 1.1.20.1.116 NativeMethods.SaveAuthenticationCertificateEx Method

**C#**

```
public static void SaveAuthenticationCertificateEx(string fileName, int readerNumber);
```

**Remarks**

This is SaveAuthenticationCertificateEx, a member of class NativeMethods.

## 1.1.20.1.117 NativeMethods.SaveCaCertificateEx Method

**C#**

```
public static void SaveCaCertificateEx(string fileName, int readerNumber);
```

**Remarks**

This is SaveCaCertificateEx, a member of class NativeMethods.

## 1.1.20.1.118 NativeMethods.SaveCardToToXMLStreamExA Method

**C#**

```
public static bool SaveCardToToXMLStreamExA(int readerNumber, IntPtr buffer);
```

**Remarks**

This is SaveCardToToXMLStreamExA, a member of class NativeMethods.

## 1.1.20.1.119 NativeMethods.SaveCardToToXMLStreamExW Method

**C#**

```
public static bool SaveCardToToXMLStreamExW(int readerNumber, IntPtr buffer);
```

**Remarks**

This is SaveCardToToXMLStreamExW, a member of class NativeMethods.

## 1.1.20.1.120 NativeMethods.SaveCardToXmlEx Method

**C#**

```
public static bool SaveCardToXmlEx(int readerNumber, string fileName);
```

**Remarks**

This is SaveCardToXmlEx, a member of class NativeMethods.

## 1.1.20.1.121 NativeMethods.SaveCardToXmlExAnsi Method

**C#**

```
public static bool SaveCardToXmlExAnsi(int readerNumber, string fileName);
```

**Remarks**

This is SaveCardToXmlExAnsi, a member of class NativeMethods.

## 1.1.20.1.122 NativeMethods.SaveContainer Method

**C#**

```
public static bool SaveContainer(IntPtr container, string fileName);
```

**Remarks**

This is SaveContainer, a member of class NativeMethods.

## 1.1.20.1.123 NativeMethods.SaveNonRepudiationCertificateEx Method

**C#**

```
public static void SaveNonRepudiationCertificateEx(string fileName, int readerNumber);
```

**Remarks**

This is SaveNonRepudiationCertificateEx, a member of class NativeMethods.

## 1.1.20.1.124 NativeMethods.SavePersonCsvToStreamA Method

**C#**

```
public static bool SavePersonCsvToStreamA(int readerNumber, IntPtr buffer);
```

**Remarks**

This is SavePersonCsvToStreamA, a member of class NativeMethods.

## 1.1.20.1.125 NativeMethods.SavePersonCsvToStreamW Method

**C#**

```
public static bool SavePersonCsvToStreamW(int readerNumber, IntPtr buffer);
```

**Remarks**

This is SavePersonCsvToStreamW, a member of class NativeMethods.

## 1.1.20.1.126 NativeMethods.SavePersonToCsvEx Method

**C#**

```
public static bool SavePersonToCsvEx(int readerNumber, string fileName);
```

**Remarks**

This is SavePersonToCsvEx, a member of class NativeMethods.

## 1.1.20.1.127 NativeMethods.SavePersonToCsvExAnsi Method

**C#**

```
public static bool SavePersonToCsvExAnsi(int readerNumber, string fileName);
```

**Remarks**

This is SavePersonToCsvExAnsi, a member of class NativeMethods.

## 1.1.20.1.128 NativeMethods.SavePhotoAsBitmapEx Method

**C#**

```
public static bool SavePhotoAsBitmapEx(int readerNumber, string fileName);
```

**Remarks**

This is SavePhotoAsBitmapEx, a member of class NativeMethods.

## 1.1.20.1.129 NativeMethods.SavePhotoAsJpegEx Method

**C#**

```
public static bool SavePhotoAsJpegEx(int readerNumber, string fileName);
```

**Remarks**

This is SavePhotoAsJpegEx, a member of class NativeMethods.

## 1.1.20.1.130 NativeMethods.SaveRootCaCertificateEx Method

**C#**

```
public static void SaveRootCaCertificateEx(string fileName, int readerNumber);
```

**Remarks**

This is SaveRootCaCertificateEx, a member of class NativeMethods.

## 1.1.20.1.131 NativeMethods.SaveRrnCertificateEx Method

**C#**

```
public static void SaveRrnCertificateEx(string fileName, int readerNumber);
```

**Remarks**

This is SaveRrnCertificateEx, a member of class NativeMethods.

## 1.1.20.1.132 NativeMethods.SelectReader Method

**C#**

```
public static bool SelectReader(int readerNumber);
```

**Remarks**

This is SelectReader, a member of class NativeMethods.

## 1.1.20.1.133 NativeMethods.SelectReaderByName Method

**C#**

```
public static bool SelectReaderByName(string ReaderName);
```

**Remarks**

This is SelectReaderByName, a member of class NativeMethods.

## 1.1.20.1.134 NativeMethods.SendAPDU Method

**C#**

```
public static bool SendAPDU(int readerNumber, byte[] apdu, int apduLen, byte[] result, ref
int len);
```

**Remarks**

This is SendAPDU, a member of class NativeMethods.

## 1.1.20.1.135 NativeMethods.SetCallback Method

**C#**

```
public static void SetCallback(ReaderCallbackDelegate callback, IntPtr userContext);
```

**Remarks**

This is SetCallback, a member of class NativeMethods.

## 1.1.20.1.136 NativeMethods.SetStartup Method

**C#**

```
public static void SetStartup(string AppName, string AppPath);
```

**Remarks**

This is SetStartup, a member of class NativeMethods.

## 1.1.20.1.137 NativeMethods.ShellCopyFile Method

**C#**

```
public static void ShellCopyFile(string oldName, string NewName);
```

**Remarks**

This is ShellCopyFile, a member of class NativeMethods.

## 1.1.20.1.138 NativeMethods.ShowError Method

**C#**

```
public static void ShowError(int errorCode);
```

**Remarks**

This is ShowError, a member of class NativeMethods.

## 1.1.20.1.139 NativeMethods.ShutdownWindows Method

**C#**

```
public static bool ShutdownWindows(int Flags);
```

**Remarks**

This is ShutdownWindows, a member of class NativeMethods.

## 1.1.20.1.140 NativeMethods.StartEngine Method

**C#**

```
public static bool StartEngine();
```

**Remarks**

This is StartEngine, a member of class NativeMethods.

## 1.1.20.1.141 NativeMethods.StopEngine Method

**C#**

```
public static void StopEngine();
```

**Remarks**

This is StopEngine, a member of class NativeMethods.

## 1.1.20.1.142 NativeMethods.StripFileName Method

**C#**

```
public static void StripFileName(string fileName, StringBuilder sb);
```

**Remarks**

This is StripFileName, a member of class NativeMethods.

## 1.1.20.1.143 NativeMethods.SuspendWindows Method

**C#**

```
public static bool SuspendWindows();
```

**Remarks**

This is SuspendWindows, a member of class NativeMethods.

## 1.1.20.1.144 NativeMethods.TurnMonitorOff Method

**C#**

```
public static void TurnMonitorOff();
```

**Remarks**

This is TurnMonitorOff, a member of class NativeMethods.

## 1.1.20.1.145 NativeMethods.TurnMonitorOn Method

**C#**

```
public static void TurnMonitorOn();
```

**Remarks**

This is TurnMonitorOn, a member of class NativeMethods.

## 1.1.20.1.146 NativeMethods.VerifyPinEx Method

**C#**

```
public static bool VerifyPinEx(int readerNumber, string Value);
```

**Remarks**

This is VerifyPinEx, a member of class NativeMethods.

## 1.1.20.1.147 NativeMethods.VerifySignature Method

**C#**

```
public static bool VerifySignature([In, Out] EIDCertificate Certificate, byte[] dataHash,
int hashSize, byte[] signature, int signatureSize);
```

**Remarks**

This is VerifySignature, a member of class NativeMethods.

# 1.1.21 Network Class

**Inheritance Hierarchy**

Swelio.Engine.Network

**C#**

```
public static class Network;
```

**File**

Network.cs (see page 117)

**Remarks**

This is class Swelio.Engine.Network.

**Members**

**Network Methods**

| | Name | Description |
|---|---|---|
| S | PortAvailable (see page 98) | Ports the available. |

**Network Properties**

| | Name | Description |
|---|---|---|
| R S | CurrentIPAddress (see page 99) | Gets the current IP address. |
| R S | IsConnectedToInternet (see page 99) | Gets a value indicating whether PC is connected to internet. |

**Network Methods**

| | Name | Description |
|---|---|---|
| S | PortAvailable (see page 98) | Ports the available. |

**Network Properties**

| | Name | Description |
|---|---|---|
| R S | CurrentIPAddress (see page 99) | Gets the current IP address. |
| R S | IsConnectedToInternet (see page 99) | Gets a value indicating whether PC is connected to internet. |

## 1.1.21.1 Network Methods

### 1.1.21.1.1 Network.PortAvailable Method

Ports the available.

**C#**

```
public static bool PortAvailable(int portNumber);
```

**Parameters**

| Parameters | Description |
|---|---|
| int portNumber | The port number. |

## 1.1.21.2 Network Properties

### 1.1.21.2.1 Network.CurrentIPAddress Property

Gets the current IP address.

**C#**

```
public static string CurrentIPAddress;
```

### 1.1.21.2.2 Network.IsConnectedToInternet Property

Gets a value indicating whether PC is connected to internet.

**C#**

```
public static bool IsConnectedToInternet;
```

**Description**

`true` if this PC is connected to internet; otherwise, `false`.

## 1.1.22 PostalAddress Class

**Inheritance Hierarchy**

Swelio.Engine.PostalAddress

**C#**

```
public class PostalAddress;
```

**File**

PostalAddress.cs (see page 118)

**Remarks**

This is class Swelio.Engine.PostalAddress.

**Members**

**Methods**

| | Name | Description |
|---|---|---|
| | PostalAddress (see page 100) | This is PostalAddress, a member of class PostalAddress. |

**PostalAddress Properties**

| | Name | Description |
|---|---|---|
| | House (see page 100) | This is House, a member of class PostalAddress. |
| | Municipality (see page 100) | This is Municipality, a member of class PostalAddress. |
| | Street (see page 100) | This is Street, a member of class PostalAddress. |
| | Zip (see page 100) | This is Zip, a member of class PostalAddress. |

**Methods**

| | Name | Description |
|---|---|---|
| | PostalAddress (see page 100) | This is PostalAddress, a member of class PostalAddress. |

**PostalAddress Properties**

| | Name | Description |
|---|---|---|
| | House (see page 100) | This is House, a member of class PostalAddress. |

| | | |
|---|---|---|
| | Municipality (see page 100) | This is Municipality, a member of class PostalAddress. |
| | Street (see page 100) | This is Street, a member of class PostalAddress. |
| | Zip (see page 100) | This is Zip, a member of class PostalAddress. |

# 1.1.22.1 PostalAddress.PostalAddress Constructor

**C#**

```
public PostalAddress(Address value);
```

**Remarks**

This is PostalAddress, a member of class PostalAddress.

# 1.1.22.2 PostalAddress Properties

## 1.1.22.2.1 PostalAddress.House Property

**C#**

```
public string House;
```

**Remarks**

This is House, a member of class PostalAddress.

## 1.1.22.2.2 PostalAddress.Municipality Property

**C#**

```
public string Municipality;
```

**Remarks**

This is Municipality, a member of class PostalAddress.

## 1.1.22.2.3 PostalAddress.Street Property

**C#**

```
public string Street;
```

**Remarks**

This is Street, a member of class PostalAddress.

## 1.1.22.2.4 PostalAddress.Zip Property

**C#**

```
public string Zip;
```

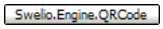
**Remarks**

This is Zip, a member of class PostalAddress.

# 1.1.23 QRCode Class

QR Code encoder class

**Inheritance Hierarchy**




**C#**

```
public static class QRCode;
```

**File**

QRCode.cs (see page 118)

**Members**

**QRCode Methods**

| | Name | Description |
|---|---|---|
| S | GenerateBitmap (see page 101) | Generates the bitmap. |
| S | GeneratePng (see page 102) | Generates the PNG. |
| S | GetBitmap (see page 102) | Gets the bitmap. |

**QRCode Methods**

| | Name | Description |
|---|---|---|
| S | GenerateBitmap (see page 101) | Generates the bitmap. |
| S | GeneratePng (see page 102) | Generates the PNG. |
| S | GetBitmap (see page 102) | Gets the bitmap. |

## 1.1.23.1 QRCode Methods

### 1.1.23.1.1 GenerateBitmap Method

#### 1.1.23.1.1.1 QRCode.GenerateBitmap Method (string, string, ErrorCorrectionLevel)

Generates the bitmap.

**C#**

```
public static void GenerateBitmap(string fileName, string text, ErrorCorrectionLevel level);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| string text | The text. |

#### 1.1.23.1.1.2 QRCode.GenerateBitmap Method (string, string, int, int, ErrorCorrectionLevel)

Generates the bitmap file with encoded string

**C#**

```
public static void GenerateBitmap(string fileName, string text, int margin, int pixelSize,
```

```
ErrorCorrectionLevel level);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| string text | The text to encode. |
| int margin | The margin. |
| int pixelSize | Size of the pixel. |

## 1.1.23.1.2 GeneratePng Method

### 1.1.23.1.2.1 QRCode.GeneratePng Method (string, string, ErrorCorrectionLevel)

Generates the PNG.

**C#**

```
public static void GeneratePng(string fileName, string text, ErrorCorrectionLevel level);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| string text | The text. |

### 1.1.23.1.2.2 QRCode.GeneratePng Method (string, string, int, int, ErrorCorrectionLevel)

Generates the PNG.

**C#**

```
public static void GeneratePng(string fileName, string text, int margin, int pixelSize,
ErrorCorrectionLevel level);
```

**Parameters**

| Parameters | Description |
|---|---|
| string fileName | Name of the file. |
| string text | The text. |
| int margin | The margin. |
| int pixelSize | Size of the pixel. |

## 1.1.23.1.3 GetBitmap Method

### 1.1.23.1.3.1 QRCode.GetBitmap Method (string, ErrorCorrectionLevel)

Gets the bitmap.

**C#**

```
public static Bitmap GetBitmap(string text, ErrorCorrectionLevel level);
```

**Parameters**

| Parameters | Description |
|---|---|
| string text | The text. |

### 1.1.23.1.3.2 QRCode.GetBitmap Method (string, int, int, ErrorCorrectionLevel)

Gets the bitmap.

**C#**

```
public static Bitmap GetBitmap(string text, int margin, int pixelSize, ErrorCorrectionLevel level);
```

**Parameters**

| Parameters | Description |
|---|---|
| string text | The text. |
| int margin | The margin. |
| int pixelSize | Size of the pixel. |

# 1.1.24 RecycleBin Class

Send (see page 104) files directly to the recycle bin.

**Inheritance Hierarchy**

Swelio.Engine.RecycleBin

**C#**

```
public static class RecycleBin;
```

**File**

RecycleBin.cs (see page 118)

**Members**

**RecycleBin Methods**

| | Name | Description |
|---|---|---|
| S | Clear (see page 104) | This is Clear, a member of class RecycleBin. |
| S | Send (see page 104) | Send file to recycle bin. Display dialog, display warning if files are too big to fit |
| S | SendSilent (see page 104) | Send (see page 104) file silently to recycle bin. Surpress dialog, surpress errors, delete if too large. |

**RecycleBin Properties**

| | Name | Description |
|---|---|---|
| R S | IsEmpty (see page 104) | Gets a value indicating whether thee recycle bin is empty. |

**RecycleBin Methods**

| | Name | Description |
|---|---|---|
| S | Clear (see page 104) | This is Clear, a member of class RecycleBin. |
| S | Send (see page 104) | Send file to recycle bin. Display dialog, display warning if files are too big to fit |
| S | SendSilent (see page 104) | Send (see page 104) file silently to recycle bin. Surpress dialog, surpress errors, delete if too large. |

**RecycleBin Properties**

| | Name | Description |
|---|---|---|
| R S | IsEmpty (see page 104) | Gets a value indicating whether thee recycle bin is empty. |

# 1.1.24.1 RecycleBin Methods

### 1.1.24.1.1 RecycleBin.Clear Method

**C#**

```
public static void Clear();
```

**Remarks**

This is Clear, a member of class RecycleBin.

### 1.1.24.1.2 RecycleBin.Send Method

Send file to recycle bin. Display dialog, display warning if files are too big to fit

**C#**

```
public static void Send(string path);
```

**Parameters**

| Parameters | Description |
|---|---|
| string path | Location of directory or file to recycle |

### 1.1.24.1.3 RecycleBin.SendSilent Method

Send (see page 104) file silently to recycle bin. Surpress dialog, surpress errors, delete if too large.

**C#**

```
public static void SendSilent(string path);
```

**Parameters**

| Parameters | Description |
|---|---|
| string path | Location of directory or file to recycle |

## 1.1.24.2 RecycleBin Properties

### 1.1.24.2.1 RecycleBin.IsEmpty Property

Gets a value indicating whether thee recycle bin is empty.

**C#**

```
public static bool IsEmpty;
```

**Description**

`true` if recycle bin is empty; otherwise, `false`.

# 1.1.25 WindowsOperations Class

Windows power management class

**Inheritance Hierarchy**

Swelio.Engine.WindowsOperations

**C#**

```
public static class WindowsOperations;
```

**File**

WindowsOperations.cs (see page 118)

**Members**

**WindowsOperations Methods**

| | Name | Description |
|---|---|---|
| | Hibernate (see page 105) | Hibernate computer |
| | Shutdown (see page 105) | Shutdown Computer |
| | Suspend (see page 105) | Suspends computer |

**WindowsOperations Methods**

| | Name | Description |
|---|---|---|
| | Hibernate (see page 105) | Hibernate computer |
| | Shutdown (see page 105) | Shutdown Computer |
| | Suspend (see page 105) | Suspends computer |

## 1.1.25.1 WindowsOperations Methods

### 1.1.25.1.1 WindowsOperations.Hibernate Method

Hibernate computer

**C#**

```
public static void Hibernate();
```

### 1.1.25.1.2 WindowsOperations.Shutdown Method

Shutdown Computer

**C#**

```
public static void Shutdown(ShutdownOption option);
```

**Parameters**

| Parameters | Description |
|---|---|
| ShutdownOption option | The option. |

### 1.1.25.1.3 WindowsOperations.Suspend Method

Suspends computer

**C#**

```
public static void Suspend();
```

# 1.1.26 Swelio.Engine.CardEvent Enumeration

The code of the card event

**C#**

```
internal enum CardEvent : int {
  CardInsert = 1,
  CardRemove = 2,
  ReadersChange = 3,
  UnknownEvent = 0
}
```

**File**

CardEvent.cs (see page 113)

## 1.1.27 Swelio.Engine.CertificateType Enumeration

The type of the cerificate

**C#**

```
public enum CertificateType {
  CaCertificate,
  RootCaCertificate,
  RrnCertificate,
  AuthenticationCertificate,
  NonRepudiationCertificate
}
```

**File**

CertificateType.cs (see page 114)

## 1.1.28 Swelio.Engine.ChangeWindowMessageFilterFlags Enumeration

**C#**

```
internal enum ChangeWindowMessageFilterFlags : int {
  Add = 1,
  Remove = 2
}
```

**File**

NativeTypes.cs (see page 117)

**Remarks**

This is record Swelio.Engine.ChangeWindowMessageFilterFlags.

## 1.1.29 Swelio.Engine.DocumentType Enumeration

**C#**

```
public enum DocumentType {
  Unknown,
  BelgianCitizen,
  EuropeanCommunity,
  NonEuropeanCommunity,
```

```
  KidsCard,
  BootstrapCard,
  HabilitationCard,
  ForeignerCard
}
```

**File**

DocumentType.cs (see page 114)

**Remarks**

This is record Swelio.Engine.DocumentType.

## 1.1.30 Swelio.Engine.ErrorCorrectionLevel Enumeration

**C#**

```
public enum ErrorCorrectionLevel : int {
  LowQuality,
  MediumQuality,
  StandardQuality,
  HighQuality
}
```

**File**

QRCode.cs (see page 118)

**Remarks**

This is record Swelio.Engine.ErrorCorrectionLevel.

## 1.1.31 Swelio.Engine.ExportFormat Enumeration

**C#**

```
public enum ExportFormat {
  Csv,
  Xml
}
```

**File**

ExportFormat.cs (see page 115)

**Remarks**

This is record Swelio.Engine.ExportFormat.

## 1.1.32 Swelio.Engine.FileEncoding Enumeration

**C#**

```
public enum FileEncoding {
  Ansi,
  Utf16,
  Utf8
}
```

**File**

FileEncoding.cs (see page 115)

**Remarks**

This is record Swelio.Engine.FileEncoding.

## 1.1.33 Swelio.Engine.Gender Enumeration

**C#**

```
public enum Gender {
  Male,
  Female
}
```

**File**

Gender.cs (see page 116)

**Remarks**

This is record Swelio.Engine.Gender.

## 1.1.34 Swelio.Engine.PhotoFormat Enumeration

**C#**

```
public enum PhotoFormat {
  Bmp,
  Jpeg
}
```

**File**

PhotoFormat.cs (see page 118)

**Remarks**

This is record Swelio.Engine.PhotoFormat.

## 1.1.35 Swelio.Engine.ShutdownOption Enumeration

Windows shutdown options

**C#**

```
public enum ShutdownOption {
  LogOff = 0,
  PowerOff = 8,
  Reboot = 2,
  Shutdown = 1,
  Suspend = -1,
  Hibernate = -2
}
```

**File**

WindowsOperations.cs (see page 118)

**Members**

| Members | Description |
|---|---|
| LogOff = 0 | Shuts down all processes running in the security context of the process that called the ExitWindowsEx function. Then it logs the user off. |
| PowerOff = 8 | Shuts down the system and turns off the power. The system must support the power-off feature. |
| Reboot = 2 | Shuts down the system and then restarts the system. |
| Shutdown = 1 | Shuts down the system to a point at which it is safe to turn off the power. All file buffers have been flushed to disk, and all running processes have stopped. If the system supports the power-off feature, the power is also turned off. |
| Suspend = -1 | Suspends the system. |
| Hibernate = -2 | Hibernates the system. |

## 1.1.36 Swelio.Engine.ReaderCallbackDelegate Type

**C#**

```
[UnmanagedFunctionPointer(CallingConvention.Winapi, CharSet = CharSet.Unicode)]
public delegate void ReaderCallbackDelegate(ref int readerNumber, ref int eventCode, IntPtr
userContext);
```

**File**

NativeMethods.cs (see page 117)

**Remarks**

This is type Swelio.Engine.ReaderCallbackDelegate.

# 1.2 Files

The following table lists files in this documentation.

**Files**

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**Namespaces**

| Name | Description |
|---|---|
| Swelio.Engine (see page 1) | This is namespace Swelio.Engine. |

## 1.2.2 Card.cs

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**Namespaces**

| Name | Description |
|---|---|
| Swelio.Engine (see page 1) | This is namespace Swelio.Engine. |

## 1.2.3 CardDate.cs

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**Namespaces**

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|---|---|
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## 1.2.4 CardEvent.cs

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**Namespaces**

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|---|---|
| Swelio.Engine (see page 1) | This is namespace Swelio.Engine. |

## 1.2.5 CardEventArgs.cs

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## 1.2.6 CardNumber.cs

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## 1.2.7 CardReader.cs

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## 1.2.8 CertificateType.cs

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**Namespaces**

| Name | Description |
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## 1.2.9 DocumentContainer.cs

This is file DocumentContainer.cs.

**Namespaces**

| Name | Description |
|---|---|
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## 1.2.10 DocumentType.cs

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## 1.2.11 Encryption.cs

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## 1.2.12 ExportFormat.cs

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## 1.2.13 FileEncoding.cs

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## 1.2.14 FileOperations.cs

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## 1.2.15 Gender.cs

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## 1.2.16 Identity.cs

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## 1.2.17 InteropHelper.cs

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## 1.2.18 Manager.cs

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## 1.2.19 NationalNumber.cs

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## 1.2.20 NativeMethods.cs

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## 1.2.21 NativeTypes.cs

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## 1.2.22 Network.cs

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## 1.2.23 PhotoFormat.cs

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**Namespaces**

| Name | Description |
|---|---|
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## 1.2.24 PostalAddress.cs

This is file PostalAddress.cs.

**Namespaces**

| Name | Description |
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## 1.2.25 QRCode.cs

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## 1.2.26 RecycleBin.cs

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## 1.2.27 WindowsOperations.cs

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# Index


## A

Add enumeration member 106
Address class 2
- about Address class 2
- Address 3
- Municipality 3
- Parse 3
- Street 4
- Zip 4

Address.cs 112
Ansi enumeration member 107
AuthenticationCertificate enumeration member 106

## B

BelgianCitizen enumeration member 106
Bmp enumeration member 108
BootstrapCard enumeration member 106

## C

CaCertificate enumeration member 106
Card class 4
- about Card class 4
- AppletVersion 9
- Card 5
- DisplayCertificate 6
- Dispose 6
- EidCard 10
- EncodeCertificate 6
- EncodePhoto 7
- Inserted 10
- ReadAddress 7
- ReadCertificate 7
- ReadIdentity 7
- ReadPhoto 8
- Save 8
- SaveCertificate 8
- SavePhoto 9
- SaveQRCode 9
- SerialNumber 10
- Tag 10
- VerifyPinCode 9

Card.cs 113
CardDate class 10
- about CardDate class 10
- ToDate 11
- ToLongDateString 11
- ToShortDateString 12

CardDate.cs 113
CardEvent.cs 113
CardEventArgs class 12
- about CardEventArgs class 12
- CardEventArgs 12
- ReaderNumber 13

CardEventArgs.cs 113
CardInsert enumeration member 105
CardNumber class 13
- about CardNumber class 13
- ToString 13

CardNumber.cs 114
CardReader class 14
- about CardReader class 14
- ActivateCard 16
- card 15
- CardActivated 17
- CardPresent 17
- CardReader 15
- DeactivateCard 16
- Dispose 16
- GetCard 16
- Index 17
- Name 17
- Selected 17
- Tag 17

CardReader.cs 114
CardRemove enumeration member 105
CertificateType.cs 114
Csv enumeration member 107

## D

DocumentContainer class 18
about DocumentContainer class 18
AddFile 19
Close 19
DocumentContainer 18
Save 19
SelectCertificate 19, 20
DocumentContainer.cs 114
DocumentType.cs 114

## E

EIDAddress class 20
about EIDAddress class 20
municipality 21
street 21
zip 21
EIDCertificate class 21
about EIDCertificate class 21
certificate 22
certificateLength 22
EIDDataLength class 22
about EIDDataLength class 22
EidMaxBirthDateLen 24
EidMaxBirthplaceLen 24
EidMaxBrexitMention1Len 24
EidMaxBrexitMention2Len 24
EidMaxCardNumberLen 24
EidMaxCertLen 24
EidMaxChipNumberLen 25
EidMaxDateAndCountryOfProtectionLen 25
EidMaxDateBeginLen 25
EidMaxDateEndLen 25
EidMaxDeliveryMunicipalityLen 25
EidMaxDocumentTypeLen 25
EidMaxDuplicateLen 25
EidMaxFirstNameLen 25
EidMaxMemberOfFamilyLen 26
EidMaxMiddleNameLen 26
EidMaxMunicipalityName 26
EidMaxNameLen 26
EidMaxNationalityLen 26
EidMaxNationalNumberLen 26
EidMaxNobleConditionLen 26
EidMaxPictureLen 27
EidMaxRegionalFileNumberLen 27
EidMaxSexLen 27
EidMaxSpecialOrganizationLen 27
EidMaxSpecialStatusLen 27
EidMaxStreetNameLen 27
EidMaxVAT1Len 27
EidMaxVAT2Len 28
EidMaxWorkPermitTypeLen 28
EidMaxZipLen 28
EIDError class 28
about EIDError class 28
Code 29
Description 29
EIDIdentity class 29
about EIDIdentity class 29
birthDate 31
birthLocation 31
brexitMention1 31
brexitMention2 31
cardNumber 31
chipNumber 31
dateAndCountryOfProtection 32
documentType 32
duplicate 32
extendedMinority 32
firstName1 32
firstName2 32
memberOfFamily 33
municipality 33
name 33
nationality 33
nationalNumber 33
nobleCondition 33
regionalFileNumber 33
sex 34

specialOrganization 34
validityDateBegin 34
validityDateEnd 34
vat1 34
vat2 34
whiteCane 35
workPermitType 35
yellowCane 35
EIDPicture class 35
about EIDPicture class 35
picture 36
pictureLength 36
Encryption class 36
about Encryption class 36
CheckFileMD5 37
CheckFileSHA1 37
CheckMD5 37, 38
CheckSHA1 38
DecryptFile 38, 39
EncryptFile 39
GenerateAuthenticationSignature 39
GenerateNonRepudiationSignature 40
GetFileMD5 40
GetFileSHA1 40
GetMD5 40, 41
GetSHA1 41
VerifySignature 41
Encryption.cs 115
EuropeanCommunity enumeration member 106
ExportFormat.cs 115


## F


Female enumeration member 108
FileEncoding.cs 115
FileOperations class 42
about FileOperations class 42
ClearFileAttributes 44
CopyFile 44
CreateUnicodeFile 44
DeleteFile 44
DeleteToRecycleBin 44
DirectoryExists 45
ExcludeTrailingPathDelimiter 45
FileExists 45
FileExtensionIs 45
FileIsAnimatedGif 45
FileIsExe 45
FileIsIcon 46
FileIsImage 46
FileIsLink 46
FileOrFolderExists 46
FullPath 46
GetExtension 47
GetFilesCount 47
GetFileSize 47
IncludeTrailingPathDelimiter 47
IsDirectory 47
IsUnicodeFile 48
IsValidFileName 48
IsValidPathName 48
RenameFile 48
ShellCopyFile 48
StripFileName 48
UnExpandPath 49
FileOperations.cs 115
Files 109
ForeignerCard enumeration member 106


## G


Gender.cs 116


## H


HabilitationCard enumeration member 106
Hibernate enumeration member 108
HighQuality enumeration member 107


## I


Identity class 49
about Identity class 49
BirthDate 51
BirthLocation 51

BrexitMention1 51
BrexitMention2 51
CardNumber 52
ChipNumber 52
Clear 51
DateAndCountryOfProtection 52
DocumentType 52
Duplicate 52
ExtendedMinority 52
FirstName1 53
FirstName2 53
MemberOfFamily 53
Municipality 53
Nationality 53
NationalityISO 53
NationalNumber 54
NobleCondition 54
RegionalFileNumber 54
Sex 54
SpecialOrganization 54
Surname 54
ValidityDateBegin 54
ValidityDateEnd 55
Vat1 55
Vat2 55
WhiteCane 55
WorkPermitType 55
YellowCane 55
Identity.cs 116
InteropHelper class 56
about InteropHelper class 56
ClearUnusedMemory 57
GetStartup 57
IsMediaCenter 58
IsMultiTouchReady 58
IsTabletPC 58
MetroActive 59
RunningOnVista 59
RunningOnWin10 59
RunningOnWin7 59
RunningOnWin8 59
RunningOnXP 59
SetStartup 57
ShowScreenKeyboard 57
ShowWindowsError 58
TurnMonitorOff 58
TurnMonitorOn 58
InteropHelper.cs 116

## J

Jpeg enumeration member 108

## K

KidsCard enumeration member 106

## L

LogOff enumeration member 108
LowQuality enumeration member 107

## M

Male enumeration member 108
Manager class 60
about Manager class 60
Active 63
CardInserted 64
CardRemoved 64
Dispose 61
GetReader 61, 62
GetReaderName 62
ListReaders 62
Manager 61
ReaderCount 63
ReadersListChanged 64
ReloadReadersList 62
SendAPDU 62
SetDefaultReader 62
TraceEvents 63
TraceHardwareEvents 63
TraceServiceEvents 63
Manager.cs 116
MediumQuality enumeration member 107

## N

NationalNumber class 64
about NationalNumber class 64
ToString 65
NationalNumber.cs 117
NativeMethods class 65
about NativeMethods class 65
ActivateCardEx 73
AddFileToContainer 73
CardDecryptFile 73
CardEncryptFile 74
CardSignCadesT 74
CardSignCadesTEx 74
CardSignCMS 74
CardSignCMSEx 74
CertSignCadesT 74
CertSignCadesTData 75
CertSignCadesTEx 75
CertSignCMS 75
CertSignCMSData 75
CertSignCMSEx 75
CheckMD5 75
CheckSHA1 76
ClearFileAttributes 76
ClearUnusedMemory 76
ContainerCertificate 76
ContainerEidCertificate 76
ContainerPickCertificate 76
CreateCardBuffer 77
CreateUnicodeFile 77
CurrentIPAddress 77
DeactivateCardEx 77
DecryptFileAES 77
DeleteCardBuffer 77
DeleteObject 78
DeleteToRecycleBin 78
DirectoryExists 78
DisplayCertificate 78
EmptyRecycleBin 78
EncodeCertificate 78
EncodePhoto 79
EncryptFileAES 79
FileCopy 79
FileDelete 79
FileExists 79
FileExtensionIs 79
FileGetSize 80
FileIsExe 80
FileIsIcon 80
FileIsImage 80
FileIsLink 80
FileOrFolderExists 80
FileRename 81
FreeContainer 81
FullPath 81
GenerateAuthenticationSignatureEx 81
GenerateBMP 81
GenerateNonRepudiationSignatureEx 81
GeneratePNG 82
GenerateQRCodeEx 82
GetCardBufferA 82
GetCardBufferSize 82
GetCardBufferW 82
GetCardSerialNumber 82
GetCardVersion 83
GetEncodedCertificateSize 83
GetEncodedPhotoSize 83
GetFileMD5 83
GetFilesCount 83
GetFileSHA1 83
GetHBitmap 84
GetISOCode 84
GetMD5 84
GetReaderIndex 84
GetReaderName 84
GetReaderNameLen 84
GetReadersCount 85
GetSelectedReaderIndex 85
GetSHA1 85
GetStartup 85
HibernateWindows 85

IgnoreHardwareEvents 85
IgnoreServiceEvents 86
InitializeContainer 86
IsAnimatedGIF 86
IsCardActivatedEx 86
IsCardPresentEx 86
IsCardStillInsertedEx 86
IsConnectedToInternet 87
IsDirectory 87
IsEIDCardEx 87
IsEngineActive 87
IsMale 87
IsMediaCenter 87
IsMetroActive 88
IsMultiTouchReady 88
IsTabletPC 88
IsUnicodeFile 88
IsValidFileName 88
IsValidPathName 88
IsWindows10 89
IsWindows7 89
IsWindows8 89
IsWindowsVista 89
IsWindowsXP 89
IsWOW64 89
LoadCertificate 90
LoadPhoto 90
PathUnExpandEnvStrings 90
PortAvailable 90
ReadAddressEx 90
ReadAuthenticationCertificate 90
ReadCaCertificate 91
ReadIdentityEx 91
ReadNonRepudiationCertificate 91
ReadPhotoEx 91
ReadRootCaCertificate 91
ReadRrnCertificate 91
RecycleBinEmpty 92
ReloadReadersList 92
RemoveCallback 92
RemoveStartup 92
SaveAuthenticationCertificateEx 92
SaveCaCertificateEx 92
SaveCardToToXMLStreamExA 93
SaveCardToToXMLStreamExW 93
SaveCardToXmlEx 93
SaveCardToXmlExAnsi 93
SaveContainer 93
SaveNonRepudiationCertificateEx 93
SavePersonCsvToStreamA 94
SavePersonCsvToStreamW 94
SavePersonToCsvEx 94
SavePersonToCsvExAnsi 94
SavePhotoAsBitmapEx 94
SavePhotoAsJpegEx 94
SaveRootCaCertificateEx 95
SaveRrnCertificateEx 95
SelectReader 95
SelectReaderByName 95
SendAPDU 95
SetCallback 95
SetStartup 96
ShellCopyFile 96
ShowError 96
ShutdownWindows 96
StartEngine 96
StopEngine 96
StripFileName 97
SuspendWindows 97
TurnMonitorOff 97
TurnMonitorOn 97
VerifyPinEx 97
VerifySignature 97
NativeMethods.cs 117
NativeTypes.cs 117
Network class 98
about Network class 98
CurrentIPAddress 99
IsConnectedToInternet 99
PortAvailable 98
Network.cs 117
NonEuropeanCommunity enumeration member 106

NonRepudiationCertificate enumeration member 106

## P

PhotoFormat.cs 118
PostalAddress class 99
about PostalAddress class 99
House 100
Municipality 100
PostalAddress 100
Street 100
Zip 100
PostalAddress.cs 118
PowerOff enumeration member 108

## Q

QRCode class 101
about QRCode class 101
GenerateBitmap 101
GeneratePng 102
GetBitmap 102
QRCode.cs 118

## R

ReadersChange enumeration member 105
Reboot enumeration member 108
RecycleBin class 103
about RecycleBin class 103
Clear 104
IsEmpty 104
Send 104
SendSilent 104
RecycleBin.cs 118
Remove enumeration member 106
RootCaCertificate enumeration member 106
RrnCertificate enumeration member 106

## S

Shutdown enumeration member 108
StandardQuality enumeration member 107
Suspend enumeration member 108
Swelio.Engine 1
Swelio.Engine namespace 1
Swelio.Engine.Address 2
Swelio.Engine.Address.Address 3
Swelio.Engine.Address.Municipality 3
Swelio.Engine.Address.Parse 3
Swelio.Engine.Address.Street 4
Swelio.Engine.Address.Zip 4
Swelio.Engine.Card 4
Swelio.Engine.Card.AppletVersion 9
Swelio.Engine.Card.Card 5
Swelio.Engine.Card.DisplayCertificate 6
Swelio.Engine.Card.Dispose 6
Swelio.Engine.Card.EidCard 10
Swelio.Engine.Card.EncodeCertificate 6
Swelio.Engine.Card.EncodePhoto 7
Swelio.Engine.Card.Inserted 10
Swelio.Engine.Card.ReadAddress 7
Swelio.Engine.Card.ReadCertificate 7
Swelio.Engine.Card.ReadIdentity 7
Swelio.Engine.Card.ReadPhoto 8
Swelio.Engine.Card.Save 8
Swelio.Engine.Card.SaveCertificate 8
Swelio.Engine.Card.SavePhoto 9
Swelio.Engine.Card.SaveQRCode 9
Swelio.Engine.Card.SerialNumber 10
Swelio.Engine.Card.Tag 10
Swelio.Engine.Card.VerifyPinCode 9
Swelio.Engine.CardDate 10
Swelio.Engine.CardDate.ToDate 11
Swelio.Engine.CardDate.ToLongDateString 11
Swelio.Engine.CardDate.ToShortDateString 12
Swelio.Engine.CardEvent 105
Swelio.Engine.CardEvent enumeration 105
Swelio.Engine.CardEventArgs 12
Swelio.Engine.CardEventArgs.CardEventArgs 12
Swelio.Engine.CardEventArgs.ReaderNumber 13
Swelio.Engine.CardNumber 13
Swelio.Engine.CardNumber.ToString 13
Swelio.Engine.CardReader 14
Swelio.Engine.CardReader.ActivateCard 16

Swelio.Engine.CardReader.card 15
Swelio.Engine.CardReader.CardActivated 17
Swelio.Engine.CardReader.CardPresent 17
Swelio.Engine.CardReader.CardReader 15
Swelio.Engine.CardReader.DeactivateCard 16
Swelio.Engine.CardReader.Dispose 16
Swelio.Engine.CardReader.GetCard 16
Swelio.Engine.CardReader.Index 17
Swelio.Engine.CardReader.Name 17
Swelio.Engine.CardReader.Selected 17
Swelio.Engine.CardReader.Tag 17
Swelio.Engine.CertificateType 106
Swelio.Engine.CertificateType enumeration 106
Swelio.Engine.ChangeWindowMessageFilterFlags 106
Swelio.Engine.ChangeWindowMessageFilterFlags enumeration 106
Swelio.Engine.DocumentContainer 18
Swelio.Engine.DocumentContainer.AddFile 19
Swelio.Engine.DocumentContainer.Close 19
Swelio.Engine.DocumentContainer.DocumentContainer 18
Swelio.Engine.DocumentContainer.Save 19
Swelio.Engine.DocumentContainer.SelectCertificate 19, 20
Swelio.Engine.DocumentType 106
Swelio.Engine.DocumentType enumeration 106
Swelio.Engine.EIDAddress 20
Swelio.Engine.EIDAddress.municipality 21
Swelio.Engine.EIDAddress.street 21
Swelio.Engine.EIDAddress.zip 21
Swelio.Engine.EIDCertificate 21
Swelio.Engine.EIDCertificate.certificate 22
Swelio.Engine.EIDCertificate.certificateLength 22
Swelio.Engine.EIDDataLength 22
Swelio.Engine.EIDDataLength.EidMaxBirthDateLen 24
Swelio.Engine.EIDDataLength.EidMaxBirthplaceLen 24
Swelio.Engine.EIDDataLength.EidMaxBrexitMention1Len 24
Swelio.Engine.EIDDataLength.EidMaxBrexitMention2Len 24
Swelio.Engine.EIDDataLength.EidMaxCardNumberLen 24
Swelio.Engine.EIDDataLength.EidMaxCertLen 24
Swelio.Engine.EIDDataLength.EidMaxChipNumberLen 25
Swelio.Engine.EIDDataLength.EidMaxDateAndCountryOfProtectionLen 25
Swelio.Engine.EIDDataLength.EidMaxDateBeginLen 25
Swelio.Engine.EIDDataLength.EidMaxDateEndLen 25
Swelio.Engine.EIDDataLength.EidMaxDeliveryMunicipalityLen 25
Swelio.Engine.EIDDataLength.EidMaxDocumentTypeLen 25
Swelio.Engine.EIDDataLength.EidMaxDuplicateLen 25
Swelio.Engine.EIDDataLength.EidMaxFirstNameLen 25
Swelio.Engine.EIDDataLength.EidMaxMemberOfFamilyLen 26
Swelio.Engine.EIDDataLength.EidMaxMiddleNameLen 26
Swelio.Engine.EIDDataLength.EidMaxMunicipalityName 26
Swelio.Engine.EIDDataLength.EidMaxNameLen 26
Swelio.Engine.EIDDataLength.EidMaxNationalityLen 26
Swelio.Engine.EIDDataLength.EidMaxNationalNumberLen 26
Swelio.Engine.EIDDataLength.EidMaxNobleConditionLen 26
Swelio.Engine.EIDDataLength.EidMaxPictureLen 27
Swelio.Engine.EIDDataLength.EidMaxRegionalFileNumberLen 27
Swelio.Engine.EIDDataLength.EidMaxSexLen 27
Swelio.Engine.EIDDataLength.EidMaxSpecialOrganizationLen 27
Swelio.Engine.EIDDataLength.EidMaxSpecialStatusLen 27
Swelio.Engine.EIDDataLength.EidMaxStreetNameLen 27
Swelio.Engine.EIDDataLength.EidMaxVAT1Len 27
Swelio.Engine.EIDDataLength.EidMaxVAT2Len 28
Swelio.Engine.EIDDataLength.EidMaxWorkPermitTypeLen 28
Swelio.Engine.EIDDataLength.EidMaxZipLen 28
Swelio.Engine.EIDError 28
Swelio.Engine.EIDError.Code 29
Swelio.Engine.EIDError.Description 29
Swelio.Engine.EIDIdentity 29
Swelio.Engine.EIDIdentity.birthDate 31
Swelio.Engine.EIDIdentity.birthLocation 31
Swelio.Engine.EIDIdentity.brexitMention1 31
Swelio.Engine.EIDIdentity.brexitMention2 31
Swelio.Engine.EIDIdentity.cardNumber 31
Swelio.Engine.EIDIdentity.chipNumber 31
Swelio.Engine.EIDIdentity.dateAndCountryOfProtection 32
Swelio.Engine.EIDIdentity.documentType 32
Swelio.Engine.EIDIdentity.duplicate 32
Swelio.Engine.EIDIdentity.extendedMinority 32

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Swelio.Engine.EIDIdentity.firstName1 32
Swelio.Engine.EIDIdentity.firstName2 32
Swelio.Engine.EIDIdentity.memberOfFamily 33
Swelio.Engine.EIDIdentity.municipality 33
Swelio.Engine.EIDIdentity.name 33
Swelio.Engine.EIDIdentity.nationality 33
Swelio.Engine.EIDIdentity.nationalNumber 33
Swelio.Engine.EIDIdentity.nobleCondition 33
Swelio.Engine.EIDIdentity.regionalFileNumber 33
Swelio.Engine.EIDIdentity.sex 34
Swelio.Engine.EIDIdentity.specialOrganization 34
Swelio.Engine.EIDIdentity.validityDateBegin 34
Swelio.Engine.EIDIdentity.validityDateEnd 34
Swelio.Engine.EIDIdentity.vat1 34
Swelio.Engine.EIDIdentity.vat2 34
Swelio.Engine.EIDIdentity.whiteCane 35
Swelio.Engine.EIDIdentity.workPermitType 35
Swelio.Engine.EIDIdentity.yellowCane 35
Swelio.Engine.EIDPicture 35
Swelio.Engine.EIDPicture.picture 36
Swelio.Engine.EIDPicture.pictureLength 36
Swelio.Engine.Encryption 36
Swelio.Engine.Encryption.CheckFileMD5 37
Swelio.Engine.Encryption.CheckFileSHA1 37
Swelio.Engine.Encryption.CheckMD5 37, 38
Swelio.Engine.Encryption.CheckSHA1 38
Swelio.Engine.Encryption.DecryptFile 38, 39
Swelio.Engine.Encryption.EncryptFile 39
Swelio.Engine.Encryption.GenerateAuthenticationSignature 39
Swelio.Engine.Encryption.GenerateNonRepudiationSignature 40
Swelio.Engine.Encryption.GetFileMD5 40
Swelio.Engine.Encryption.GetFileSHA1 40
Swelio.Engine.Encryption.GetMD5 40, 41
Swelio.Engine.Encryption.GetSHA1 41
Swelio.Engine.Encryption.VerifySignature 41
Swelio.Engine.ErrorCorrectionLevel 107
Swelio.Engine.ErrorCorrectionLevel enumeration 107
Swelio.Engine.ExportFormat 107
Swelio.Engine.ExportFormat enumeration 107
Swelio.Engine.FileEncoding 107
Swelio.Engine.FileEncoding enumeration 107
Swelio.Engine.FileOperations 42
Swelio.Engine.FileOperations.ClearFileAttributes 44
Swelio.Engine.FileOperations.CopyFile 44
Swelio.Engine.FileOperations.CreateUnicodeFile 44
Swelio.Engine.FileOperations.DeleteFile 44
Swelio.Engine.FileOperations.DeleteToRecycleBin 44
Swelio.Engine.FileOperations.DirectoryExists 45
Swelio.Engine.FileOperations.ExcludeTrailingPathDelimiter 45
Swelio.Engine.FileOperations.FileExists 45
Swelio.Engine.FileOperations.FileExtensionIs 45
Swelio.Engine.FileOperations.FileIsAnimatedGif 45
Swelio.Engine.FileOperations.FileIsExe 45
Swelio.Engine.FileOperations.FileIsIcon 46
Swelio.Engine.FileOperations.FileIsImage 46
Swelio.Engine.FileOperations.FileIsLink 46
Swelio.Engine.FileOperations.FileOrFolderExists 46
Swelio.Engine.FileOperations.FullPath 46
Swelio.Engine.FileOperations.GetExtension 47
Swelio.Engine.FileOperations.GetFilesCount 47
Swelio.Engine.FileOperations.GetFileSize 47
Swelio.Engine.FileOperations.IncludeTrailingPathDelimiter 47
Swelio.Engine.FileOperations.IsDirectory 47
Swelio.Engine.FileOperations.IsUnicodeFile 48
Swelio.Engine.FileOperations.IsValidFileName 48
Swelio.Engine.FileOperations.IsValidPathName 48
Swelio.Engine.FileOperations.RenameFile 48
Swelio.Engine.FileOperations.ShellCopyFile 48
Swelio.Engine.FileOperations.StripFileName 48
Swelio.Engine.FileOperations.UnExpandPath 49
Swelio.Engine.Gender 108
Swelio.Engine.Gender enumeration 108
Swelio.Engine.Identity 49
Swelio.Engine.Identity.BirthDate 51
Swelio.Engine.Identity.BirthLocation 51
Swelio.Engine.Identity.BrexitMention1 51
Swelio.Engine.Identity.BrexitMention2 51
Swelio.Engine.Identity.CardNumber 52
Swelio.Engine.Identity.ChipNumber 52
Swelio.Engine.Identity.Clear 51
Swelio.Engine.Identity.DateAndCountryOfProtection 52

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Swelio.Engine.Identity.DocumentType 52
Swelio.Engine.Identity.Duplicate 52
Swelio.Engine.Identity.ExtendedMinority 52
Swelio.Engine.Identity.FirstName1 53
Swelio.Engine.Identity.FirstName2 53
Swelio.Engine.Identity.MemberOfFamily 53
Swelio.Engine.Identity.Municipality 53
Swelio.Engine.Identity.Nationality 53
Swelio.Engine.Identity.NationalityISO 53
Swelio.Engine.Identity.NationalNumber 54
Swelio.Engine.Identity.NobleCondition 54
Swelio.Engine.Identity.RegionalFileNumber 54
Swelio.Engine.Identity.Sex 54
Swelio.Engine.Identity.SpecialOrganization 54
Swelio.Engine.Identity.Surname 54
Swelio.Engine.Identity.ValidityDateBegin 54
Swelio.Engine.Identity.ValidityDateEnd 55
Swelio.Engine.Identity.Vat1 55
Swelio.Engine.Identity.Vat2 55
Swelio.Engine.Identity.WhiteCane 55
Swelio.Engine.Identity.WorkPermitType 55
Swelio.Engine.Identity.YellowCane 55
Swelio.Engine.InteropHelper 56
Swelio.Engine.InteropHelper.ClearUnusedMemory 57
Swelio.Engine.InteropHelper.GetStartup 57
Swelio.Engine.InteropHelper.IsMediaCenter 58
Swelio.Engine.InteropHelper.IsMultiTouchReady 58
Swelio.Engine.InteropHelper.IsTabletPC 58
Swelio.Engine.InteropHelper.MetroActive 59
Swelio.Engine.InteropHelper.RunningOnVista 59
Swelio.Engine.InteropHelper.RunningOnWin10 59
Swelio.Engine.InteropHelper.RunningOnWin7 59
Swelio.Engine.InteropHelper.RunningOnWin8 59
Swelio.Engine.InteropHelper.RunningOnXP 59
Swelio.Engine.InteropHelper.SetStartup 57
Swelio.Engine.InteropHelper.ShowScreenKeyboard 57
Swelio.Engine.InteropHelper.ShowWindowsError 58
Swelio.Engine.InteropHelper.TurnMonitorOff 58
Swelio.Engine.InteropHelper.TurnMonitorOn 58
Swelio.Engine.Manager 60
Swelio.Engine.Manager.Active 63
Swelio.Engine.Manager.CardInserted 64
Swelio.Engine.Manager.CardRemoved 64
Swelio.Engine.Manager.Dispose 61
Swelio.Engine.Manager.GetReader 61, 62
Swelio.Engine.Manager.GetReaderName 62
Swelio.Engine.Manager.ListReaders 62
Swelio.Engine.Manager.Manager 61
Swelio.Engine.Manager.ReaderCount 63
Swelio.Engine.Manager.ReadersListChanged 64
Swelio.Engine.Manager.ReloadReadersList 62
Swelio.Engine.Manager.SendAPDU 62
Swelio.Engine.Manager.SetDefaultReader 62
Swelio.Engine.Manager.TraceEvents 63
Swelio.Engine.Manager.TraceHardwareEvents 63
Swelio.Engine.Manager.TraceServiceEvents 63
Swelio.Engine.NationalNumber 64
Swelio.Engine.NationalNumber.ToString 65
Swelio.Engine.NativeMethods 65
Swelio.Engine.NativeMethods.ActivateCardEx 73
Swelio.Engine.NativeMethods.AddFileToContainer 73
Swelio.Engine.NativeMethods.CardDecryptFile 73
Swelio.Engine.NativeMethods.CardEncryptFile 74
Swelio.Engine.NativeMethods.CardSignCadesT 74
Swelio.Engine.NativeMethods.CardSignCadesTEx 74
Swelio.Engine.NativeMethods.CardSignCMS 74
Swelio.Engine.NativeMethods.CardSignCMSEx 74
Swelio.Engine.NativeMethods.CertSignCadesT 74
Swelio.Engine.NativeMethods.CertSignCadesTData 75
Swelio.Engine.NativeMethods.CertSignCadesTEx 75
Swelio.Engine.NativeMethods.CertSignCMS 75
Swelio.Engine.NativeMethods.CertSignCMSData 75
Swelio.Engine.NativeMethods.CertSignCMSEx 75
Swelio.Engine.NativeMethods.CheckMD5 75
Swelio.Engine.NativeMethods.CheckSHA1 76
Swelio.Engine.NativeMethods.ClearFileAttributes 76
Swelio.Engine.NativeMethods.ClearUnusedMemory 76
Swelio.Engine.NativeMethods.ContainerCertificate 76
Swelio.Engine.NativeMethods.ContainerEidCertificate 76
Swelio.Engine.NativeMethods.ContainerPickCertificate 76
Swelio.Engine.NativeMethods.CreateCardBuffer 77
Swelio.Engine.NativeMethods.CreateUnicodeFile 77

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Swelio.Engine.NativeMethods.CurrentIPAddress 77
Swelio.Engine.NativeMethods.DeactivateCardEx 77
Swelio.Engine.NativeMethods.DecryptFileAES 77
Swelio.Engine.NativeMethods.DeleteCardBuffer 77
Swelio.Engine.NativeMethods.DeleteObject 78
Swelio.Engine.NativeMethods.DeleteToRecycleBin 78
Swelio.Engine.NativeMethods.DirectoryExists 78
Swelio.Engine.NativeMethods.DisplayCertificate 78
Swelio.Engine.NativeMethods.EmptyRecycleBin 78
Swelio.Engine.NativeMethods.EncodeCertificate 78
Swelio.Engine.NativeMethods.EncodePhoto 79
Swelio.Engine.NativeMethods.EncryptFileAES 79
Swelio.Engine.NativeMethods.FileCopy 79
Swelio.Engine.NativeMethods.FileDelete 79
Swelio.Engine.NativeMethods.FileExists 79
Swelio.Engine.NativeMethods.FileExtensionIs 79
Swelio.Engine.NativeMethods.FileGetSize 80
Swelio.Engine.NativeMethods.FileIsExe 80
Swelio.Engine.NativeMethods.FileIsIcon 80
Swelio.Engine.NativeMethods.FileIsImage 80
Swelio.Engine.NativeMethods.FileIsLink 80
Swelio.Engine.NativeMethods.FileOrFolderExists 80
Swelio.Engine.NativeMethods.FileRename 81
Swelio.Engine.NativeMethods.FreeContainer 81
Swelio.Engine.NativeMethods.FullPath 81
Swelio.Engine.NativeMethods.GenerateAuthenticationSignatureEx 81
Swelio.Engine.NativeMethods.GenerateBMP 81
Swelio.Engine.NativeMethods.GenerateNonRepudiationSignatureEx 81
Swelio.Engine.NativeMethods.GeneratePNG 82
Swelio.Engine.NativeMethods.GenerateQRCodeEx 82
Swelio.Engine.NativeMethods.GetCardBufferA 82
Swelio.Engine.NativeMethods.GetCardBufferSize 82
Swelio.Engine.NativeMethods.GetCardBufferW 82
Swelio.Engine.NativeMethods.GetCardSerialNumber 82
Swelio.Engine.NativeMethods.GetCardVersion 83
Swelio.Engine.NativeMethods.GetEncodedCertificateSize 83
Swelio.Engine.NativeMethods.GetEncodedPhotoSize 83
Swelio.Engine.NativeMethods.GetFileMD5 83
Swelio.Engine.NativeMethods.GetFilesCount 83
Swelio.Engine.NativeMethods.GetFileSHA1 83
Swelio.Engine.NativeMethods.GetHBitmap 84
Swelio.Engine.NativeMethods.GetISOCode 84
Swelio.Engine.NativeMethods.GetMD5 84
Swelio.Engine.NativeMethods.GetReaderIndex 84
Swelio.Engine.NativeMethods.GetReaderName 84
Swelio.Engine.NativeMethods.GetReaderNameLen 84
Swelio.Engine.NativeMethods.GetReadersCount 85
Swelio.Engine.NativeMethods.GetSelectedReaderIndex 85
Swelio.Engine.NativeMethods.GetSHA1 85
Swelio.Engine.NativeMethods.GetStartup 85
Swelio.Engine.NativeMethods.HibernateWindows 85
Swelio.Engine.NativeMethods.IgnoreHardwareEvents 85
Swelio.Engine.NativeMethods.IgnoreServiceEvents 86
Swelio.Engine.NativeMethods.InitializeContainer 86
Swelio.Engine.NativeMethods.IsAnimatedGIF 86
Swelio.Engine.NativeMethods.IsCardActivatedEx 86
Swelio.Engine.NativeMethods.IsCardPresentEx 86
Swelio.Engine.NativeMethods.IsCardStillInsertedEx 86
Swelio.Engine.NativeMethods.IsConnectedToInternet 87
Swelio.Engine.NativeMethods.IsDirectory 87
Swelio.Engine.NativeMethods.IsEIDCardEx 87
Swelio.Engine.NativeMethods.IsEngineActive 87
Swelio.Engine.NativeMethods.IsMale 87
Swelio.Engine.NativeMethods.IsMediaCenter 87
Swelio.Engine.NativeMethods.IsMetroActive 88
Swelio.Engine.NativeMethods.IsMultiTouchReady 88
Swelio.Engine.NativeMethods.IsTabletPC 88
Swelio.Engine.NativeMethods.IsUnicodeFile 88
Swelio.Engine.NativeMethods.IsValidFileName 88
Swelio.Engine.NativeMethods.IsValidPathName 88
Swelio.Engine.NativeMethods.IsWindows10 89
Swelio.Engine.NativeMethods.IsWindows7 89
Swelio.Engine.NativeMethods.IsWindows8 89
Swelio.Engine.NativeMethods.IsWindowsVista 89
Swelio.Engine.NativeMethods.IsWindowsXP 89
Swelio.Engine.NativeMethods.IsWOW64 89
Swelio.Engine.NativeMethods.LoadCertificate 90
Swelio.Engine.NativeMethods.LoadPhoto 90
Swelio.Engine.NativeMethods.PathUnExpandEnvStrings 90
Swelio.Engine.NativeMethods.PortAvailable 90

*Made with **Doc-O-Matic**.*

Swelio.Engine.NativeMethods.ReadAddressEx 90
Swelio.Engine.NativeMethods.ReadAuthenticationCertificate 90
Swelio.Engine.NativeMethods.ReadCaCertificate 91
Swelio.Engine.NativeMethods.ReadIdentityEx 91
Swelio.Engine.NativeMethods.ReadNonRepudiationCertificate 91
Swelio.Engine.NativeMethods.ReadPhotoEx 91
Swelio.Engine.NativeMethods.ReadRootCaCertificate 91
Swelio.Engine.NativeMethods.ReadRrnCertificate 91
Swelio.Engine.NativeMethods.RecycleBinEmpty 92
Swelio.Engine.NativeMethods.ReloadReadersList 92
Swelio.Engine.NativeMethods.RemoveCallback 92
Swelio.Engine.NativeMethods.RemoveStartup 92
Swelio.Engine.NativeMethods.SaveAuthenticationCertificateEx 92
Swelio.Engine.NativeMethods.SaveCaCertificateEx 92
Swelio.Engine.NativeMethods.SaveCardToToXMLStreamExA 93
Swelio.Engine.NativeMethods.SaveCardToToXMLStreamExW 93
Swelio.Engine.NativeMethods.SaveCardToXmlEx 93
Swelio.Engine.NativeMethods.SaveCardToXmlExAnsi 93
Swelio.Engine.NativeMethods.SaveContainer 93
Swelio.Engine.NativeMethods.SaveNonRepudiationCertificateEx 93
Swelio.Engine.NativeMethods.SavePersonCsvToStreamA 94
Swelio.Engine.NativeMethods.SavePersonCsvToStreamW 94
Swelio.Engine.NativeMethods.SavePersonToCsvEx 94
Swelio.Engine.NativeMethods.SavePersonToCsvExAnsi 94
Swelio.Engine.NativeMethods.SavePhotoAsBitmapEx 94
Swelio.Engine.NativeMethods.SavePhotoAsJpegEx 94
Swelio.Engine.NativeMethods.SaveRootCaCertificateEx 95
Swelio.Engine.NativeMethods.SaveRrnCertificateEx 95
Swelio.Engine.NativeMethods.SelectReader 95
Swelio.Engine.NativeMethods.SelectReaderByName 95
Swelio.Engine.NativeMethods.SendAPDU 95
Swelio.Engine.NativeMethods.SetCallback 95
Swelio.Engine.NativeMethods.SetStartup 96
Swelio.Engine.NativeMethods.ShellCopyFile 96
Swelio.Engine.NativeMethods.ShowError 96
Swelio.Engine.NativeMethods.ShutdownWindows 96
Swelio.Engine.NativeMethods.StartEngine 96
Swelio.Engine.NativeMethods.StopEngine 96
Swelio.Engine.NativeMethods.StripFileName 97
Swelio.Engine.NativeMethods.SuspendWindows 97
Swelio.Engine.NativeMethods.TurnMonitorOff 97
Swelio.Engine.NativeMethods.TurnMonitorOn 97
Swelio.Engine.NativeMethods.VerifyPinEx 97
Swelio.Engine.NativeMethods.VerifySignature 97
Swelio.Engine.Network 98
Swelio.Engine.Network.CurrentIPAddress 99
Swelio.Engine.Network.IsConnectedToInternet 99
Swelio.Engine.Network.PortAvailable 98
Swelio.Engine.PhotoFormat 108
Swelio.Engine.PhotoFormat enumeration 108
Swelio.Engine.PostalAddress 99
Swelio.Engine.PostalAddress.House 100
Swelio.Engine.PostalAddress.Municipality 100
Swelio.Engine.PostalAddress.PostalAddress 100
Swelio.Engine.PostalAddress.Street 100
Swelio.Engine.PostalAddress.Zip 100
Swelio.Engine.QRCode 101
Swelio.Engine.QRCode.GenerateBitmap 101
Swelio.Engine.QRCode.GeneratePng 102
Swelio.Engine.QRCode.GetBitmap 102
Swelio.Engine.ReaderCallbackDelegate 109
Swelio.Engine.ReaderCallbackDelegate type 109
Swelio.Engine.RecycleBin 103
Swelio.Engine.RecycleBin.Clear 104
Swelio.Engine.RecycleBin.IsEmpty 104
Swelio.Engine.RecycleBin.Send 104
Swelio.Engine.RecycleBin.SendSilent 104
Swelio.Engine.ShutdownOption 108
Swelio.Engine.ShutdownOption enumeration 108
Swelio.Engine.WindowsOperations 104
Swelio.Engine.WindowsOperations.Hibernate 105
Swelio.Engine.WindowsOperations.Shutdown 105
Swelio.Engine.WindowsOperations.Suspend 105

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## U

Unknown enumeration member 106
UnknownEvent enumeration member 105
Utf16 enumeration member 107
Utf8 enumeration member 107

## W

WindowsOperations class 104
- about WindowsOperations class 104
- Hibernate 105
- Shutdown 105
- Suspend 105

WindowsOperations.cs 118

## X

Xml enumeration member 107